

The weighted Dyck Newton triangle (OEIS A258219, A258220 and A292692)

Bala's Riccati and continued-fraction conjecture, Kotěšovec's asymptotic
form and growth rate for the Newton diagonal, all fixed orders on the interior,
and the upper-edge transition

Three research reports of one external research session
(bundle Reports 141, 142 and 143), merged into one ProveIt report

Parts I and II: 2 October 2026; Part III: 3 October 2026 merged: 5 October 2026

Abstract

Give a peak at (a, b) of a Dyck path the weight $k + a/b$ and let $p_N(k)$ be the total weight of the paths of semilength N (OEIS A258219). Its falling-factorial (Newton) coefficients $P_{N,m} = [k^m]p_N(k)$ form the triangle A258220, whose diagonal $a_n = P_{2n,n}$ is A292692. This report prints three manuscripts as Parts I–III, in the order in which each depends on the one before. Part I proves the Riccati identity $F - 1 = kxF + xF^2 + 2x^2F_x$ for the generating function and its Stieltjes continued fraction, conjectured by Peter Bala on A258219 (July 28, 2022), through an exact identity for all weighted prefixes; it then proves $a_n \sim Cd^n n!/\sqrt{n}$ with $d = (51\sqrt{17} - 107)/4$ and $C = \sqrt{2 + 2/\sqrt{17}}/\pi^{3/2}$, which establishes the asymptotic form stated numerically by Václav Kotěšovec on A292692 (December 1, 2017) and his conjectured rate (March 17, 2024), with the amplitude in closed form, and adds the first relative correction and a Lambert- W threshold bracket. Part II proves every fixed order of the normalized Newton ratio $R_{N,m}$ uniformly on every compact subset of $0 < m/N < 1$, removing Part I's contraction restriction, with coefficients $\sqrt{1-s^2}$ times rational functions of the saddle parameter, all diagonal coefficients in $\mathbb{Q}(\sqrt{17})$, the second one explicitly, and a Lambert- W inverse at every fixed order. Part III treats the upper edge $m = N - q$: exact two-sided bounds for every $q \leq N/2$, a connected profile uniformly for $q = o(\sqrt{N})$ built from the four-regular-map sequence A292186, its expansion to every fixed order, its strict monotonicity, and a ceiling-safe inverse obtained by polynomial reversion, without Lambert W . Asymptotics in the range $\sqrt{N} \lesssim q = o(N)$ between Parts II and III and at the lower edge $m = o(N)$, effective constants and convergence of the expansions remain open. The report is unrefereed, and nothing in it is formalized.

Contents

Guide to this report	4
What is proved, and where	5
The OEIS statements, and the inverses	6
Notation across the three Parts	7
Provenance and merge decisions	8
What was checked, and what was not	9

Relation to neighbouring reports and formal projects	10
I A weighted Dyck path Newton diagonal	11
1 Definitions and results	11
2 All weighted prefixes and the Riccati equation	13
3 The formal hypergeometric identity and continued fraction	15
4 Positive Newton transforms	16
5 A uniform large power local limit estimate	17
6 Endpoint and middle estimates for products	18
7 A positive contraction proves the moving ratio limit	21
8 The diagonal constants	23
9 The first correction with a uniform remainder	23
10 Lambert W inversion with integer rounding	29
11 Exact computations and reproducibility	31
12 Further questions	31
II All orders for the weighted Dyck Newton diagonal	33
13 Definitions and main results	33
14 Exact transforms and the positive convolution	35
15 A uniform shifted saddle expansion	36
16 A positive estimate away from the endpoints	38
17 Polynomial coefficients at both endpoints	39
18 The exact coefficient algorithm and rationality	42
19 The uniform remainder on the full interior	43
20 Diagonal normalization and the second correction	46
21 All orders of the integer safe inverse	48
22 Reproducibility and scope	49
III Quantified upper edge transition of a weighted Dyck Newton triangle	51
23 The model and the contribution	51
24 Exact bounds and the connected profile	53

25 A nonlinear bound uniform in both indices	55
26 Positive bounds for the Newton conversion	57
27 Assembly and explicit error bounds	58
28 The classical sequence with a controlled remainder	59
29 Binomial smoothing and the profile correction	61
30 All fixed orders of the edge profile	62
31 An increasing profile and a ceiling-safe inverse	65
32 Comparison with the compact interior	67
33 Further questions and precise limitations	68
34 Reproducibility and proof status	68
35 Further questions and research (all three Parts)	69

Guide to this report

The three Parts are three manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit e85586b7c (batch 105). They study one object, the weighted Dyck polynomials $p_N(k)$ and their Newton triangle:

Part	Bundle report	Delivered title	Labels
I	141	A weighted Dyck path Newton diagonal	<code>wdn:dia:</code>
II	142 (base)	All orders for the weighted Dyck Newton diagonal	<code>wdn:int:</code>
III	143	Quantified upper edge transition of a weighted Dyck Newton triangle	<code>wdn:edge:</code>

Arrangement. The base of the merge is Report 142: it proves every analytic statement of Report 141 under weaker hypotheses (every compact subset of the whole interior $0 < m/N < 1$ instead of Report 141’s contraction range, every fixed order instead of the first correction), and its `Report142.tex` was staged as this `article.tex`. It is printed second, as Part II, because it is not self-contained: it takes the exact weighted-path bridge and the convolution identity from Report 141, which its package carries unchanged. Report 141 is therefore printed in full, as Part I; it is also the only Part that proves Bala’s conjecture and the continued fraction, and its contraction proof of the leading and first-corrected ratio is a different route from Part II’s. Report 143 treats a different regime, the upper edge $m = N - q$; it starts from the recurrence that Part I proves and compares its result with Part II’s interior coefficients, so it is printed last, as Part III. The three Parts thus appear in dependency order, which is also the order of writing.

Embedded copies. Report 142’s archive carries Report 141’s source and PDF, and Report 143’s carries those of Reports 141 and 142; all six files are byte-identical to the standalone deliveries (checked at placement and again at the write). They are printed once each, as Parts I and II, and are not shipped.

Numbering. Sections are numbered continuously and statements within sections, so a statement number moves with its section only. Part I keeps Report 141’s Sections 1–12. Report 142’s Section k is Section $k + 12$ here (Sections 13–22) and its Theorem $k.j$ is Theorem $(k + 12).j$: so its Theorem 1.1 is Theorem 13.1 and its Corollary 1.2 is Corollary 13.2. Report 143’s Section k is Section $k + 22$ here (Sections 23–34): its Theorems 2.1, 2.2, 2.4, 8.1 and 9.1 are Theorems 24.1, 24.2, 24.4, 30.1 and 31.1, and its Corollary 2.3 is Corollary 24.3. The manuscripts number equations consecutively through each document; each Part keeps its manuscript’s equation numbers, prefixed in Parts II and III: Report 142’s equation (k) is (II. k) here and Report 143’s is (III. k). Subsection 22.1 and Section 35 were added in the write; no delivered number moved.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (5 October 2026). They give cross-references between the Parts, mark statements that a later Part supersedes or answers, mark results proved twice, and record what the write checked. Superseded statements and answered questions are printed as delivered, with a dated note. No delivered statement was changed, and no delivered symbol was renamed. Each Part has a section of open questions: Part I its last, Section 12; Part II its last, Subsection 22.1 (added in the write); Part III its Section 33, followed by its reproducibility Section 34; Section 35 gathers them for the whole report under Vladimir’s standing rule of 4 October 2026, with their sources, sketches and what is missing. (Wording corrected after the independent check of 5 October 2026, recorded at the end of Section 35; the first wording read “Each Part ends with open questions: Part I with its Section 12, Part II with Subsection 22.1 (added in the write), Part III with its Section 33”).

What is proved, and where

Here $\alpha = m/N = 2s^2/(1+s)$, $t = 1 - s^2$, $R_{N,m} = P_{N,m}/(2NQ_{N,m})$ is the connected ratio, $Q_{N,m} = [k^m]z_N(k)$ with $z_N(k) = \gamma_N \prod_{i < N} (k + 2i + 1)$ and $\gamma_N = 4^{-N} \binom{2N}{N}$, and $q = N - m$ is the deficit.

Statement	Where
Riccati identity $F - 1 = kxF + xF^2 + 2x^2F_x$, the recurrence $p_N = (2N - 2 + k)p_{N-1} + \sum p_j p_{N-1-j}$, and the S-fraction with numerators $(k+1)x, 3x, (k+3)x, 5x, \dots$ (Bala's conjecture on A258219); the prefix identity $H_{2n+y}(y) = [x^n]G^{y+1}$	Part I, Theorem 1.1, Proposition 2.2, Proposition 3.2
$F = 1 + 2xZ_x/Z$ with $Z = {}_2F_0(\frac{k+1}{2}, \frac{1}{2}; ; 2x)$; $2Nz_N = \sum_{j \geq 1} p_j z_{N-j}$; $0 \leq R_{N,m} \leq 1$	Part I, Lemma 3.1, (20); restated in Part II, Section 14, and Part III, Section 24
$a_n \sim Cd^n n!/\sqrt{n}$, $d = (51\sqrt{17} - 107)/4$, $C = \sqrt{2 + 2/\sqrt{17}}/\pi^{3/2}$ (Kotěšovec's form and rate on A292692)	Part I, Theorem 1.2; every fixed order in Part II, Corollary 13.2
$R_{N,m} \rightarrow \sqrt{1 - s^2}$ on compact subsets of $\text{int}[9/20, 49/85]$; first correction $r_1(s)$ for $\alpha < 6 - 3\sqrt{3}$;	Part I, Theorems 7.1 and 9.1 (positive contraction); superseded by Part II
$b_1 = -\frac{989}{2176} - \frac{907\sqrt{17}}{36992}$ Every fixed order $R_{N,m} = \sum_{j < K} R_j(s)N^{-j} + O(N^{-K})$ on every compact $J \subset (0, 1)$, $R_j = \sqrt{1 - s^2} r_j$ with $r_j \in \mathbb{Q}(s)$, by an explicit finite algorithm; r_2 explicitly	Part II, Theorem 13.1, Sections 18–20
$b_j \in \mathbb{Q}(\sqrt{17})$ for all j ; $b_2 = -(30800095 + 3912441\sqrt{17})/80494592$	Part II, Corollary 13.2
Threshold $\mathcal{N}(X) = \min\{n : a_n \geq X\}$ inside two ceilings around $x_0 = T/W(dT/e)$ (Lambert W), first-corrected and at every fixed order	Part I, Theorem 10.1 and Corollary 10.3; Part II, Section 21
Top coefficients $c_a(n) = [k^{n-a}]p_n$: polynomial in n divisible by n^a (fixed a); $L_a(n - 2a)_+^{2a} \leq c_a(n) \leq L_a(n + a)^{2a}$ for all $a \geq 1$, $n \geq 0$	Part II, Lemma 17.1; Part III, Theorem 24.1
Exact sandwiches $K_q(N - 2q)^{2q} \leq P_{N,N-q} \leq K_q(N + q)^{2q}$ and for $Q_{N,N-q}$, $1 \leq q \leq N/2$;	Part III, Theorem 24.2, Corollary 24.3
$e^{-8q^2/N} \leq \sqrt{N}R_{N,N-q}/F_q \leq e^{(10q^2+1/8)/N}$ for $q \leq N/4$; profile uniformly for $q = o(\sqrt{N})$	
$F_q = 2\sqrt{q/3}\{1 - 7/(16q) + \dots\}$ to every fixed order ($f_2, \dots, f_5$ explicit); F_q strictly increasing; ceiling-safe inverse $J_M(x) = x + \frac{7}{8} + \frac{203}{128x} + \dots$, $x = 3Y^2/4$	Part III, Theorems 24.4, 30.1 and 31.1
The four-regular-map sequence A292186: Riccati equation, logarithmic derivative, $V_n \sim (2\sqrt{2}/\pi)4^n n!$ (prior art, re-derived), with the correction $1 - 3/(8n)$	Part III, Section 28
$R_{N,N-q} \asymp \sqrt{(q+1)/N}$ uniformly for $q \leq c\sqrt{N}$	[write] Remark 35.1

Open in all three Parts

The range $\sqrt{N} \lesssim q = o(N)$ between the interior and the edge theorems, and a limit profile at $q \asymp \sqrt{N}$; the lower edge $m = o(N)$; a bridge uniform across all ratios; effective constants, onsets and certified thresholds; convergence or resummation of the expansions and uniformity in a growing order; combinatorial meanings of $T(N, m)$, $\sqrt{t_0}$ and K_q ; a sharper nonlinear comparison. See Section 35.

The finite computations of the three companions (exact integer, rational and $\mathbb{Q}(\sqrt{17})$ arithmetic) check identities, normalizations and coefficient algebra; no asymptotic statement rests on them.

The leading inverses of Parts I and II are instances of the transseries volume’s factorial core, which becomes its Lambert core after taking logarithms, and Part III’s is a polynomial reversion; see “The OEIS statements, and the inverses” below. (Wording corrected after the independent check of 5 October 2026, recorded at the end of Section 35; the first wording read “The inverses of Parts I and II are instances of the transseries volume’s Lambert core”).

The OEIS statements, and the inverses

The entries as read by the write on 5 October 2026 (A258219 revision #30 of August 15, 2022; A292692 revision #12 of March 17, 2024; A292186 revision #57 of November 14, 2025):

- **A258219** (Alois P. Heinz, May 23, 2015): the square array $A(n, k)$, the sum over Dyck paths of semilength n of the products over peaks of $(x_p + ky_p)/y_p$, that is of $k + x_p/y_p$; so $A(n, k) = p_n(k)$. A comment signed *Peter Bala, Jul 28 2022*, introduced by “Conjecture”, states that the generating function $G(k, x)$ of the k -th column satisfies $2x^2 dG/dx + 1 + (kx - 1)G + xG^2 = 0$ “and hence, by Stokes 1982”, has the Stieltjes continued fraction $1/(1 - (k + 1)x/(1 - 3x/(1 - (k + 3)x/(1 - 5x/\cdots))))$. The differential equation is (4) rearranged. Theorem 1.1 proves it with k an indeterminate, hence for every column, and Proposition 3.2 proves the continued fraction directly.
- **A292692** (Alois P. Heinz, Sep 20, 2017), $a(n) = A258220(2n, n)$. A formula signed *Vaclav Kotesovec, Dec 01 2017* states $a(n) \sim c d^n n!/\sqrt{n}$ with $d = 25.81959672662517201\dots$ and $c = 0.28310309251343879\dots$, numerically and without proof; a formula signed *Vaclav Kotesovec, Mar 17 2024* states “Conjecture: $d = (51\sqrt{17} - 107)/4$ ”. Theorem 1.2 proves the form with this d and with $c = C$ in closed form; at 40 digits $C = 0.28310309251343879373\dots$ and $d = 25.81959672662517201022\dots$, agreeing with every stated digit (computed by the write). Part I’s text dates only the rate conjecture to 2024, correctly; the companions’ 141-diagonal-SOURCES.md and 142-interior-SOURCES.md date “the numerical asymptotic form and conjectural algebraic rate” both to 2024-03-17, which is right for the rate only.
- **A292186** (Sasha Kolpakov, Sep 11, 2017): rooted unlabeled connected four-regular maps on a compact closed oriented surface with n vertices; the entry notes that it is Martin and Kearney’s $S(4, -6, 1)$ sequence with offset 1. A formula block signed *Peter Bala, Aug 22 2023* gives the Riccati equation $4x^2 A' = -1 + (1 - 2x)A - xA^2$ and the S-fraction with numerators $3x, 5x, 7x, \dots$. Part III’s (III.34) is this equation rearranged; Section 28 re-derives it and the leading asymptotic, which are prior art, as the manuscript says.

The inverses. The threshold theorems of Parts I and II solve $\Phi(x) = L$ with $\Phi(x) = x(\log x + \log d - 1) + c$. Writing $T = L - c$, this is $x(\kappa \log x + d') = T$ with $\kappa = 1$ and $d' = \log d - 1$, the factorial core `p0:prop:factorial-core` of the transseries volume [15], whose solution $\exp(W_0(Te^{d'}) - d')$ is $T/W(dT/e) = x_0$; after taking logarithms it is the Lambert core `p0:thm:lambert-core` with $a = b = 1$, in the variable $\log x + d'$ and with right-hand side $\log T + d'$. (Wording corrected after the independent check of 5 October 2026, recorded at the end of Section 35; the first wording read “that proposition is the Lambert core `p0:thm:lambert-core` in the variable $\log x$ ”). The two-ceiling brackets are the separation pattern of `p0:thm:staircase` (1)–(2), proved here directly at the integers without choosing an interpolation of a_n . Part III’s inverse has no Lambert W : after squaring, $\frac{3}{4}F_q^2 = q + g_0 + g_1q^{-1} + \dots$ has a linear core (the case $b = 0$ of `p0:thm:lambert-core`), inverted by formal reversion, with the same ceiling pattern. None of the manuscripts claims novelty

for the inversion; Part I calls it “a standard consequence”.

Notation across the three Parts

Common to all Parts: $p_N(k)$; the Newton coefficient $\mathcal{L}_m P = [k^m]P = \Delta^m P(0)/m!$ (Part III writes $[k^m]$ without $\mathcal{L}$); $P_{N,m} = \mathcal{L}_m p_N$ (Part I also writes $T(N, m)$); $\gamma_N, z_N, Q_{N,m}, R_{N,m}$; $f(t) = (1-t)^{-1/2}$, $h = f - 1$, $C_{N,m} = [t^N]fh^m$ (Parts I, II); $\alpha = m/N = 2s^2/(1+s)$, $t = 1 - s^2$, $s_0 = (1 + \sqrt{17})/8$ (Parts I, II); $c_a(n) = [k^{n-a}]p_n(k)$ (Parts II, III, same meaning). Part I’s generating function $F(x, k)$ is Part III’s $G(x, k)$. The symbols below change meaning between the Parts or inside one; each Part’s text fixes its meaning there, and each Part opens with a short table of its own readings. No symbol was renamed.

Symbol	Meanings
C	The diagonal amplitude $C = \sqrt{2 + 2/\sqrt{17}}/\pi^{3/2}$ (Parts I, II); $C_{N,m} = [t^N]fh^m$ (Parts I, II); $C_\ell(y)$ an array (Part I, Section 2); $C_a(n) = c_a(n)/n^a$ (Part II, (II.32)); $C_q = (3/2)^q/q!$ (Part III, (III.7)); unnamed constants. <i>Tempting false reading</i> : Part II’s polynomial $C_a(n)$ and Part III’s number C_q are unrelated, although both are indexed by a top-coefficient or deficit index.
$B, \mathcal{B}$	Part I: $B(t) = t\mu'(t)$ the saddle variance; $\mathcal{B}_{N,m}$ the opposite endpoint (63). Part II: B the same variance; $B_{N,m}$ the opposite endpoint (II.20) (without calligraphy), $B(\varepsilon, s)$ and $B_q(s)$ its expansion, $\mathcal{B}_s$ the inverse-kernel operator (II.53), $B_s(u, v)$ its symbol, $\mathcal{B}_{2v}$ Bernoulli numbers. Part III: $B_{N,q} = (1 + q/N)^{2q}$; B a Bernoulli variable (Section 31); $\mathcal{B}_j(x)$ Bernoulli polynomials.
d	The rate $d = (51\sqrt{17} - 107)/4$ (Parts I, II). Part I: $d_0 = \ell - r$. Part II: $d = \ell - r$ (Section 17) and a monomial degree. Part III: $d_N = 1/(\sqrt{\pi N}\gamma_N)$ and a moment order d .
h	$h(t) = f(t) - 1$ (Parts I, II; Part II also writes $h = (1 - s)/s$ for $h(t)$). Part III: $h_n = n!L_n$ ((III.43)) and $h = 1/n$ (Section 30). <i>False reading</i> : Part III’s h_A is the sequence h_n at a binomial index, not the function h .
K	Part I: an integer evaluation point $K \approx m/(1 - s)$ (Lemma 6.2), K_0 a constant. Part II: K the expansion order; the integer point is H ; $\mathcal{K}_N, \mathcal{K}_x$ the exact kernel. Part III: K_q the profile numerator (III.7); K the order in Theorem 30.1.
L	Part I: $L = \log X$ (Section 10), $L_N = \lfloor A \log N \rfloor$, $L = \lfloor N/2 \rfloor$ (Section 9), the moments L_1, L_2 . Part II: $L = \lfloor N/2 \rfloor$, $L_N, L = \log X$. Part III: the rational constants L_a (III.5), with $V_a = (2a)!L_a$.
W	The Lambert function (Parts I, II). Part II: $W_{d,r}(x)$ a monomial Newton series (II.34). Part III: $W(x) = \sum w_n x^n$ a hypergeometric series (III.33); Part III uses no Lambert function.
w	Part I: $w_\ell(s) = \gamma_\ell s^{2\ell}$, the limiting kernel masses. Part III: $w_n = 4^n(1/4)_n(3/4)_n/n!$.
λ	Part II: λ_j the inverse coefficients (II.80), $\lambda_1 = b_1 + 1/12$ (Part I’s β_1). Part III: λ_r the Stirling (Bernoulli-polynomial) coefficients (III.62). <i>False reading</i> : Part III’s $\lambda_1 = -1/16$ is not Part II’s λ_1 .
b, β	b_j the diagonal coefficients (Parts I, II). Part I: $b_{N,m,\ell,r}$ the exact kernel (35), $b_{\ell,r}^0$ its limit, $\beta_1 = b_1 + 1/12$, $\beta = \ell - \mu r$. Part II: the exact kernel is $a_{N,m,\ell,r}$, its limit $a_s(\ell, r)$ (Part I’s $b_{\ell,r}^0$), $b_s(\ell, r)$ the inverse-kernel weights, $\beta_{j,r}$ opposite-endpoint weights. Part III: b, c summation indices.
a	a_n the diagonal A292692 (Parts I, II). Parts II, III: a the top-coefficient index of $c_a(n)$. Part III: (a, b) peak coordinates.
μ	$\mu = N/m = 1/\alpha$ (Parts I, II). Part III: $\mu = 2q/3$, the binomial mean.
r	A Newton-index shift (Parts I, II); Part I: $r(\alpha) = \sqrt{1 - s(\alpha)^2}$, $r_1(s)$; Part II: $r_j \in \mathbb{Q}(s)$. Part III: $r_n = w_n/w_{n-1}$ ((III.35)), and Report 142’s r_1 in Section 32.

Symbol	Meanings
R	The connected ratio $R_{N,m}$ (all). Part I: also $R_N(k)$, the polynomials defined by the recurrence (Section 2), $R(a, b; z)$ a hypergeometric ratio, and $R_0(\alpha), R_1(\alpha)$ the coefficient functions (Part II's R_0, R_1).
q	Part I: $q_0(s) = t^{-1/2} - 1$ the kernel mass (also Part II), q and $\bar{q}$ ratio bounds. Part II: q a ratio bound and an expansion order, q_* . Part III: the deficit $q = N - m$.
$D, \mathcal{D}$	Part I: $D = t d/dt$ (Section 9), $D_1(\ell, r)$, $D_0, D_{\ker}$ exponents, $D = 1 - vuF$ in Section 2. Part II: $\mathcal{D} = t d/dt$ (the same operator), $D_j(\ell, r; s)$ the saddle coefficients (D_1 is Part I's), $\mathcal{D}_x$ a perturbation. Part III: $D_\ell(h)$ a product (III.56).
E	Part I: $E = g_1 - k_3/(2B)$ (64), $E_{N,m}$ the discarded mass. Part II: $E = R - H_K$ the error. Part III: $E_a(N) = e_a(1, 3, \dots, 2N - 1)$, E_n an error, and E_n in a coefficient recurrence.
$H, \mathcal{G}$	Part I: $H_\ell(y)$ prefix weights, $H(v, u)$, and $H(N, m)$ the first-order approximation. Part II: H the integer point, $H(\varepsilon, \alpha)$, H_K ; $\mathcal{G}(x, y) = (1 - s^2x(1 - s + sy))^{-1/2}$. Part III: $H(x) = \sum V_n x^n$, $H_K(n)$; $\mathcal{G}(z)$ the squared profile used for reversion.
M	Part I: $M(\alpha_0)$ an envelope, $M = m(1 + h^{-1})$, M, M_N constants. Part II: M_K exponents and witness constants, $M = m(1 + h^{-1})$, $\mathcal{M}$. Part III: $M_d(q)$ binomial central moments, $M = N - a$, M the inverse order, M_K .
κ	Parts I, II: $\kappa_j = D^j \log h$ the saddle cumulants; Part I also $\kappa_{\ell, r}$ the kernel coefficient. Part III: $\kappa_j(p)$ the Bernoulli cumulants.
U, V	Part I: $U(a, b; z)$ a hypergeometric series, $U_{N,m,\ell}$ a product ratio (also Part II), U a sum of uniforms (also Part II), V a random variable, V_{j,d_0} . Part II: $U_{\ell,r}(x)$, $V_{\ell,r}(x)$ ratio series. Part III: $U_{n,\ell} = w_\ell w_{n-\ell}/w_n$; V_a the A292186 terms.
A	A logarithmic cutoff constant (Parts I, II); $\mathcal{A}_s$ (Part II). Part III: A a binomial variable, $A_{N,q}$.
$\mathcal{N}, \mathcal{Q}$	Part I: $\mathcal{N}(X)$ the threshold; Part II writes it $N(X)$. Part III: $\mathcal{Q}(Y)$ the profile threshold.

Provenance and merge decisions

Three manuscripts of the session bundle of Reports 1–243. Their author lines read “Report 141”, “Report 142” and “Report 143”; none names an author, a tool or an addressee (each sets an empty PDF author field), none carries “prepared for private review” wording, and none pins a ProveIt commit or names a repository path. They arrived in commit 60f54ea06 and were placed by e85586b7c (batch 105).

- **Part I**, bundle Report 141, archive A292692_Weighted_Dyck_Diagonal_Asymptotics_and_Inversion_Source.zip (Report141.tex, 1314 lines, 21 pages, dated October 2, 2026). Contributes the prefix identity and Bala’s conjecture, the continued fraction, the hypergeometric logarithmic derivative, the Newton transform of the auxiliary, the local limit estimate, the positive contraction, the leading equivalent with its constants, the first correction and the first two inverse brackets.
- **Part II**, bundle Report 142, archive A292692_All_Orders_Interior_Asymptotics_and_Inversion_Source.zip (Report142.tex, 1041 lines, 16 pages, dated October 2, 2026); the base. Its foundation141/ holds Report 141’s source and PDF byte for byte. Contributes the shifted saddle expansion to every order, the polynomial top coefficients, both endpoint expansions to every order, the coefficient algorithm and its rationality, the frozen inverse with the nested-compact bootstrap, b_2 and r_2 , and the inverse at every fixed order.
- **Part III**, bundle Report 143, archive Weighted_Dyck_Upper_Edge_Transition_All_Orders_and_Inverse_Source.zip (Report143.tex, 1092 lines, 17 pages, dated October 3, 2026).

Its `foundation141/` and `comparison142/` hold the sources and PDFs of Reports 141 and 142 byte for byte. Contributes the simultaneous top-coefficient comparison, the exact sandwiches, the near-diagonal Stirling and elementary-symmetric bounds, the edge profile with explicit errors, its fixed-order expansion, monotonicity and inverse, and the comparison with Part II at $s \rightarrow 1$.

Where the merge had to choose:

1. *Order and base.* Dependency order I–II–III, with the base in the middle (see the Guide). Part I’s superseded statements stay, with dated notes; Part I is kept in full for its unique theorems (the bridge, Bala’s conjecture, the continued fraction) and as an independent elementary route.
2. *Restatements.* In their own words, Part II’s Section 14 restates Part I’s convolution, the positivity of Newton extraction, the auxiliary transform (Lemma 4.1), the multiplication rule and the exact split; Part III’s Subsection 24.1 restates the equivalence of the recurrence with the Riccati equation, the logarithmic derivative, the convolution and the bound $0 \leq R_{N,m} \leq 1$, which it strengthens to $0 < R_{N,m}$ for $0 \leq m \leq N$ (Report 142’s Sections 2–5 share 16 word 8-grams, 0.3%, with Report 141); they are printed in full, because they fix the statements the later Parts use, with notes pointing to Part I. No passage of one manuscript repeats another verbatim. (Lists corrected after the independent check of 5 October 2026, recorded at the end of Section 35; the first wording attributed “Part I’s convolution, Newton transform, multiplication rule and logarithmic derivative” to both Parts, but Part II does not restate the logarithmic derivative, and Part III restates neither the multiplication rule nor the auxiliary transform.)
3. *Front matter and bibliography.* The three title blocks and abstracts are printed at the heads of their Parts; one table of contents replaces three. The three bibliographies are merged into one, keeping every detail of every entry; where two manuscripts annotate an entry differently both annotations are kept. Report 142’s only reference, Report 141, and Report 143’s entries for Reports 141 and 142 now point to Parts I and II. Report 142 cites the OEIS and the literature only in its `SOURCES.md` (shipped as `142-interior-SOURCES.md`); those entries are in the merged bibliography through Parts I and III. The write added the entry for the transseries volume [15].
4. *Write additions.* This front matter, the reading-conventions tables of the Parts, Subsection 22.1, Section 35 with Remark 35.1, the dated notes, labels for unlabelled sections, and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 105, 5 October 2026) the three manuscripts were read in full and no step was found false. Checked by hand there: Part I’s prefix recurrence matching (Proposition 2.2, including $y = 0$ and unreachable states) and the peak correction $k - 1 + \ell/(y + 1)$; the constants of Lemma 6.2 ($K/N \leq 98/51$, the ratio bound $4/5$, $q_0 \leq 10/\sqrt{51} - 1 < 1$, $\alpha_* = 6 - 3\sqrt{3}$); that Part II’s Lemma 19.1 applies the frozen inverse at the base state only, bounds the neighbouring parameter difference by the s -Lipschitz norm and starts from $E = O(1)$; Part III’s integral comparisons, forest injection, exponential inequalities, $d_N \geq 1$ and $\log d_N \leq 1/(8N)$, the ratio bound $2/3$, the first profile correction and the reversion coefficients ξ_0, ξ_1, ξ_2 . Independent exact computations (integers from the recurrence at integer k ,

Newton coefficients by finite differences, 60-digit arithmetic) reproduced A292692’s first terms; gave bounded residuals $n^3(a_n/(Cd^n n!/\sqrt{n}) - 1 - b_1/n - b_2/n^2) = -3.69, -2.96, -2.80, -2.73$ at $n = 25, 50, 75, 100$; matched λ_1, λ_2 to 20 digits; gave interior residuals after r_1, r_2 of stable size at $\alpha = 0.3, 0.5$ and 0.9 (the last beyond Part I’s $\alpha_* \approx 0.804$; “of stable size” first read “bounded”, corrected after the independent check of 5 October 2026); found no violation of Theorem 24.1 for $n, a \leq 40$ or of (III.9) for $N \leq 200, q \leq 40$; and gave bounded q^6 residuals after $f_0, \dots, f_5$ for $q \leq 1200$. The delivered suites pass on copies (README; Report 143’s self-test needs a POSIX host). The write recomputed C and d to 40 digits and the lower-edge ratios quoted in Section 35.

What was *not* done: the analytic estimates (the local limit and shifted-saddle remainders, the endpoint expansions, the nested-compact bootstrap, the binomial-smoothing remainders) were read and their displayed algebra checked, but they are the manuscripts’ proofs, corroborated numerically, not an independent verification. External inputs: the write located and read Martin–Kearney’s Section 2.4, Theorem 3 (a leading-order Gamma-type asymptotic for simply represented sequences), Borinsky’s Section 6.4.1 (QED) and Table 10 (whose first rows are 1, 1, 4, 25, 208, 2146, the sequence A005411), and Ciobanu–Kolkpakov’s Theorem 4.1 and Examples 4.4 and 5.3 (the hypermap asymptotic shape, the quadrangulation counts A292186, a Riccati equation) in their arXiv versions, and DLMF 5.11.8 (the expansion of $\ln \Gamma(z + h)$ with Bernoulli polynomials, which gives (III.62)). It did *not* read Hsu and Shiue’s Theorem 5 and Section 5.2 (the publisher’s page was not accessible), Stokes (1982) or Arratia–DeSalvo; those citations are as the manuscripts give them. An independent adversarial check made after the write is recorded at the end of Section 35 (dated note, 5 October 2026).

Relation to neighbouring reports and formal projects

No other report of the collection treats A258219, A258220, A292692, A292186 or the weighted Dyck polynomials (repository search, 5 October 2026); the Dyck- and Catalan-path reports of `oeis-sequence-asymptotics` share no statement with this one. Two articles filed under `Analysis/Transseries/docs/series-and-transseries/` use the same mechanism for other self-convolutive recurrences: `Factorial_Transseries_OEIS_A006014` (an exact ${}_2F_0$ logarithmic-derivative linearization of the A006014 recurrence, a positive normalized ratio and an all-orders factorial expansion) and `Late_Coefficients_Factorially_Forced_Catalan_Recurrence` (A229741 and A260879, positive Stirling transforms and safe integer threshold enclosures). Neither cites, or is cited by, the manuscripts here; the parallel is of method (Part I’s Lemma 3.1, Part III’s Section 28), not of statements. The leading inverses of Parts I and II are instances of the transseries volume’s factorial core; their ceiling brackets follow the pattern of its staircase theorem but are proved directly at the integers, without the interpolation that theorem presupposes (above). (Wording corrected after the independent check of 5 October 2026, recorded at the end of Section 35; the first wording read “The inverses are instances of the transseries volume’s apparatus (above).”) Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats these sequences, and no statement of this report is formalized.

Part I

A weighted Dyck path Newton diagonal

Bundle Report 141, archive A292692_Weighted_Dyck_Diagonal_Asymptotics_and_Inversion_Source.zip, delivered as Report141.tex (21 pages). Delivered title block “A weighted Dyck path Newton diagonal / Report 141 / October 2, 2026”. Printed as delivered, with [write] notes; section, statement and equation numbers are those of the delivery.

Reading conventions of Part I	Elsewhere in the report
$F(x, k) = \sum p_N x^N$; $R_N(k)$ the recurrence polynomials (Section 2)	Part III: $G(x, k)$ for F ; $R_{N,m}$ is always the ratio
$b_{N,m,\ell,r}$, $b_{\ell,r}^0$ the kernel and its limit; $\mathcal{B}_{N,m}$ the opposite endpoint	Part II: $a_{N,m,\ell,r}$, $a_s(\ell, r)$, $B_{N,m}$; $b_s(\ell, r)$ inverse weights
$D = t d/dt$; $D_1(\ell, r)$, $c_1(s)$	Part II: $\mathcal{D}$, $D_1(\ell, r; s)$, $D_1(0, 0; s)$
K an integer evaluation point; $q_0(s)$ the kernel mass	Part II: H the integer point, K an order; Part III: q the deficit, K_q
$w_\ell(s) = \gamma_\ell s^{2\ell}$; W Lambert	Part III: w_n , $W(x)$ hypergeometric
$\beta_1 = b_1 + 1/12$; $\mathcal{N}(X)$	Part II: λ_1 ; $\mathcal{N}(X)$

Abstract of this Part (as delivered). Give a peak at coordinates (a, b) in a Dyck path the weight $k + a/b$, and let $p_N(k)$ be the total weight of paths of semilength N . We prove the Riccati identity and Stieltjes continued fraction conjectured for these polynomials in OEIS A258219. The proof is an exact identity for all weighted nonnegative prefixes. We then study the falling-factorial coefficient $a_n = [k^n]p_{2n}(k)$, the sequence A292692, and prove

$$a_n \sim \frac{\sqrt{2 + 2/\sqrt{17}}}{\pi^{3/2}} \left(\frac{51\sqrt{17} - 107}{4} \right)^n \frac{n!}{\sqrt{n}}.$$

The essential analytic step is a positive transformed convolution with strictly contractive endpoint mass. A uniform lattice local limit estimate and an integer-evaluation middle bound justify the reduction. We include all normalization constants, a local uniform theorem for a moving Newton index, and the first relative correction with an $O(n^{-2})$ remainder. A Lambert W threshold bracket has a real error before integer rounding of $O(x_0^{-2}/\log x_0)$, with $x_0 \sim \log X/\log \log X$. The proof does not depend on numerical fitting or on fixed-parameter Feynman diagram asymptotics. A reproducible exact-arithmetic companion checks finite identities and source normalization, separately from the mathematical proof.

1 Definitions and results

A Dyck path of semilength N starts at $(0, 0)$, ends at $(2N, 0)$, uses $U = (1, 1)$ and $D = (1, -1)$, and never has negative height. A peak is the vertex between consecutive steps UD . For an indeterminate k , define

$$p_N(k) = \sum_P \prod_{(a,b) \text{ a peak of } P} \left(k + \frac{a}{b} \right), \quad p_0(k) = 1. \quad (1)$$

The sum ranges over these Dyck paths; every peak has $b \geq 1$. This is the polynomial model of A258219 [1]. Although individual path weights are rational polynomials, we will prove $p_N \in \mathbb{Z}_{\geq 0}[k]$.

For any polynomial P , its Newton coefficient is

$$\mathcal{L}_m P = [k^{\underline{m}}]P(k) = \frac{\Delta^m P(0)}{m!} = \frac{1}{m!} \sum_{i=0}^m (-1)^{m-i} \binom{m}{i} P(i), \quad k^{\underline{m}} = k(k-1) \cdots (k-m+1). \quad (2)$$

We set $k^0 = 1$. This normalization is essential. The triangle $T(N, m) = \mathcal{L}_m p_N$ is A258220 [2]; its diagonal

$$a_n = T(2n, n), \quad a_0 = 1, \quad (3)$$

is A292692 [3]. Its initial terms are

$$1, 6, 217, 13997, 1283817, 151778430, 21902350480.$$

Theorem 1.1 (Exact weighted path identity). *The formal series $F(x, k) = \sum_{N \geq 0} p_N(k) x^N$ is the unique element of $\mathbb{Q}[k][[x]]$ with constant term 1 satisfying*

$$F - 1 = kxF + xF^2 + 2x^2F_x. \quad (4)$$

Equivalently,

$$p_N = (2N - 2 + k)p_{N-1} + \sum_{j=0}^{N-1} p_j p_{N-1-j} \quad (N \geq 1). \quad (5)$$

Its formal Stieltjes fraction has successive numerators $(k+1)x, 3x, (k+3)x, 5x, (k+5)x, 7x, \dots$

Theorem 1.2 (Leading diagonal equivalent). *As $n \rightarrow \infty$,*

$$a_n \sim C d^n \frac{n!}{\sqrt{n}}, \quad d = \frac{51\sqrt{17} - 107}{4}, \quad C = \frac{\sqrt{2 + 2/\sqrt{17}}}{\pi^{3/2}}. \quad (6)$$

In particular, $d = 25.8195967266 \dots$ and $C = 0.283103092513 \dots$

The first theorem resolves the Riccati and continued-fraction conjecture recorded by Peter Bala on July 28, 2022. The second proves the displayed asymptotic shape and the algebraic rate conjectured on A292692, with an exact expression for the formerly numerical amplitude. The rate conjecture there is attributed to Václav Kotěšovec, March 17, 2024. These statements refer to the source pages inspected on October 2, 2026; a site-wide footer is not evidence of a particular sequence's last revision date.

[write] **The OEIS statements, 5 October 2026.** The front matter (“The OEIS statements, and the inverses”) quotes the three entries as read by the write. Bala’s comment on A258219 (July 28, 2022) states the Riccati equation for the generating function of each integer column $k \geq 0$ and deduces the continued fraction from Stokes [8]; Theorem 1.1 proves both for indeterminate k . On A292692 the numerical form $a(n) \sim c d^n n! / \sqrt{n}$ is signed Václav Kotěšovec, December 1, 2017, and the closed form of d is his conjecture of March 17, 2024, as the paragraph above says. C and d agree with all 17 and 19 digits stated there. Part II extends Theorem 1.2 to every fixed order (Corollary 13.2).

Scope and prior work. The recurrence family and its fixed-parameter asymptotics are classical in this setting. Martin and Kearney [6, Section 2.4, Theorem 3] give a Gamma-type asymptotic for a self-convolutive recurrence with fixed parameters. Borinsky [7, Section 6.4.1 and Table 10] develops extensive asymptotic expansions for zero-dimensional quantum field theories; A005411 appears in this literature and is the $k = 0$ specialization here for positive indices [4]. These results

are relevant prior art, not new claims of this report. Extracting a Newton coefficient whose order grows proportionally to N is a different limiting operation. A fixed- k asymptotic alone does not justify it. We do not identify a_n itself with a particular class of ordinary QED diagrams without a separate interpretation. A bounded literature check located no proof of this precise moving diagonal; no worldwide priority or exhaustive novelty claim is made. Theorem 9.1 strengthens the equivalent by giving its first relative correction and a uniform second-order remainder. General all-orders claims are outside the scope of this report.

[write] **Scope, 5 October 2026.** This scope sentence is Report 141's. Part II proves all fixed orders on the whole interior (Theorem 13.1) and on the diagonal (Corollary 13.2).

Formal and analytic variables. The series F and its hypergeometric auxiliary Z below are formal series in x and are typically factorially divergent. We never evaluate them at a nonzero complex x . The saddle argument uses a different variable t in convergent functions on $|t| < 1$. All manipulations of F and Z take place coefficientwise in $\mathbb{Q}[k][[x]]$.

2 All weighted prefixes and the Riccati equation

Let $H_\ell(y)$ be the total weight of nonnegative prefixes of length ℓ ending at height y . Only completed UD peaks carry their weight; a final up-step alone has not yet completed a peak. Set $H_\ell(y) = 0$ for $y < 0$, $H_{-1}(y) = 0$, and

$$H_0(y) = \mathbf{1}_{y=0}, \quad H_1(y) = \mathbf{1}_{y=1}.$$

The array vanishes unless $0 \leq y \leq \ell$ and $\ell - y$ is even.

Lemma 2.1 (Exact last-step recurrence). *For $\ell \geq 1$ and $y \geq 0$,*

$$H_{\ell+1}(y) = H_\ell(y-1) + H_\ell(y+1) + \left(k-1 + \frac{\ell}{y+1}\right) H_{\ell-1}(y). \quad (7)$$

Proof. A final up-step contributes $H_\ell(y-1)$. Provisionally giving every final down-step weight 1 contributes $H_\ell(y+1)$. The prefixes in this second term whose previous step was up are obtained by appending U to a prefix counted by $H_{\ell-1}(y)$. The final D completes a peak at $(\ell, y+1)$. Its provisional factor 1 must be replaced by $k + \ell/(y+1)$, yielding the correction in (7). At $y = 0$ the first term is zero, and $y+1$ is positive, so the same argument applies without a boundary exception. $\square$

Independently of paths, define $R_0(k) = 1$ and use the right side of (5) inductively to define $R_N(k)$. The formal series $G(x) = \sum_{N \geq 0} R_N(k)x^N$ exists uniquely and satisfies

$$G - 1 = kxG + xG^2 + 2x^2G'. \quad (8)$$

There is no circular use of a path identity in this construction.

Proposition 2.2 (Stronger prefix identity). *For every $n, y \geq 0$,*

$$H_{2n+y}(y) = [x^n]G(x)^{y+1}. \quad (9)$$

Proof. Multiply (8) by G^y and use the formal product rule to obtain

$$G^{y+1} - G^y = kxG^{y+1} + xG^{y+2} + \frac{2x^2}{y+1}(G^{y+1})'.$$

Taking the coefficient of x^n , with negative indices interpreted as zero, gives

$$\begin{aligned} [x^n]G^{y+1} &= [x^n]G^y + [x^{n-1}]G^{y+2} \\ &\quad + \left(k + \frac{2n-2}{y+1}\right)[x^{n-1}]G^{y+1}. \end{aligned} \tag{10}$$

Define $C_\ell(y) = 0$ for unreachable (ℓ, y) and $C_\ell(y) = [x^{(\ell-y)/2}]G^{y+1}$ otherwise. The arrays C_0, C_1 equal H_0, H_1 . For a reachable target of length $\ell + 1$, set $n = (\ell + 1 - y)/2$, so $\ell = 2n + y - 1$. When $y \geq 1$, the first term of (10) is $C_\ell(y - 1)$. When $y = 0$, reachability at positive length gives $n \geq 1$, so that term is $[x^n]1 = 0 = C_\ell(-1)$. The second term is $C_\ell(y + 1)$, and the last coefficient equals $k - 1 + \ell/(y + 1)$, as required in (7). This includes $n = 0$ since all coefficients of x^{-1} vanish. At unreachable targets the recurrence is zero by range or parity. Thus C obeys the same initial conditions and recurrence as H . Induction on ℓ proves $C = H$. $\square$

At $y = 0$, complete prefixes are exactly Dyck paths. Consequently $p_N = R_N$, proving the Riccati assertion of Theorem 1.1. Recurrence (5) also proves the promised nonnegative integral ordinary coefficients. For example,

$$\begin{aligned} p_1(k) &= k + 1, \\ p_2(k) &= k^2 + 5k + 4, \\ p_3(k) &= k^3 + 12k^2 + 36k + 25. \end{aligned}$$

Individual rational peak factors have disappeared from the total polynomial.

2.1 A bivariate generating function and a boundary check

The prefix theorem can also be encoded as

$$H(v, u) := \sum_{\ell, y \geq 0} H_\ell(y) v^\ell u^y = \frac{F(v^2, k)}{1 - vuF(v^2, k)}. \tag{11}$$

This is a formal identity; each coefficient of v^ℓ is a polynomial in u . For completeness, summing (7), including its valid $\ell = 0$ version, gives

$$[u - v(u^2 + 1) - v^2(k - 1)u]H = u - vH(v, 0) + v^2 \int_0^u (vH_v(v, w) + H(v, w)) dw.$$

Formal differentiation yields

$$[u - v(u^2 + 1) - v^2(k - 1)u]H_u - v^3H_v + (1 - 2vu - kv^2)H = 1. \tag{12}$$

This differentiated equation still determines H uniquely in $\mathbb{Q}[k, u][[v]]$: its coefficient at v^ℓ has leading operator $u d/du + 1$ on $[v^\ell]H$, and the remaining terms involve only smaller indices. The

operator multiplies u^y by $y + 1$ and is invertible. If $x = v^2$ and $D = 1 - vuF(x)$, the right side of (11) has derivatives

$$H_u = \frac{vF^2}{D^2}, \quad H_v = \frac{2vF' + uF^2}{D^2}.$$

Substitution into (12) gives numerator

$$F - kxF - xF^2 - 2x^2F' - 2vuF + v^2u^2F^2 = 1 - 2vuF + v^2u^2F^2 = D^2.$$

This independently verifies the zero-height closure in the generating function presentation. The direct recurrence proof above already establishes all prefix identities without a PDE uniqueness assumption.

3 The formal hypergeometric identity and continued fraction

Put

$$\gamma_N = \frac{1}{4^N} \binom{2N}{N}, \quad z_N(k) = \gamma_N \prod_{i=0}^{N-1} (k + 2i + 1), \quad Z(x, k) = \sum_{N \geq 0} z_N(k) x^N. \quad (13)$$

Empty products and z_0 equal 1. In rising-factorial notation,

$$Z(x, k) = {}_2F_0 \left(\frac{k+1}{2}, \frac{1}{2}; ; 2x \right) = \sum_{N \geq 0} \frac{((k+1)/2)_N (1/2)_N}{N!} (2x)^N. \quad (14)$$

The symbol ${}_2F_0$ in this report denotes precisely this formal series.

Lemma 3.1 (Logarithmic derivative). *In $\mathbb{Q}[k][[x]]$,*

$$F = 1 + 2x \frac{Z_x}{Z}, \quad 2Nz_N = \sum_{j=1}^N p_j z_{N-j} \quad (N \geq 1). \quad (15)$$

Proof. The product formula gives $2Nz_N = (2N-1)(k+2N-1)z_{N-1}$. Coefficient extraction then proves

$$2x^2 Z'' + ((k+4)x - 1)Z' + \frac{k+1}{2}Z = 0. \quad (16)$$

Formal division by Z is legitimate since $Z(0) = 1$. Let $\tilde{F} = 1 + 2xZ'/Z$. Writing $q = Z'/Z$ and using (16),

$$2x^2(q' + q^2) + ((k+4)x - 1)q + (k+1)/2 = 0$$

is exactly the Riccati identity for $\tilde{F}$ after substitution and cancellation. Its constant term is 1, so uniqueness gives $\tilde{F} = F$. Multiplying by Z and extracting x^N proves the second identity. The $j = 0$ term has already canceled, explaining the lower limit 1. $\square$

Proposition 3.2 (Formal Stieltjes fraction). *One has*

$$F(x, k) = \frac{1}{1 - \frac{(k+1)x}{1 - \frac{3x}{1 - \frac{(k+3)x}{1 - \frac{5x}{1 - \frac{(k+5)x}{1 - \ddots}}}}}}. \quad (17)$$

The infinite fraction is interpreted coefficientwise: any coefficient is unchanged after sufficiently many levels.

Proof. For formal parameters a, b, z , write $U(a, b; z) = \sum_{N \geq 0} (a)_N (b)_N z^N / N!$. Direct coefficient comparison gives

$$U(a, b+1; z) - U(a, b; z) = az U(a+1, b+1; z).$$

If $R(a, b; z) = U(a, b+1; z)/U(a, b; z)$, symmetry in a, b yields

$$R(a, b; z) = \frac{1}{1 - azR(b+1, a; z)}.$$

Iterating this identity gives a fraction with numerators $az, (b+1)z, (a+1)z, (b+2)z, \dots$. Every U and every ratio has constant term 1, and each level introduces a factor of z , so the iteration is valid in the formal topology. Finally, for $b = 1/2$,

$$U(a, 3/2; z) = U(a, 1/2; z) + 2z \frac{\partial}{\partial z} U(a, 1/2; z).$$

Taking $a = (k+1)/2$ and $z = 2x$, Lemma 3.1 identifies the ratio with F , and the stated numerators follow. $\square$

This supplies a direct formal proof of the continued fraction, in addition to the general Riccati continued-fraction theory cited by the OEIS entry [8].

4 Positive Newton transforms

For ordinary coefficientwise order, write $P \preceq Q$ when every coefficient of $Q - P$ is nonnegative. Both p_N and z_N have nonnegative coefficients, so isolating the $j = N$ term of (15) gives

$$0 \preceq p_N \preceq 2Nz_N \quad (N \geq 1). \quad (18)$$

Newton extraction preserves this order. Indeed,

$$k^j = \sum_{m=0}^j \left\{ \begin{matrix} j \\ m \end{matrix} \right\} k^{\underline{m}}, \quad \mathcal{L}_m k^j = \left\{ \begin{matrix} j \\ m \end{matrix} \right\} \geq 0,$$

where the braces denote Stirling numbers of the second kind. Set

$$P_{N,m} = \mathcal{L}_m p_N, \quad Q_{N,m} = \mathcal{L}_m z_N, \quad R_{N,m} = \frac{P_{N,m}}{2NQ_{N,m}} \quad (1 \leq m \leq N). \quad (19)$$

We will see immediately that $Q_{N,m} > 0$ in this range. Consequently

$$0 \leq R_{N,m} \leq 1. \quad (20)$$

Define the analytic functions

$$f(t) = (1-t)^{-1/2}, \quad h(t) = f(t) - 1, \quad C_{N,m} = [t^N] f(t) h(t)^m. \quad (21)$$

They are analytic for $|t| < 1$, using the branch with $f(0) = 1$. All coefficients of f , and all positive-degree coefficients of h , are strictly positive.

Lemma 4.1 (Exact auxiliary transform). *With the convention $(-1)!! = 1$,*

$$Q_{N,m} = \frac{(2N-1)!!}{m!} C_{N,m} \quad (0 \leq m \leq N). \quad (22)$$

Proof. The product formula gives the coefficientwise identity

$$\sum_{N \geq 0} \frac{z_N(k)}{(2N-1)!!} t^N = (1-t)^{-(k+1)/2} = f(t)(1+h(t))^k.$$

Expand $(1+h)^k = \sum_{m \geq 0} k^{\underline{m}} h^m / m!$ and extract t^N and the Newton coefficient. Only $m \leq N$ contributes, so these operations are finite at each degree. Positivity of $C_{N,m}$ proves the assertion following (19). $\square$

The factor $1/m!$ distinguishes a falling-factorial coefficient from a plain finite difference. Omitting it would change both the growth scale and the answer.

5 A uniform large power local limit estimate

We work on the concrete compact parameter interval

$$\frac{3}{5} \leq s \leq \frac{7}{10}, \quad \alpha := \frac{m}{N} = \frac{2s^2}{1+s} \in I := \left[\frac{9}{20}, \frac{49}{85} \right], \quad t = 1 - s^2. \quad (23)$$

The diagonal $m/N = 1/2$ lies in the interior. Constants in uniform estimates may depend on this interval and, when used, a fixed cutoff constant A . The positive saddle satisfies

$$\mu(t) := \frac{th'(t)}{h(t)} = \frac{1+s}{2s^2} = \frac{N}{m}, \quad B(t) := t\mu'(t) = \frac{t(s+2)}{4s^4} > 0. \quad (24)$$

Here $t \in [51/100, 16/25]$, and $h(t)$ and $B(t)$ are uniformly bounded above and away from zero.

Lemma 5.1 (Uniform local limit with logarithmic shifts). *Uniformly for $m/N \in I$,*

$$C_{N,m} = \frac{f(t)h(t)^m t^{-N}}{\sqrt{2\pi m B(t)}} (1 + o(1)). \quad (25)$$

For every fixed $A > 0$, uniformly for integers $0 \leq r \leq \ell \leq A \log N$,

$$\frac{C_{N-\ell, m-r}}{C_{N,m}} = t^\ell h(t)^{-r} \left(1 + O_A \left(\frac{\log N}{\sqrt{N}} \right) \right). \quad (26)$$

The saddle on the right is the original one from (N, m) .

Proof. At a fixed saddle t , define independent integer random variables by

$$\mathbb{P}(X = j) = \frac{[u^j]h(u)t^j}{h(t)} \quad (j \geq 1), \quad \mathbb{P}(V = j) = \frac{[u^j]f(u)t^j}{f(t)} \quad (j \geq 0).$$

The variable X has span one since all its positive integer probabilities are nonzero. Its mean is $\mu = N/m$ and variance is B . Both distributions have exponential moments uniformly on our interval:

choose $\eta > 0$ so that $te^\eta < 1$ for every permitted t . For a sum S_M of M independent copies of X , the exact coefficient identity is

$$[u^q]f(u)h(u)^M = f(t)h(t)^M t^{-q} \mathbb{P}(S_M + V = q). \quad (27)$$

We claim that, for $\delta = q - M\mu = O(\log M)$,

$$\mathbb{P}(S_M + V = q) = \frac{1}{\sqrt{2\pi MB}} \left(1 + O\left(\frac{1 + |\delta|}{\sqrt{M}}\right) \right) \quad (28)$$

uniformly on the compact saddle interval.

Here are the Fourier details. Let $\phi_t(\theta) = \mathbb{E}e^{i\theta X}$ and $\psi_t(\theta) = \mathbb{E}e^{i\theta V}$. Uniform exponential moments give, on a fixed small arc,

$$\log \phi_t(\theta) = i\mu\theta - \frac{1}{2}B\theta^2 + O(|\theta|^3), \quad \psi_t(\theta) = 1 + O(|\theta|),$$

with uniform Taylor constants. After decreasing that arc if necessary, $|\phi_t(\theta)| \leq e^{-c\theta^2}$ there, with $c > 0$ independent of t . On a fixed outer arc $\varepsilon \leq |\theta| \leq \pi$, span one and compactness give $|\phi_t(\theta)| \leq \rho < 1$. Fourier inversion splits into $|\theta| \leq M^{-2/5}$, the intermediate arc $M^{-2/5} < |\theta| < \varepsilon$, and the outer arc. Their latter two contributions are $O(e^{-cM^{1/5}})$ and $O(\rho^M)$, respectively.

On the central arc put $u = \sqrt{M}\theta$. The centered characteristic function becomes $e^{-Bu^2/2}$ with logarithmic remainder $O(|u|^3/\sqrt{M})$; the target phase differs from 1 by at most $|\delta u|/\sqrt{M}$, and the V factor differs from 1 by $O(|u|/\sqrt{M})$. Since $|u| \leq M^{1/10}$, the cubic remainder is small and can be absorbed into a uniform Gaussian majorant. Integrating the differences against this majorant bounds the relative error by $O((1 + |\delta|)/\sqrt{M})$. The Gaussian integral is $\sqrt{2\pi/B}$; including the Fourier factor $1/(2\pi\sqrt{M})$ proves (28). In particular, the mean of V affects the error, not the leading constant.

Take $(M, q) = (m, N)$ in (27) to obtain (25). For $(M, q) = (m - r, N - \ell)$ the mean displacement at the same original saddle is $\delta = -\ell + r\mu = O_A(\log N)$. Also $M \asymp N$ and $\sqrt{m/(m - r)} = 1 + O_A(\log N/N)$. Dividing the two probabilities proves (26). This treats both moving indices; no unproved saddle interchange is needed. $\square$

6 Endpoint and middle estimates for products

Define, for $1 \leq \ell < N$,

$$U_{N,m,\ell} = \frac{\mathcal{L}_m(z_\ell z_{N-\ell})}{Q_{N,m}}, \quad w_\ell(s) = \gamma_\ell s^{2\ell}. \quad (29)$$

Two different estimates are needed: a sharp endpoint estimate with no polynomial loss, and a rough exponentially decreasing middle bound.

6.1 Newton multiplication near an endpoint

The identity $kk^{\underline{m}} = k^{\underline{m+1}} + mk^{\underline{m}}$ implies

$$\mathcal{L}_m((k + c)P) = (m + c)\mathcal{L}_m P + \mathcal{L}_{m-1}P. \quad (30)$$

Applying it to the ℓ factors in z_ℓ gives exact coefficients $c_{\ell,r}(m)$ such that

$$\mathcal{L}_m(z_\ell P) = \sum_{r=0}^{\ell} c_{\ell,r}(m) \mathcal{L}_{m-r}P. \quad (31)$$

In the endpoint range below $m > \ell$, so no negative Newton indices arise. All these coefficients are nonnegative.

Lemma 6.1 (Uniform endpoint kernel). *For every fixed $A > 0$, uniformly for $0 \leq r \leq \ell \leq A \log N$,*

$$c_{\ell,r}(m) = \gamma_\ell \binom{\ell}{r} m^{\ell-r} (1 + O(\ell^2/m)), \quad (32)$$

$$\frac{Q_{N-\ell,m-r}}{Q_{N,m}} = (2N)^{-\ell} m^r t^\ell h^{-r} (1 + O_A(\varepsilon_N)), \quad (33)$$

where $\varepsilon_N = \log N / \sqrt{N} + (\log N)^2 / N$. Consequently,

$$U_{N,m,\ell} = w_\ell(s) (1 + O_A(\varepsilon_N)), \quad (34)$$

and the exact nonnegative coefficients

$$b_{N,m,\ell,r} = \frac{N-\ell}{N} c_{\ell,r}(m) \frac{Q_{N-\ell,m-r}}{Q_{N,m}} \quad (35)$$

satisfy

$$b_{N,m,\ell,r} = w_\ell(s) \binom{\ell}{r} s^r (1-s)^{\ell-r} (1 + O_A(\varepsilon_N)). \quad (36)$$

Proof. In the iterated multiplication rule, choose which r of the ℓ factors shift the Newton index. There are $\binom{\ell}{r}$ paths. Every stay factor has form $m - q + 2i + 1$ with $0 \leq q \leq \ell$ and $0 \leq i < \ell$, hence is $m + O(\ell)$. If $m \geq 4\ell$, the product of the $\ell - r$ stay factors is $m^{\ell-r} \exp(O(\ell^2/m))$. The estimate is uniform over all paths; positivity permits summing their relative bounds without a combinatorial loss. At $r = \ell$ the product is empty and the formula is exact. This proves (32).

By (22), the required ratio is a product of the coefficient ratio in (26), a double-factorial ratio, and $m!/(m-r)!$. Direct finite products give

$$\frac{(2N - 2\ell - 1)!!}{(2N - 1)!!} = (2N)^{-\ell} (1 + O(\ell^2/N)), \quad \frac{m!}{(m-r)!} = m^r (1 + O(r^2/m)),$$

proving (33). Notice the positive power m^r .

Multiply these estimates, sum in r , and use

$$\frac{\alpha t}{2} = s^2(1-s), \quad h^{-1} = \frac{s}{1-s}. \quad (37)$$

The individual leading summand is $\gamma_\ell (\alpha t/2)^\ell \binom{\ell}{r} h^{-r}$, which equals $w_\ell(s) \binom{\ell}{r} s^r (1-s)^{\ell-r}$. Its sum is $w_\ell(s)$. All terms are positive, so the uniform relative errors remain uniform on summation. The factor $(N-\ell)/N = 1 + O_A(\log N/N)$ gives (36). $\square$

In particular, uniformly throughout this logarithmic endpoint range,

$$U_{N,m,\ell} \leq 2w_\ell(s) \quad (38)$$

for sufficiently large N .

6.2 An integer evaluation controls the middle

Lemma 6.2 (Uniform middle bound). *There exists a constant $K_0 < \infty$ such that, for $m/N \in I$ and $1 \leq \ell < N$,*

$$U_{N,m,\ell} \leq K_0 N (4/5)^{\min(\ell, N-\ell)}. \quad (39)$$

Proof. Choose an integer K nearest to $m/(1-s)$. Uniformly,

$$K > m \quad \text{eventually,} \quad \frac{K}{N} = \frac{2s^2}{1-s^2} + O(N^{-1}) \leq \frac{98}{51} + O(N^{-1}) < 2.$$

The polynomial $z_\ell z_{N-\ell}$ has nonnegative ordinary, and therefore nonnegative Newton, coefficients. At an integer K , every falling factorial is nonnegative or zero; degrees larger than K vanish. Thus

$$\mathcal{L}_m(z_\ell z_{N-\ell}) \leq \frac{z_\ell(K) z_{N-\ell}(K)}{K^m}. \quad (40)$$

A nonintegral evaluation would not provide this argument.

The normalization from (22) is exactly

$$\frac{z_N(K)}{K^m Q_{N,m}} = \frac{[u^N] f(u) (1+h(u))^K}{\binom{K}{m} C_{N,m}}. \quad (41)$$

Positive-coefficient evaluation bounds its numerator by $f(t)(1+h(t))^K t^{-N}$. Lemma 5.1 bounds the denominator from below by a fixed multiple of $\binom{K}{m} f(t) h(t)^m t^{-N} / \sqrt{m}$. Set $p = h/(1+h) = 1-s \in [3/10, 2/5]$. The resulting ratio is at most

$$\frac{C_1 \sqrt{m}}{\binom{K}{m} p^m (1-p)^{K-m}}.$$

Here $m - Kp = O(1)$ and $K \asymp N$. Uniform elementary Stirling estimates for the central binomial probability give a lower bound $c_1/\sqrt{K}$ for its denominator. Hence (41) is at most $C_2 N$. No asymptotic formula for the numerator is required.

For $\ell \leq N/2$, the product definition gives

$$\frac{z_\ell(K) z_{N-\ell}(K)}{z_N(K)} = \frac{\gamma_\ell \gamma_{N-\ell}}{\gamma_N} \prod_{i=0}^{\ell-1} \frac{K+2i+1}{K+2(N-\ell+i)+1}. \quad (42)$$

The first factor is bounded by $e^{1/2}$: use $\gamma_\ell \leq 1$ and

$$\log \frac{\gamma_{N-\ell}}{\gamma_N} = \sum_{j=N-\ell+1}^N \log \left(1 + \frac{1}{2j-1} \right) \leq \frac{\ell}{2(N-\ell)+1} < \frac{1}{2}.$$

Each linear ratio increases with i and is at most

$$\frac{K+2\ell-1}{K+2N-1} \leq \frac{K+N}{K+2N-1} \leq \frac{3N}{4N-1} \leq \frac{4}{5}$$

for all sufficiently large N . Combine this with (40) and (41). Symmetry treats $\ell > N/2$, and a larger constant absorbs the finitely many smaller N . $\square$

The factor N in (39) is harmless only away from the endpoints. It will never replace the sharper bound (38) near them.

7 A positive contraction proves the moving ratio limit

Theorem 7.1 (Local uniform Newton ratio). *Uniformly on every compact subinterval J of the interior of I ,*

$$R_{N,m} - \sqrt{1 - s(m/N)^2} \longrightarrow 0 \quad (N \rightarrow \infty, m/N \in J), \quad (43)$$

where $s(\alpha)$ is the positive solution of $\alpha = 2s^2/(1+s)$. Thus, with $t = 1 - s^2$,

$$P_{N,m} \sim \frac{2N(2N-1)!!}{m!} \frac{\sqrt{t} f(t) h(t)^m t^{-N}}{\sqrt{2\pi m B(t)}} \quad (44)$$

uniformly on such J .

Proof. Apply $\mathcal{L}_m$ to the convolution in (15), isolate $j = N$, and write $\ell = N - j$. The exact equation is

$$R_{N,m} = 1 - \sum_{\ell=1}^{N-1} \frac{\mathcal{L}_m(p_{N-\ell} z_\ell)}{2NQ_{N,m}}. \quad (45)$$

All subtracted terms are nonnegative. Fix $A = 20$ and $L_N = \lfloor A \log N \rfloor$. For large N , split the sum into the disjoint ranges $1 \leq \ell \leq L_N$, $L_N < \ell < N - L_N$, and $N - L_N \leq \ell \leq N - 1$.

In the middle, (18) and Lemma 6.2 give a total $O(N^2(4/5)^{L_N}) = o(1)$; the chosen A exceeds $3/(-\log(4/5))$. At the opposite endpoint put $j = N - \ell \leq L_N$. Equations (18) and (38) give a total at most

$$\frac{1}{N} \sum_{j=1}^{L_N} j U_{N,m,j} \leq \frac{2}{N} \sum_{j \geq 1} j \gamma_j s^{2j} = O(N^{-1}) \quad (46)$$

uniformly. At the principal endpoint, (31) and (35) give exactly

$$\frac{\mathcal{L}_m(p_{N-\ell} z_\ell)}{2NQ_{N,m}} = \sum_{r=0}^{\ell} b_{N,m,\ell,r} R_{N-\ell,m-r}.$$

Consequently,

$$R_{N,m} = 1 - \sum_{\ell=1}^{L_N} \sum_{r=0}^{\ell} b_{N,m,\ell,r} R_{N-\ell,m-r} - E_{N,m}, \quad 0 \leq E_{N,m} = o(1), \quad (47)$$

uniformly on I .

The total limiting kernel mass is

$$q_0(s) = \sum_{\ell \geq 1} w_\ell(s) = \frac{1}{\sqrt{1-s^2}} - 1 \leq \frac{10}{\sqrt{51}} - 1 < 1. \quad (48)$$

The scalar solution of $r = 1 - q_0(s)r$ is

$$r(\alpha) = \sqrt{1 - s(\alpha)^2}. \quad (49)$$

Finding this scalar solution does not alone prove convergence. We provide the necessary moving-index argument.

Positivity, Lemma 6.1, and the uniformly geometric tail of w_ℓ imply

$$\sum_{\ell \leq L_N} \sum_{r \leq \ell} b_{N,m,\ell,r} = q_0(s(m/N)) + o(1). \quad (50)$$

Every predecessor has ratio displacement

$$\left| \frac{m-r}{N-\ell} - \frac{m}{N} \right| = \frac{|m\ell - rN|}{N(N-\ell)} = O(\log N/N),$$

so its reference value $r((m-r)/(N-\ell))$ differs from $r(m/N)$ by $o(1)$. This holds uniformly on a neighborhood of any interior ratio, because r is continuously differentiable there. Subtract (49) from (47), add and subtract predecessor reference values, and use (50). For $e_{N,m} = |R_{N,m} - r(m/N)|$, this gives

$$e_{N,m} \leq \sum_{\ell \leq L_N} \sum_{r \leq \ell} b_{N,m,\ell,r} e_{N-\ell,m-r} + o(1). \quad (51)$$

The error here includes the discarded mass, the kernel-mass error, and the reference-value shifts; it does not assume convergence of R .

Fix an interior ratio α_0 and define the precise local envelope

$$M(\alpha_0) = \lim_{\delta \downarrow 0} \lim_{N_0 \rightarrow \infty} \sup_{\substack{N \geq N_0 \\ |m/N - \alpha_0| < \delta}} e_{N,m}. \quad (52)$$

These monotone limits exist and are finite by (20). Choose a sequence tending to α_0 , with $N \rightarrow \infty$, whose errors tend to $M(\alpha_0)$. Along it, every predecessor in (51) has size $N - O(\log N) \rightarrow \infty$ and ratio tending to α_0 , uniformly. Hence

$$\limsup \max_{\ell \leq L_N, r \leq \ell} e_{N-\ell,m-r} \leq M(\alpha_0).$$

If this failed, choose an offending predecessor on each member of a subsequence; those predecessors themselves contradict the envelope's definition. Now (50)–(51) imply

$$M(\alpha_0) \leq q_0(s(\alpha_0))M(\alpha_0).$$

The strict bound in (48) forces $M(\alpha_0) = 0$. This proves convergence along every sequence tending to this ratio. If uniformity failed on a compact J , an offending sequence of ratios would have a convergent subsequence in J , contradicting this conclusion. Equation (44) now follows from Lemmas 4.1 and 5.1. $\square$

Why the positivity matters. The formal expression $F = 1 + 2x(\log Z)'$ tempts one to expand a logarithm and discard most of its alternating terms. Such a step needs a separate uniform remainder argument in the moving Newton regime. The proof above avoids it. The transformed convolution is positive, its middle is directly bounded at an integer K , and its endpoint mass is genuinely contractive. The distinction is what permits a rigorous moving-index equivalent.

[write] **Superseded on the whole interior, 5 October 2026.** Theorem 7.1 holds on compact subintervals of $\text{int } I = (9/20, 49/85)$. Part II, Theorem 13.1 with $K = 1$, gives the same limit $\sqrt{1-s^2}$ on every compact $J \subset (0, 1)$, by freezing the inverse of the convolution at the base state (Section 19) rather than by a contraction. The positive-contraction proof above is kept as an independent, elementary route on its range. Lemmas 5.1 and 6.2 are generalized by Part II's Lemmas 15.1 (every order) and 16.1 (every compact $J \subset (0, 1)$).

8 The diagonal constants

At $N = 2n$, $m = n$, equation (23) becomes $4s^2 = 1 + s$. Its positive root and associated parameters are

$$s_0 = \frac{1 + \sqrt{17}}{8}, \quad t_0 = 1 - s_0^2, \quad h(t_0) = \frac{1 - s_0}{s_0}, \quad B_0 = \frac{t_0(s_0 + 2)}{4s_0^4}. \quad (53)$$

For this section write s, t, B for these fixed values. Theorem 7.1 gives, with every normalization retained,

$$a_n \sim \frac{4n(4n-1)!!}{n!} \frac{h(t)^{nt-2n}}{s\sqrt{2\pi nB}} \sqrt{t}. \quad (54)$$

In particular, the local-limit variance prefactor contains $m = n$, not $N = 2n$. Using $(4n-1)!! = (4n)!/(2^{2n}(2n)!)$ and Stirling's formula gives

$$\frac{(4n-1)!!}{(n!)^2} \sim \frac{16^n}{\pi\sqrt{2}n}. \quad (55)$$

For clarity, the factorial square roots contribute $\sqrt{8\pi n}/(\sqrt{4\pi n}2\pi n) = 1/(\pi\sqrt{2}n)$; the exponential and power terms contribute 16^n . Combining (54)–(55) yields

$$d = \frac{16h(t)}{t^2} = \frac{16}{s(1-s)(1+s)^2} = \frac{51\sqrt{17} - 107}{4}, \quad (56)$$

$$C = \frac{2\sqrt{t}}{s\pi\sqrt{\pi B}} = \frac{4s}{\pi^{3/2}\sqrt{s+2}} = \frac{\sqrt{2+2/\sqrt{17}}}{\pi^{3/2}}. \quad (57)$$

All roots are positive. The final simplifications use $4s^2 = 1 + s$. For example, squaring the last amplitude identity reduces it to $16s^2/(s+2) = 2 + 2/\sqrt{17}$, an identity in $\mathbb{Q}(\sqrt{17})$. This completes the proof of Theorem 1.2. The posted numerical amplitude is consistent with it, but the proof of the constant is the exact normalization calculation, not agreement of decimal digits.

9 The first correction with a uniform remainder

We now strengthen the leading result by one relative order. This requires more than formally expanding the fixed-point equation: both endpoint errors must be summable, and a stopped contraction must create an N^{-2} bound from a bounded initial error.

Theorem 9.1 (First correction). *Put $\alpha_* = 6 - 3\sqrt{3}$. For every compact interval $J \subset (0, \alpha_*)$, uniformly for integer pairs with $m/N \in J$,*

$$R_{N,m} = \sqrt{t} \left(1 + \frac{r_1(s)}{N} + O_J(N^{-2}) \right), \quad r_1(s) = \frac{s^4 + 6s^3 + 7s^2 + 3s + 4}{4(s-1)(s+1)(s+2)^2}, \quad (58)$$

where $\alpha = 2s^2/(1+s)$ and $t = 1 - s^2$. In particular,

$$a_n = Cd^n \frac{n!}{\sqrt{n}} \left(1 + \frac{b_1}{n} + O(n^{-2}) \right), \quad b_1 = -\frac{989}{2176} - \frac{907\sqrt{17}}{36992}. \quad (59)$$

Thus $b_1 = -0.555597340031 \dots$. No coefficients beyond this first relative correction are asserted here.

[write] **Superseded and answered, 5 October 2026.** The restriction $\alpha < \alpha_* = 6 - 3\sqrt{3}$, where $q_0(s) < 1$, is removed by Part II: Theorem 13.1 with $K = 2$ gives (58) on every compact $J \subset (0, 1)$, with the same r_1 ((II.71)). The sentence “No coefficients beyond this first relative correction are asserted here” is answered by Corollary 13.2: every b_j exists and lies in $\mathbb{Q}(\sqrt{17})$, and b_2 is explicit.

9.1 Parameter domain and exact split

Choose nested compact intervals

$$J \subset \text{int } J_1 \subset \text{int } J_2 \subset (0, \alpha_*). \quad (60)$$

On J_2 , s is bounded away from 0 and bounded strictly below $\sqrt{3}/2$. This is exactly the range in which $q_0(s) < 1$, since $2(\sqrt{3}/2)^2/(1 + \sqrt{3}/2) = 6 - 3\sqrt{3}$. The local-limit and endpoint arguments above apply without change on J_2 : t, h, B and all required derivatives remain uniformly controlled. For the middle bound the integer K has K/N in a fixed positive bounded interval, so some $q < 1$ bounds $(K + N)/(K + 2N - 1)$ uniformly for large N . The proof of Lemma 6.2 therefore gives

$$U_{N,m,\ell} \leq C_2 N q^{\min(\ell, N-\ell)} \quad (61)$$

on J_2 . The special value $q = 4/5$ is no longer needed.

For valid zero Newton index use $R_{n,0} = p_n(0)/(2nz_n(0)) \in [0, 1]$. When an index is negative or exceeds the polynomial degree, its transformed summand is zero; define the corresponding kernel to be zero and assign the unused R any bounded value. This avoids undefined $0/0$ expressions. Such indices never occur in the logarithmic endpoint range for large N .

Set $L = \lfloor N/2 \rfloor$ and split (45) exactly as

$$R_{N,m} = 1 - \sum_{\ell=1}^L \sum_{r=0}^{\ell} b_{N,m,\ell,r} R_{N-\ell,m-r} - \mathcal{B}_{N,m}, \quad (62)$$

$$\mathcal{B}_{N,m} = \sum_{j=1}^{N-L-1} \frac{\mathcal{L}_m(p_j z_{N-j})}{2N Q_{N,m}}. \quad (63)$$

There is no overlap at the midpoint. Every principal predecessor has size at least $N/2$, all valid kernels are nonnegative, and $\sum_r b_{N,m,\ell,r} = (1 - \ell/N)U_{N,m,\ell}$.

9.2 A second order remainder for shifted saddle coefficients

Write $D = t d/dt = -(t/(2s))d/ds$ and put

$$g_1 = D \log f, \quad g_2 = D^2 f/f, \quad k_3 = DB, \quad k_4 = D^2 B, \quad E = g_1 - \frac{k_3}{2B} = -\frac{s^3 + s + 4}{4s^2(s+2)}. \quad (64)$$

For $\beta = \ell - \mu r$, define

$$D_1(\ell, r) = \frac{-g_2/2 - g_1\beta + (rB - \beta^2)/2}{B} + \frac{(g_1 + \beta)k_3}{2B^2} + \frac{k_4}{8B^2} - \frac{5k_3^2}{24B^3}. \quad (65)$$

For each fixed A , there is a finite exponent D_0 such that, uniformly for $0 \leq r \leq \ell \leq A \log N$ on J_2 ,

$$C_{N-\ell,m-r} = \frac{f(t)h(t)^m t^{-N} t^\ell h(t)^{-r}}{\sqrt{2\pi m B}} \times \left(1 + \frac{D_1(\ell, r)}{m} + O(m^{-2}(1 + \ell + r)^{D_0}) \right). \quad (66)$$

We give the remainder details because fixed-shift expansions do not by themselves imply this logarithmic-shift assertion.

Set

$$\psi(\theta) = \frac{h(te^{i\theta})}{h(t)}, \quad g(\theta) = \frac{f(te^{i\theta})}{f(t)}.$$

After removing the prefactor $f(t)h(t)^m t^{-N} t^\ell h(t)^{-r}$, the exact Fourier integral is

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} g(\theta) \psi(\theta)^{-r} e^{i\ell\theta} \exp\{m(\log \psi(\theta) - i\mu\theta)\} d\theta. \quad (67)$$

The logarithm is used only near zero. Away from zero use the original integrand $g(\theta)\psi(\theta)^{m-r} e^{-i(N-\ell)\theta}$. The uniform span-one and exponential-moment bounds from Section 5 apply. Choose $0 < \varepsilon < 1/6$. The fixed outer arc is exponentially small. On the intermediate arc $m^{-1/2+\varepsilon} \leq |\theta| \leq \theta_0$, the exact power $m-r \geq m/2$ gives $O(e^{-cm^{2\varepsilon}})$, smaller than any fixed inverse power.

On the central arc put $\theta = zu$, $z = m^{-1/2}$, so $|u| \leq m^\varepsilon$. After removing $e^{-Bu^2/2}$, the integrand is

$$g(zu) e^{i\ell zu - r \log \psi(zu)} \exp \left\{ \sum_{j \geq 3} \kappa_j (iu)^j z^{j-2} / j! \right\}, \quad \kappa_j = D^j \log h. \quad (68)$$

Expand this expression through degree 3 in z . Its fourth derivative, for $0 \leq z' \leq m^{-1/2}$, is bounded by

$$C(1 + \ell + r)^{D_0} (1 + |u|)^{E_0} e^{\delta u^2}, \quad 0 < \delta < B_{\min}/4,$$

with fixed finite D_0, E_0 . To justify this bound, the saddle remainder exponent is $O(z'|u|^3)$, whose coefficient relative to u^2 tends uniformly to zero. Its fixed derivatives have only polynomial growth in u . The shifted amplitude exponential is bounded by $\exp\{C(\ell + r)z'|u| + Cr(z'u)^2\}$. Young's inequality absorbs the linear term into a small multiple of u^2 plus $C'(\ell + r)^2/m = o(1)$, and the quadratic term is likewise absorbed. Fixed derivatives add only polynomial factors. The analytic functions and their needed derivatives are uniformly bounded on a common complex neighborhood. Taylor's remainder is therefore $O(m^{-2}(1 + \ell + r)^{D_0})$ after Gaussian integration. Odd powers of z are odd polynomials in u and integrate to zero. Extending the even polynomial integrals to the real line costs a superpolynomially small Gaussian tail.

The degree-two polynomial, integrated with Gaussian moments $\mathbb{E}u^{2j} = (2j-1)!!/B^j$, is exactly (65). This proves (66). In particular, the unshifted coefficient is

$$\begin{aligned} c_1(s) := D_1(0,0) &= -\frac{D^2 f}{2fB} + \frac{(Df)(DB)}{2fB^2} + \frac{D^2 B}{8B^2} - \frac{5(DB)^2}{24B^3} \\ &= \frac{s^6 - 7s^5 - 5s^4 + 32s^3 + 32s^2 - 9s - 8}{12(s-1)(s+1)(s+2)^3}. \end{aligned} \quad (69)$$

Dividing by the unshifted expansion is uniformly legitimate and gives

$$\begin{aligned} \frac{C_{N-\ell, m-r}}{C_{N, m}} &= t^\ell h^{-r} \left[1 + \frac{1}{m} \left(\frac{r}{2} - \frac{E\beta}{B} - \frac{\beta^2}{2B} \right) \right. \\ &\quad \left. + O(N^{-2}(1 + \ell + r)^{D_0}) \right]. \end{aligned} \quad (70)$$

9.3 The principal endpoint through first order

There is an exact useful version of the Newton multiplication coefficient:

$$c_{\ell, r}(m) = \frac{\Delta^r z_\ell(m-r)}{r!}. \quad (71)$$

Indeed, expand $z_\ell(k)$ in the Newton basis centered at $q = m - r$ and multiply by k^q ; the shifted falling factorial becomes k^{q+r} . Applying this to every Newton term of a polynomial proves (71).

For a uniform product estimate let $d_0 = \ell - r$, let U be a sum of r independent uniform $[0, 1]$ variables, and independently choose a uniform d_0 -subset S of $\{0, \dots, \ell - 1\}$. Repeated integration of a finite difference gives the exact identity

$$c_{\ell,r}(m) = \gamma_\ell \binom{\ell}{r} m^{d_0} \mathbb{E} \prod_{i \in S} \left(1 + \frac{2i + 1 - r + U}{m} \right). \quad (72)$$

The expectation of the sum of offsets is $d_0(\ell - r/2)$. Each offset has magnitude at most 2ℓ , so the remainder beyond the linear term is bounded in absolute value by $(2\ell^2/m)^2 e^{2\ell^2/m}/2$. Thus

$$c_{\ell,r}(m) = \gamma_\ell \binom{\ell}{r} m^{\ell-r} \left[1 + \frac{(\ell-r)(\ell-r/2)}{m} + O(\ell^4/m^2) \right] \quad (73)$$

uniformly through logarithmic shifts, including $r = 0$ and $r = \ell$.

The remaining finite products have expansions

$$\begin{aligned} \frac{(2N - 2\ell - 1)!!}{(2N - 1)!!} &= (2N)^{-\ell} \left(1 + \frac{\ell^2}{2N} + O((1 + \ell)^4/N^2) \right), \\ \frac{m!}{(m-r)!} &= m^r \left(1 - \frac{r(r-1)}{2m} + O((1+r)^4/m^2) \right). \end{aligned}$$

Combine these with (70), (73), and $(N - \ell)/N$. With

$$b_{\ell,r}^0 = \gamma_\ell s^{2\ell} \binom{\ell}{r} s^r (1-s)^{\ell-r},$$

one obtains

$$b_{N,m,\ell,r} = b_{\ell,r}^0 \left[1 + \frac{\kappa_{\ell,r}}{N} + O(N^{-2}(1+\ell)^{D_{\text{ker}}}) \right], \quad (74)$$

where D_{ker} is a fixed finite exponent and

$$\kappa_{\ell,r} = \frac{\ell^2}{2} - \ell + \mu \left[\ell^2 - \frac{3}{2}\ell r + r - \frac{E(\ell - \mu r)}{B} - \frac{(\ell - \mu r)^2}{2B} \right]. \quad (75)$$

All errors are relative to the positive b^0 . Consequently, their sum is bounded by a constant times $N^{-2} \sum_{\ell \geq 1} \gamma_\ell s^{2\ell} (1 + \ell)^{D_{\text{ker}}} = O_J(N^{-2})$. There is no leftover power of $\log N$ after summation.

9.4 The opposite endpoint and degree loss

We claim

$$\mathcal{B}_{N,m} = \frac{S_0(s)}{N} + O_J(N^{-2}), \quad S_0(s) = \frac{s^2}{2t}. \quad (76)$$

The recurrence makes every p_j monic. Its top coefficient comes only from kp_{j-1} ; all other terms in (5) have smaller degree. For $0 \leq d_0 \leq j \leq A \log N$, put $V_{j,d_0} = \mathcal{L}_m(k^{d_0} z_{N-j})/Q_{N,m}$. The multiplication coefficient for k^{d_0} is nonnegative and bounded by $\binom{d_0}{r} m^{d_0-r}$, since its stay factors do not exceed m . The uniform shifted ratio therefore gives

$$\begin{aligned} V_{j,d_0} &\leq C(2N)^{-j} m^{d_0} t^j (1 + h^{-1})^{d_0} \\ &= C s^{2j} M^{d_0-j}, \quad M = m(1 + h^{-1}) \geq cN. \end{aligned} \quad (77)$$

For the top degree, the same first-order estimates give

$$V_{j,j} = s^{2j} \left(1 + O((1+j)^{D_2}/N)\right) \quad (78)$$

for a fixed exponent D_2 .

By (18), the coefficient of k^{j-a} in p_j , $a \geq 1$, is at most $2j\gamma_j j^{2a}/a!$. This uses the elementary-symmetric bound for the positive offsets $1, 3, \dots, 2j-1$, whose sum is j^2 . Equations (77) and $\gamma_j \leq 1$ bound the sum of all lower-degree contributions by

$$Cs^{2j}j(e^{Cj^2/N} - 1) \leq C's^{2j}j^3/N \quad (79)$$

uniformly for logarithmic j . Dividing (78)–(79) by $2N$ yields

$$\frac{\mathcal{L}_m(p_j z_{N-j})}{2NQ_{N,m}} = \frac{s^{2j}}{2N} + O\left(\frac{s^{2j}(1+j)^{D_3}}{N^2}\right).$$

The error sums absolutely and uniformly. For $j > A \log N$ in the opposite half, positivity bounds its total by $CNq^{A \log N}$ using (61). Choose A so large that this and all polynomially weighted geometric tails needed here are $O(N^{-2})$. Summing $\sum_{j \geq 1} s^{2j} = s^2/t$ proves (76). This argument accounts for every degree loss and the entire opposite half, not only each fixed j .

9.5 Residual cancellation and its explicit moments

Write $R_0(\alpha) = \sqrt{t}$, $R_1(\alpha) = R_0(\alpha)r_1(s)$, and $H(N, m) = R_0(m/N) + R_1(m/N)/N$. In the short range,

$$\frac{m-r}{N-\ell} = \alpha + \frac{\alpha\ell-r}{N} + O((1+\ell)^2/N^2),$$

and smoothness on nested compact intervals gives

$$H(N-\ell, m-r) = R_0 + \frac{R_1 + (\alpha\ell-r)R'_0}{N} + O(N^{-2}(1+\ell)^{D_4}). \quad (80)$$

The prime here means differentiation in α , not in s . The zeroth-order residual in (62) is zero because $(1+q_0)R_0 = 1$. For the first order define

$$\mathcal{K} = \sum_{\ell,r} b_{\ell,r}^0 \kappa_{\ell,r}, \quad \mathcal{J} = \sum_{\ell,r} b_{\ell,r}^0 (\alpha\ell-r).$$

Its cancellation equation is

$$(1+q_0)R_1 = -R_0\mathcal{K} - R'_0\mathcal{J} - S_0. \quad (81)$$

All sums converge absolutely. For transparent exact evaluation, put

$$\begin{aligned} L_1 &= \sum_{\ell \geq 1} \ell w_\ell = \frac{s^2}{2t^{3/2}}, & L_2 &= \sum_{\ell \geq 1} \ell^2 w_\ell = L_1 + \frac{3s^4}{4t^{5/2}}, \\ \mathcal{J} &= (\alpha-s)L_1, & R'_0 &= -\frac{(1+s)^2}{2(s+2)\sqrt{t}}. \end{aligned}$$

Conditionally on ℓ , the index r is binomial with parameters (ℓ, s) . Using its first two moments in (75) gives

$$\mathcal{K} = \frac{L_2}{2} - L_1 + \mu \left[L_2 - \frac{3}{2}sL_2 + sL_1 - \frac{E}{B}(1 - \mu s)L_1 - \frac{(1 - \mu s)^2 L_2 + \mu^2 s(1 - s)L_1}{2B} \right]. \quad (82)$$

Equivalently, these moments follow by Euler differentiation of

$$\sum_{\ell \geq 1, r} b_{\ell, r}^0 x^\ell y^r = (1 - s^2 x(1 - s + sy))^{-1/2} - 1.$$

Substituting (64), (82), and S_0 into (81) gives exactly the rational function r_1 in (58). Both the drift term $R'_0 \mathcal{J}$ and the opposite-endpoint source S_0 are essential. The exact companion checks this rational identity without using the sequence values.

The local polynomial remainders are summable against w_ℓ , and the remaining tails were chosen $O(N^{-2})$. Thus H satisfies (62) with residual $O(N^{-2})$ uniformly on J_1 . For long predecessor jumps use bounded smooth extensions of R_0, R_1 agreeing with them near J_2 ; their contributions are bounded by the middle estimate. No local Taylor assertion is made outside those neighborhoods.

9.6 Stopped contraction creates the second order bound

Let $e(N, m) = R_{N, m} - H(N, m)$. Subtract the exact equation and absorb the long principal tail using boundedness and (61). For a sufficiently large logarithmic cutoff,

$$|e(N, m)| \leq \sum_{\ell \leq A \log N, r \leq \ell} b_{N, m, \ell, r} |e(N - \ell, m - r)| + CN^{-2} \quad (83)$$

uniformly on J_1 for large N . The short kernel mass is at most a fixed $\bar{q} < 1$ by its leading limit and the strict contraction on J_2 . Choose $c > 2/|\log \bar{q}|$ and iterate for $M_N = \lceil c \log N \rceil$ steps.

Each step changes either index by at most $A \log N$. Starting from J , all paths of depth at most M_N satisfy

$$N' = N - O((\log N)^2), \quad m' = m - O((\log N)^2), \quad \left| \frac{m'}{N'} - \frac{m}{N} \right| = O((\log N)^2/N).$$

The margins in (60) ensure that every such node remains in J_1 , has size at least $N/2$, and is above the uniform validity threshold. Thus (83) applies at every node. The sum of path weights at level j is at most $\bar{q}^j$, independent of the branching count. The accumulated forcing is at most $4CN^{-2} \sum_{j < M_N} \bar{q}^j = O(N^{-2})$. The terminal errors are uniformly bounded because $0 \leq R \leq 1$ and H is bounded; their weighted contribution is $O(\bar{q}^{M_N}) = O(N^{-2})$. Hence $e(N, m) = O_J(N^{-2})$. This proves (58) without assuming beforehand that $N(R - R_0)$ or $N^2(R - H)$ is bounded.

9.7 The diagonal coefficient

At $N = 2n, m = n$, the correction from (58) is $r_1(s_0)/(2n)$, whereas the saddle correction in (66) is $c_1(s_0)/n$. The factorial factor (55) has logarithmic, and therefore relative, first coefficient

$$\frac{1}{48} - \frac{1}{24} - \frac{2}{12} = -\frac{3}{16}.$$

It follows that

$$b_1 = -\frac{3}{16} + c_1(s_0) + \frac{r_1(s_0)}{2} = -\frac{989}{2176} - \frac{907\sqrt{17}}{36992}. \quad (84)$$

This is an exact identity in $\mathbb{Q}(\sqrt{17})$. Each of the three factors has a controlled relative $O(n^{-2})$ remainder, by the stopped contraction, the unshifted case of (66), and Stirling's expansion. Their finite product retains that remainder, proving (59).

10 Lambert W inversion with integer rounding

Define the integer threshold

$$\mathcal{N}(X) = \min\{n \geq 0 : a_n \geq X\}.$$

It exists for large X by Theorem 1.2. Let W denote the positive real Lambert function, characterized by $W(z)e^{W(z)} = z$ for $z > 0$. Put

$$L = \log X, \quad c = \log(C\sqrt{2\pi}), \quad T = L - c, \quad x_0(X) = \frac{T}{W(dT/e)}. \quad (85)$$

All these expressions are used only for X sufficiently large that $T > 0$.

Theorem 10.1 (Rounded threshold bracket). *There exists a positive function $\epsilon(X)$ satisfying $\epsilon(X) = o(1/\log \log X)$ such that*

$$\lceil x_0(X) - \epsilon(X) \rceil \leq \mathcal{N}(X) \leq \lceil x_0(X) + \epsilon(X) \rceil. \quad (86)$$

In particular, $\mathcal{N}(X) \sim L/W(dL/e)$.

Proof. The leading equivalent and Stirling's formula give

$$\log a_n = \Phi(n) + e_n, \quad \Phi(x) = x(\log x + \log d - 1) + c, \quad e_n \rightarrow 0. \quad (87)$$

The $\frac{1}{2} \log n$ in Stirling cancels the $n^{-1/2}$ in the forward formula; therefore no $\log n$ term remains in (87). The equivalent also gives

$$\frac{a_{n+1}}{da_n} \rightarrow 1,$$

so a_n is eventually strictly increasing.

Writing $w = W(dT/e)$ gives $x_0 = T/w = (e/d)e^w$, hence $x_0 \log(dx_0/e) = T$. Thus $\Phi(x_0) = L$ exactly and $\Phi'(x) = \log(dx)$. Also $\log x_0 \sim \log L$; for example $W(z) = \log z - \log \log z + o(1)$ follows directly from $W(z) + \log W(z) = \log z$.

Define the finite tail bound

$$\eta(X) = \sup_{n \geq \lfloor x_0/2 \rfloor} |e_n| + \frac{1}{x_0}, \quad \epsilon(X) = \frac{2\eta(X)}{\log(dx_0/2)}.$$

For sufficiently large X , the denominator is positive and $\eta(X) \rightarrow 0$, so $\epsilon = o(1/\log x_0) = o(1/\log L)$. If an integer $n \geq x_0/2$ satisfies $n < x_0 - \epsilon$, the mean value theorem gives $\Phi(n) < L - 2\eta$, and hence $\log a_n < L$. For $n_+ = \lceil x_0 + \epsilon \rceil$, the same derivative lower bound gives $\Phi(n_+) \geq L + 2\eta$, so $\log a_{n_+} > L$.

Integers below $x_0/2$ cause no difficulty: $\Phi(x_0/2)$ differs from $\Phi(x_0)$ by a quantity tending to infinity, $e_n \rightarrow 0$, and eventual monotonicity controls all sufficiently large earlier indices. The finitely

many initial a_n are also below X eventually. Thus every integer less than $x_0 - \epsilon$ fails the threshold, whereas n_+ reaches it, proving (86) without choosing any interpolation of a_n . Finally, replacing $T = L - c$ by L changes only lower-order terms, proving the leading scale. $\square$

Remark 10.2 (What the error does and does not mean). The $o(1/\log \log X)$ error applies to the *real endpoints before ceilings*. In general the integer $\mathcal{N}(X) - x_0(X)$ need not tend to zero. Near an integer boundary the bracket can contain two adjacent integers. The function ϵ above is an existence bound from a qualitative asymptotic, not an effective numerical certificate for a specified X . A sharper computable threshold would require explicit forward error bounds. Lambert W inversion itself is a standard consequence, not a separate novelty claim.

Corollary 10.3 (First corrected threshold bracket). *With x_0 as in (85), put*

$$\beta_1 = b_1 + \frac{1}{12}, \quad x_1 = x_0 - \frac{\beta_1}{x_0 \log(dx_0)}. \quad (88)$$

There is a fixed constant $M > 0$ such that, for all sufficiently large X ,

$$\left\lceil x_1 - \frac{M}{x_0^2 \log(dx_0)} \right\rceil \leq \mathcal{N}(X) \leq \left\lceil x_1 + \frac{M}{x_0^2 \log(dx_0)} \right\rceil. \quad (89)$$

The constant is an existence constant inherited from the forward remainder; no effective value for a prescribed finite threshold is claimed.

Proof. Theorem 9.1 and Stirling's formula give

$$\log a_n = \Psi(n) + O(n^{-2}), \quad \Psi(x) = \Phi(x) + \frac{\beta_1}{x},$$

where Φ is defined in (87). In particular, the $1/12$ comes from the factorial, not from b_1 alone. By definition $\Phi(x_0) = L$. The shift $\delta = x_1 - x_0$ is $O(1/(x_0 \log x_0))$, and Taylor's theorem yields

$$\Psi(x_1) - L = O(x_0^{-3}/\log x_0), \quad \Psi'(x) = \log(dx) - \frac{\beta_1}{x^2}.$$

This derivative is positive and at least a fixed multiple of $\log x_0$ on $[x_0/2, \infty)$ when X is large. Choose D so that the forward logarithmic error is at most D/n^2 for all sufficiently large n . For $n \geq x_0/2$ this is at most $4D/x_0^2$. Increasing M if necessary, the mean value theorem shows that every integer below $x_1 - M/(x_0^2 \log(dx_0))$ in this range has $\log a_n < L$. The integer ceiling of the upper endpoint has $\log a_n \geq L$. Earlier integers are below threshold by eventual monotonicity and the same finite-initial-segment argument as in Theorem 10.1. These integer witnesses prove (89). $\square$

The real half-width in (89) is $O(x_0^{-2}/\log x_0)$, substantially smaller than the leading-only existence width. The ceilings remain indispensable.

[write] **Instances of the transseries apparatus; superseded, 5 October 2026.** Theorem 10.1 solves $\Phi(x) = L$, that is $x(\log x + \log d - 1) = T$. This is Proposition `p0:prop:factorial-core` of the transseries volume [15] with $\kappa = 1$ and its d equal to $\log d - 1$; its solution $\exp(W_0(Te^{\log d - 1}) - \log d + 1)$ is $T/W(dT/e) = x_0$, and after taking logarithms the proposition is the Lambert core `p0:thm:lambert-core` with $a = b = 1$, in the variable $\log x + \log d - 1$ and with right-hand side $\log T + \log d - 1$. (Wording corrected after the independent check of 5 October 2026, recorded at the end of Section 35; the first wording read “and the proposition is the Lambert core `p0:thm:lambert-core` in the variable $\log x$ ”.) The brackets (86) and (89) have the two-ceiling form of `p0:thm:staircase` (1)–(2); the proofs here work directly at the integers. Remark 10.2 already disclaims novelty for the inversion. Part II's theorem in Section 21 gives brackets of half-width $O(x_0^{-K}/\log x_0)$ for every fixed K and supersedes both statements; its λ_1 is β_1 here.

11 Exact computations and reproducibility

The companion separates exact finite checks from analytic claims. The proof of the asymptotic and its first correction is contained in Sections 5–9; no finite sample proves an $o(1)$ remainder or a local limit theorem. The exact checks independently construct the weighted prefixes with a last-step-state method, compare them with powers of the Riccati series, verify the auxiliary differential and logarithmic-derivative identities, and compare both Stirling-number and finite-difference Newton extraction. They check the source terms, the factor $1/m!$, and the quadratic-field identities for $s_0, d, \pi^3 C^2, b_1$. The companion independently regenerates the Gaussian correction and kernel moment algebra, and verifies the inverse correction algebra.

All stated finite checks use exact integers or rational numbers. Decimal approximations, if displayed, are only normalization diagnostics. They are not interval certificates. Public provenance identifies the OEIS entries and prior literature; the package contains intended computational sources and typed fixtures, not exploratory notes. The README specifies clean output directories and how to rerun exact verification and a deterministic PDF and ZIP build. It also describes the validation and fault tests. Source and output hashes identify the precise delivered bytes; matching a hash establishes byte identity, not a mathematical theorem.

[write] **The companion here, 5 October 2026.** Report 141’s companion is shipped as `141-diagonal-SOURCES.md`, `code/141-diagonal-*.py` and `data/141-diagonal-checks-fixtures.json`; its README, PDF and manifest are not shipped. Its verifier runs only in the delivered layout; the report README gives the rerun route from the arrival commit.

12 Further questions

Higher relative orders. The proved first correction suggests continuing in descending powers of n . Further orders need their own uniform coefficient estimates, kernel equations, and propagated remainders. This report makes no all-orders claim. A formal symbolic recursion alone would not certify that continuation.

[write] **Answered by Part II, 5 October 2026.** Part II proves every fixed order: Theorem 13.1 on every compact $J \subset (0, 1)$ and Corollary 13.2 on the diagonal, each coefficient produced by the finite algorithm of Section 18 and backed by a propagated remainder (Section 19). Convergence of the series and uniformity for a growing order remain open (Section 35, item 5).

Beyond the contraction interval. The elementary kernel mass satisfies $q_0(s) < 1$ exactly when $s < \sqrt{3}/2$. The first-correction theorem applies on every compact subinterval of this strict contraction range. Can the same limiting ratio $\sqrt{1-s^2}$ be proved throughout the full range $0 < \alpha < 1$, including where this direct absolute-error contraction fails? A failed sufficient contraction test is not evidence of a phase transition. Additional structure of the positive convolution may be needed.

[write] **Answered by Part II, 5 October 2026.** Yes. Part II, Theorem 13.1, proves $R_{N,m} \rightarrow \sqrt{1-s^2}$, with every fixed order, on every compact subset of $0 < \alpha < 1$. The additional structure is the exact inverse $(I - \mathcal{B}_s)(I + \mathcal{A}_s) = I$ of the frozen convolution, with $\|\mathcal{B}_s\|_1 = 1 - \sqrt{t} < 1$ for every $s \in (0, 1)$ ((II.56)–(II.57)). At intake the interior residuals $N^3(R_{N,m}/\sqrt{t} - 1 - r_1/N - r_2/N^2)$ were also found of stable size at $\alpha = 0.9$, beyond $\alpha_* \approx 0.804$ (–5565, –7052 at $N = 100, 200$; –5496 at $N = 300$ in an independent check). (Wording corrected after the independent check of 5 October 2026, recorded at the end of Section 35; the first wording read “were also found bounded”; three values cannot show boundedness.)

Effective errors and thresholds. Quantifying the convergence of the kernel recurrence could produce explicit forward error constants, which would convert the qualitative Lambert W bracket into a numerical threshold certificate. Care is needed near integer rounding boundaries even with a very accurate real approximation.

[write] **Still open, 5 October 2026.** Not settled by Parts II or III; see Section 35, item 4.

Combinatorial meaning of the Newton triangle. The recurrence proves nonnegative integral ordinary coefficients, and Stirling conversion proves $T(N, m) \in \mathbb{Z}_{\geq 0}$. It is natural to seek a direct combinatorial interpretation for the falling-factorial coefficient, especially the diagonal $T(2n, n)$, that explains the limiting large-component factor $\sqrt{t_0}$ without an analytic transformation.

[write] **Still open, 5 October 2026.** Part III asks the analogous question for its profile numerator K_q ; see Section 35, item 6.

Other moving regimes. The boundaries $m = o(N)$ and $N - m = o(N)$ approach different saddle limits. Uniform transition estimates there would clarify how the Newton diagonal connects to fixed-order transforms and to the top coefficients of the polynomial. Neither boundary regime is covered by the compact-ratio theorem.

[write] **Partly answered by Part III, 5 October 2026.** The upper edge $N - m = q$ is treated by Part III: exact sandwiches for every $q \leq N/2$ (Theorem 24.2) and the connected profile $\sqrt{N}R_{N, N-q} \sim F_q$ uniformly for $q = o(\sqrt{N})$ (Corollary 24.3), which connects to the top coefficients of the polynomial as asked. The range $\sqrt{N} \lesssim q = o(N)$ and the lower edge $m = o(N)$ remain open: Section 35, items 1 and 2.

Part II

All orders for the weighted Dyck Newton diagonal

Bundle Report 142, archive A292692_All_Orders_Interior_Asymptotics_and_Inversion_Source.zip, delivered as Report142.tex (16 pages), the base of the merge. Delivered title block “All orders for the weighted Dyck Newton diagonal / Report 142 / October 2, 2026”. Printed as delivered, with [write] notes. Delivered Section k is Section $k + 12$ here, its statement $k.j$ is $(k + 12).j$, and its equation (k) is (II. k). Subsection 22.1 is added in the write.

Reading conventions of Part II	Elsewhere in the report
“Report 141”, [13]	Part I
$a_{N,m,\ell,r}$, $a_s(\ell, r)$ the kernel; $B_{N,m}$, B_q the opposite endpoint; $\mathcal{A}_s$, $\mathcal{B}_s$, $b_s(\ell, r)$ the frozen operators	Part I: $b_{N,m,\ell,r}$, $b_{\ell,r}^0$, $\mathcal{B}_{N,m}$
$\mathcal{D} = t d/dt$; $D_j(\ell, r; s)$; $C_a(n) = c_a(n)/n^a$	Part I: D ; Part III: $C_q = (3/2)^q/q!$, $D_\ell(h)$
K the order; H the integer point; H_K the truncation	Part I: K the integer point; Part III: K_q , $H(x)$
λ_j inverse coefficients; $c = \log(C\sqrt{2\pi})$	Part III: λ_r Stirling coefficients, $c = 2\sqrt{2/\pi}$
$\mathcal{G}(x, y)$ moment generating function	Part III: $\mathcal{G}(z)$ the squared profile

Abstract of this Part (as delivered). We prove a complete fixed-order asymptotic expansion for the Newton coefficients of the weighted Dyck polynomials underlying OEIS A292692. The moving ratio $\alpha = m/N$ may lie in any compact subset of $(0, 1)$. Every normalized coefficient has the form $\sqrt{1-s^2}$ times a rational function of the saddle parameter s , where $\alpha = 2s^2/(1+s)$. An explicit finite algorithm determines the coefficients. The full interior range follows by freezing a convolution inverse at the base state, controlling the change of saddle at neighboring states, and bootstrapping a bounded error through finitely many nested compact sets. On the diagonal, all relative coefficients belong to $\mathbb{Q}(\sqrt{17})$; in particular the second is $-(30800095 + 3912441\sqrt{17})/80494592$. We also obtain an all-orders Lambert W and Newton inversion algorithm, with the real uncertainty retained inside the integer ceilings. Report 141 supplies the exact weighted-path foundation; all analytic arguments required for this continuation are given here. The conclusions are fixed-order statements, with no effective remainder constants or claims at the two ratio boundaries.

13 Definitions and main results

For a Dyck path of semilength N , weight each peak at coordinates (a, b) by $k + a/b$, and let $p_N(k)$ be the total path weight. The exact bridge proved in Report 141 [13] identifies these polynomials by

$$p_0(k) = 1, \quad p_N(k) = (k + 2N - 2)p_{N-1}(k) + \sum_{i=0}^{N-1} p_i(k)p_{N-1-i}(k). \quad (\text{II.1})$$

Our analytic starting point may equivalently be this recurrence. Write

$$\mathcal{L}_m P = [k^m]P(k) = \frac{\Delta^m P(0)}{m!}, \quad \gamma_N = 4^{-N} \binom{2N}{N}, \quad (\text{II.2})$$

$$z_N(k) = \gamma_N \prod_{i=0}^{N-1} (k + 2i + 1), \quad Q_{N,m} = \mathcal{L}_m z_N, \quad R_{N,m} = \frac{\mathcal{L}_m p_N}{2N Q_{N,m}}. \quad (\text{II.3})$$

The last definition applies for $N \geq 1$ and $0 \leq m \leq N$. The sequence of interest is

$$a_n = \mathcal{L}_n p_{2n}, \quad a_0 = 1. \quad (\text{II.4})$$

The factor $1/m!$ in (II.2) is essential.

Throughout the interior ratio range, set

$$\alpha = \frac{m}{N} = \frac{2s^2}{1+s}, \quad 0 < s < 1, \quad t = 1 - s^2, \quad h = \frac{1-s}{s}, \quad \mu = \frac{1}{\alpha} = \frac{1+s}{2s^2}, \quad B = \frac{t(s+2)}{4s^4}. \quad (\text{II.5})$$

The map $s \mapsto \alpha$ is strictly increasing from $(0, 1)$ onto $(0, 1)$. When convenient, a function of s is also regarded as a function of α through this map.

Theorem 13.1 (All fixed orders on the full interior). *There are unique analytic coefficient functions R_j on $0 < s < 1$, of the form*

$$R_j(s) = \sqrt{1-s^2} r_j(s), \quad r_j \in \mathbb{Q}(s), \quad r_0 = 1, \quad (\text{II.6})$$

regular throughout this interval, with the following property. For every fixed integer $K \geq 1$ and every compact $J \subset (0, 1)$,

$$R_{N,m} = \sum_{j=0}^{K-1} R_j(s(m/N)) N^{-j} + O_{J,K}(N^{-K}) \quad (\text{II.7})$$

uniformly over integer pairs with $m/N \in J$. The finite algorithm in Section 18 determines all coefficients exactly.

Corollary 13.2 (The diagonal coefficient field). *For every fixed $K \geq 1$,*

$$a_n = C d^n \frac{n!}{\sqrt{n}} \left(\sum_{j=0}^{K-1} b_j n^{-j} + O_K(n^{-K}) \right), \quad (\text{II.8})$$

where

$$d = \frac{51\sqrt{17} - 107}{4}, \quad C = \frac{\sqrt{2 + 2/\sqrt{17}}}{\pi^{3/2}}, \quad b_0 = 1, \quad (\text{II.9})$$

$$b_j \in \mathbb{Q}(\sqrt{17}) \quad (j \geq 0), \quad b_1 = -\frac{989}{2176} - \frac{907\sqrt{17}}{36992}, \quad (\text{II.10})$$

$$b_2 = -\frac{30800095 + 3912441\sqrt{17}}{80494592}. \quad (\text{II.11})$$

The first relative correction and the constants were established in Report 141. The continuation here includes all fixed orders, the entire interior ratio range, the exact coefficient field, and an integer-safe inverse at every fixed order. No convergence of the expansion, uniformity for K growing

with N , boundary result at $\alpha = 0$ or 1 , effective onset or error constant, or priority claim is made. Finite exact checks in the companion corroborate algebra; the analytic remainder is proved below.

[write] **Reading Part II against Part I, 5 October 2026.** “Report 141” and [13] are Part I of this report. Its first relative correction and constants are Part I’s Theorems 1.2 and 9.1 (the latter for $\alpha < 6 - 3\sqrt{3}$ only). Corollary 13.2 proves Kotěšovec’s form on A292692 at every fixed order. “No boundary result at $\alpha = 0$ or 1 ” stays true of this expansion; Part III treats the upper edge by a separate argument, for deficits $q = o(\sqrt{N})$, and the range between is open (Section 35).

14 Exact transforms and the positive convolution

Report 141 proves the formal convolution identity

$$2Nz_N = \sum_{j=1}^N p_j z_{N-j}. \quad (\text{II.12})$$

All ordinary coefficients of p_N, z_N are nonnegative. Isolating the $j = N$ term therefore gives $0 \preceq p_N \preceq 2Nz_N$ in ordinary coefficientwise order. Since $\mathcal{L}_m k^d = \{ \frac{d}{m} \} \geq 0$, Newton extraction preserves this order, and hence

$$0 \leq R_{N,m} \leq 1. \quad (\text{II.13})$$

Here $Q_{N,m} > 0$ throughout the valid range, as the next formula shows.

Define analytic functions on $|z| < 1$ by

$$f(z) = (1 - z)^{-1/2}, \quad h(z) = f(z) - 1, \quad C_{N,m} = [z^N] f(z) h(z)^m. \quad (\text{II.14})$$

The product definition gives

$$\sum_{N \geq 0} \frac{z_N(k)}{(2N-1)!!} z^N = f(z)(1 + h(z))^k = f(z) \sum_{m \geq 0} \frac{k^m}{m!} h(z)^m.$$

Thus, with $(-1)!! = 1$,

$$Q_{N,m} = \frac{(2N-1)!!}{m!} C_{N,m}. \quad (\text{II.15})$$

All relevant coefficients are positive. The saddle condition $th'(t)/h(t) = N/m$ is exactly (II.5).

For any polynomial P , Newton multiplication gives

$$\mathcal{L}_m(z_\ell P) = \sum_{r=0}^{\ell} c_{\ell,r}(m) \mathcal{L}_{m-r} P, \quad c_{\ell,r}(m) = \frac{\Delta^r z_\ell(m-r)}{r!}. \quad (\text{II.16})$$

Indeed, expand $z_\ell(k)$ in the Newton basis centered at $q = m - r$, and multiply each term by k^q ; the identity $k^q(k - q)^r = k^{q+r}$ proves the assertion. Only admissible predecessor indices are included; otherwise the transformed summand is zero and its kernel is defined to be zero. For admissible predecessors the coefficients in (II.16) are nonnegative.

Put $L = \lfloor N/2 \rfloor$. Splitting (II.12) into disjoint halves yields the exact equation

$$R_{N,m} = 1 - (\mathcal{K}_N R)_{N,m} - B_{N,m}, \quad (\text{II.17})$$

$$(\mathcal{K}_N R)_{N,m} = \sum_{\ell=1}^L \sum_{r=0}^{\ell} a_{N,m,\ell,r} R_{N-\ell,m-r}, \quad (\text{II.18})$$

$$a_{N,m,\ell,r} = \left(1 - \frac{\ell}{N}\right) c_{\ell,r}(m) \frac{Q_{N-\ell,m-r}}{Q_{N,m}}, \quad (\text{II.19})$$

$$B_{N,m} = \sum_{j=1}^{N-L-1} \frac{\mathcal{L}_m(p_j z_{N-j})}{2N Q_{N,m}}. \quad (\text{II.20})$$

The possible even midpoint occurs once. Every principal predecessor has size at least $N/2$.

[write] **Restated from Part I, 5 October 2026.** Section 14 restates Part I's results in Report 142's own words: (II.12) is the second identity of Lemma 3.1; (II.13) is (20), here also for $m = 0$; (II.15) is Lemma 4.1; the formula for $c_{\ell,r}(m)$ in (II.16) is (71); and (II.17)–(II.20) is Part I's split (62)–(63), with the kernel written $a_{N,m,\ell,r}$ instead of $b_{N,m,\ell,r}$ and the opposite endpoint $B_{N,m}$ instead of $\mathcal{B}_{N,m}$.

15 A uniform shifted saddle expansion

Let S be any compact subinterval of $(0, 1)$ in the s variable. Define

$$\mathcal{D} = t \frac{d}{dt} = -\frac{1-s^2}{2s} \frac{d}{ds}, \quad \kappa_j = \mathcal{D}^j \log h(t), \quad g_j = \mathcal{D}^j \log f(t). \quad (\text{II.21})$$

These are rational functions of s ; in particular $\kappa_1 = \mu$ and $\kappa_2 = B > 0$.

Lemma 15.1 (Shifted expansion with summable remainders). *Fix $A > 0$ and $K \geq 1$. Uniformly on S and for $0 \leq r \leq \ell \leq A \log m$,*

$$C_{N-\ell,m-r} = \frac{f(t)h(t)^m t^{-N} t^\ell h^{-r}}{\sqrt{2\pi m B}} \left(\sum_{j=0}^{K-1} D_j(\ell, r; s) m^{-j} + O(m^{-K} (1 + \ell + r)^{M_K}) \right), \quad (\text{II.22})$$

for some fixed exponent M_K . Here $D_0 = 1$ and $D_j \in \mathbb{Q}(s)[\ell, r]$.

Proof. Set $\psi(\theta) = h(te^{i\theta})/h(t)$ and $g(\theta) = f(te^{i\theta})/f(t)$. After extracting the factors outside the parentheses in (II.22), the Fourier integral before its Gaussian normalization is

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} g(\theta) \psi(\theta)^{-r} e^{i\ell\theta} \exp\{m(\log \psi(\theta) - i\mu\theta)\} d\theta. \quad (\text{II.23})$$

Use the analytic logarithm only near zero. Elsewhere the exact integrand is

$$g(\theta) \psi(\theta)^{m-r} e^{-i(N-\ell)\theta}.$$

No global logarithm is needed. Moreover h has no zero on $|z| = t > 0$, since $h(z) = 0$ would imply $z = 0$.

The function ψ is the characteristic function of the integer law $\mathbb{P}(X = j) = [z^j] h(z) t^j / h(t)$, $j \geq 1$. Every positive degree has positive mass, so the law has span one. Its exponential moments are uniform on S , because t stays in a compact subset of $(0, 1)$. The mean and variance are μ and

B. Uniform aperiodicity gives $|\psi| \leq q_* < 1$ on any fixed outer arc. On a sufficiently small fixed inner arc,

$$\Re(\log \psi(\theta) - i\mu\theta) \leq -b\theta^2$$

with $b > 0$ uniform on S . Choose $0 < \eta < 1/6$. The outer arc is exponentially small; the intermediate arc $m^{-1/2+\eta} \leq |\theta| \leq \theta_0$ contributes $O(\exp(-cm^{2\eta}))$. Polynomial logarithmic shifts do not change these conclusions, since $m - r \geq m/2$ eventually.

On the central arc put $\theta = u/\sqrt{m}$ and $z = m^{-1/2}$. After removing $e^{-Bu^2/2}$, the remaining analytic factor is

$$\exp \left\{ \sum_{j \geq 1} \frac{g_j - r\kappa_j + \ell \mathbf{1}_{j=1}}{j!} (iu)^j z^j + \sum_{j \geq 3} \frac{\kappa_j}{j!} (iu)^j z^{j-2} \right\}. \quad (\text{II.24})$$

Expand in the real variable z through degree $2K - 1$. Fixed derivatives at zero are polynomials in u, ℓ, r over $\mathbb{Q}(s)$. Odd powers of z give odd polynomials in u and integrate to zero.

Here is a uniform Taylor remainder bound. For $0 \leq z' \leq m^{-1/2}$, $|u| \leq m^\eta$, the saddle remainder exponent is $O(z'|u|^3) = O(m^{-1/2+\eta})u^2$. Each required fixed derivative has polynomial u growth. The amplitude exponent has modulus bounded by

$$C(1 + \ell + r)z'|u| + C(1 + r)(z'u)^2.$$

Young's inequality bounds its linear term by $\delta u^2/2 + C'(1 + \ell + r)^2/m$, and the second summand is bounded uniformly in the logarithmic range. The quadratic term is absorbed for large m . Analytic derivatives are uniformly bounded in a fixed complex neighborhood of the compact saddle set. Thus every derivative required for the remainder is bounded by

$$C_K(1 + \ell + r)^{M_K}(1 + |u|)^{E_K} e^{\delta u^2}, \quad \delta < \frac{1}{4} \min_S B.$$

Taylor's integral remainder, multiplied by the Gaussian and integrated, is $O(m^{-K}(1 + \ell + r)^{M_K})$. Extending polynomial Gaussian moments to the whole real line costs a superpolynomially small error. This proves the asserted expansion and its uniform remainder. $\square$

The proof is also a finite coefficient algorithm: expand (II.24) through z^{2j} and apply

$$u^{2a} \mapsto \frac{(2a-1)!!}{B^a}, \quad u^{2a+1} \mapsto 0. \quad (\text{II.25})$$

The coefficient of z^{2j} is D_j . Only $\kappa_1, \dots, \kappa_{2j+2}$ and $g_1, \dots, g_{2j}$ are needed. For later use introduce the normalized ratio series

$$V_{\ell,r}(x) = \frac{\sum_{j \geq 0} D_j(\ell, r; s) x^j}{\sum_{j \geq 0} D_j(0, 0; s) x^j}. \quad (\text{II.26})$$

It has polynomial coefficients in ℓ, r over $\mathbb{Q}(s)$, satisfies $V_{0,0} = 1$ identically, and has the corresponding relative uniform remainder when truncated at a fixed order.

[write] Generalizes Part I, 5 October 2026. Lemma 15.1 extends Part I's Lemma 5.1 and (66) (first order, compact subsets of $(0, \alpha_*)$) to every order and every compact s -interval. Its $D_1(\ell, r; s)$ is Part I's $D_1(\ell, r)$ of (65), and $D_1(0, 0; s)$ is Part I's $c_1(s)$ (compare (II.72) with (69)).

16 A positive estimate away from the endpoints

Define

$$U_{N,m,\ell} = \frac{\mathcal{L}_m(z_\ell z_{N-\ell})}{Q_{N,m}}.$$

Lemma 16.1 (Uniform middle bound). *On every compact $J \subset (0, 1)$ there are constants $C < \infty$ and $0 < q < 1$ such that*

$$U_{N,m,\ell} \leq CNq^{\min(\ell, N-\ell)} \quad (1 \leq \ell < N, \ m/N \in J). \quad (\text{II.27})$$

Proof. Choose an integer $H = m/(1-s) + O(1)$. On the compact ratio range, $H > m$ eventually and $H/N = 2s^2/(1-s^2) + O(N^{-1})$ stays in a compact positive interval. Nonnegative Newton coefficients imply

$$\mathcal{L}_m(z_\ell z_{N-\ell}) \leq \frac{z_\ell(H) z_{N-\ell}(H)}{H^m}.$$

The evaluation must be at an integer: then higher falling factorials are nonnegative or zero. By (II.15),

$$\frac{z_N(H)}{H^m Q_{N,m}} = \frac{[z^N]f(z)(1+h(z))^H}{\binom{H}{m} C_{N,m}}.$$

Positive-coefficient evaluation bounds the numerator by $f(t)(1+h)^H t^{-N}$. Lemma 15.1, with no shift and $K = 1$, bounds the denominator below by a constant times $\binom{H}{m} f(t) h^m t^{-N} / \sqrt{m}$. Put $p = h/(1+h) = 1-s$. The ratio is therefore at most

$$\frac{C\sqrt{m}}{\binom{H}{m} p^m (1-p)^{H-m}} \leq C' N.$$

The last inequality follows from the uniform central binomial estimate: p stays away from 0, 1, $m - Hp = O(1)$, and Stirling's formula makes the binomial probability at least $c/\sqrt{H}$.

For $\ell \leq N/2$, the product formula gives

$$\frac{z_\ell(H) z_{N-\ell}(H)}{z_N(H)} = \frac{\gamma_\ell \gamma_{N-\ell}}{\gamma_N} \prod_{i=0}^{\ell-1} \frac{H+2i+1}{H+2(N-\ell+i)+1}.$$

The gamma ratio is uniformly bounded: $\gamma_\ell \leq 1$ and

$$\log \frac{\gamma_{N-\ell}}{\gamma_N} \leq \sum_{j=N-\ell+1}^N \frac{1}{2j-1} < \frac{1}{2}.$$

Each product factor is at most $(H+N)/(H+2N-1)$, which is bounded by some $q < 1$ on our compact range for all sufficiently large N . Symmetry treats $\ell > N/2$, and a larger constant absorbs the finite remaining sizes. This proves (II.27). $\square$

In (II.18), $\sum_r a_{N,m,\ell,r} = (1-\ell/N)U_{N,m,\ell}$. Also $p_j \preceq 2jz_j$ controls (II.20) by the same bound. Consequently discarding either endpoint beyond $A \log N$ costs at most $CN^2 q^{A \log N}$. For any prescribed fixed power saving, choose A large enough. The sharp logarithmic endpoint estimates below, rather than the extra factor N in (II.27), control the retained terms.

[write] **Generalizes Part I, 5 October 2026.** Lemma 16.1 is Part I's Lemma 6.2 and (61), by the same integer-evaluation argument, on every compact $J \subset (0, 1)$.

17 Polynomial coefficients at both endpoints

17.1 The Newton multiplication polynomial

Let $d = \ell - r$, let U be a sum of r independent uniform variables on $[0, 1]$, and let S_d be an independent uniformly chosen d -subset of $\{0, \dots, \ell - 1\}$. Repeated integration of the finite difference in (II.16) gives the exact identity

$$c_{\ell,r}(m) = \gamma_\ell \binom{\ell}{r} m^d \mathbb{E} \prod_{i \in S_d} \left(1 + \frac{2i + 1 - r + U}{m} \right). \quad (\text{II.28})$$

Each offset has absolute value at most 2ℓ . Truncating the expectation after degree $K - 1$ in m^{-1} has absolute error at most

$$\frac{(2\ell^2/m)^K}{K!} \exp(2\ell^2/m). \quad (\text{II.29})$$

Indeed the sum of the absolute elementary products is bounded by the corresponding exponential series.

Write this expectation as $U_{\ell,r}(x) = \sum_{q \geq 0} u_q(\ell, r) x^q$. For fixed q ,

$$u_q(\ell, r) = \frac{(\ell - r)^q}{\ell^q} \mathbb{E} e_q(1 - r + U, 3 - r + U, \dots, 2\ell - 1 - r + U). \quad (\text{II.30})$$

Newton identities and sums of powers show that the elementary symmetric expression is a polynomial in ℓ, r, U over $\mathbb{Q}$, of degree at most $2q$. At $\ell = 0, \dots, q - 1$ it vanishes identically in r, U . It is therefore divisible by ℓ^q . Moreover moments of U are polynomials in r , from $\mathbb{E} e^z U = ((e^z - 1)/z)^r$. Thus u_q is a genuine polynomial in ℓ, r . Polynomial continuation, together with $(\ell - r)^q$, also gives the correct value when $\ell < q$. For example

$$u_0 = 1, \quad u_1 = (\ell - r)(\ell - r/2), \quad u_2 = \frac{(\ell - r)(\ell - r - 1)(12\ell^2 - 12\ell r - 4\ell + 3r^2 + r - 4)}{24}. \quad (\text{II.31})$$

17.2 Top ordinary coefficients and their boundary zeros

Define $c_a(n) = [k^{n-a}]p_n(k)$ for $n \geq a$, and set $c_a(n) = 0$ for $0 \leq n < a$. This notation is distinct from the two-index Newton coefficient $c_{\ell,r}(m)$.

Lemma 17.1 (Polynomial top coefficients). *For every $a \geq 0$, $c_a(n)$ extends to a polynomial in $\mathbb{Q}[n]$ of degree at most $2a$. For $a \geq 1$ it is divisible by n^a . In particular,*

$$C_a(n) := \frac{c_a(n)}{n^a} \in \mathbb{Q}[n], \quad \deg C_a \leq a, \quad C_0 = 1. \quad (\text{II.32})$$

The quotient is understood polynomially at its apparent zero denominators.

Proof. Here $c_0(n) = 1$. For $a \geq 1$, coefficient extraction from (II.1) gives the constructive recurrence

$$\begin{aligned} c_a(n) - c_a(n-1) &= (2n-2)c_{a-1}(n-1) \\ &\quad + \sum_{i=0}^{n-1} \sum_{b=0}^{a-1} c_b(i) c_{a-1-b}(n-1-i), \quad c_a(0) = 0. \end{aligned} \quad (\text{II.33})$$

Assume the conclusion for smaller a . The right side is a polynomial of degree at most $2a - 1$: summing a monomial $i^v(n-1-i)^w$ has degree at most $v+w+1 \leq 2a-1$. Its discrete antiderivative with value zero at 0 is therefore a polynomial of degree at most $2a$.

For an integer $1 \leq n < a$, the first term in (II.33) vanishes. Both factors of a convolution summand can be nonzero only if $i \geq b$ and $n-1-i \geq a-1-b$, which would require $n \geq a$. Every increment at $1 \leq n < a$ is therefore zero. Together with $c_a(0) = 0$ this proves zeros at $0, \dots, a-1$, and hence divisibility by n^a . The same coefficient extraction, including the zero convention, shows that this polynomial agrees with the actual top coefficients. The asserted degree of the quotient follows. $\square$

For example $c_1(n) = (3n^2 - n)/2$. The boundary zeros in this lemma are needed for the opposite-endpoint expansion; a degree bound alone would not justify the coefficient algorithm at small integer indices.

[write] **The same top coefficients in Part III, 5 October 2026.** Part III's Theorem 24.1 bounds the same $c_a(n)$ (its zero convention for $a > n$ is this one) uniformly in both indices: $L_a(n-2a)_+^{2a} \leq c_a(n) \leq L_a(n+a)^{2a}$. Since c_a is a polynomial of degree at most $2a$ by Lemma 17.1 and $c_a(n)/n^{2a} \rightarrow L_a$, its coefficient of n^{2a} is Part III's L_a . Both Parts give $c_1(n) = (3n^2 - n)/2$.

17.3 The opposite endpoint to every fixed order

For a monomial k^d and $0 \leq r \leq d$,

$$\frac{\Delta^r k^d(m-r)}{r!} = \binom{d}{r} m^{d-r} W_{d,r}(m^{-1}), \quad W_{d,r}(x) = \sum_{q \geq 0} \binom{d-r}{q} \mathbb{E}(-r+U)^q x^q. \quad (\text{II.34})$$

At every fixed order its coefficients are polynomial in d, r . For valid integer indices it is an exact finite sum, with a remainder bound of the form (II.29). Outside that range below, only its polynomial continuation multiplied by a vanishing falling factorial is used.

Put $\varepsilon = N^{-1}$ and define formal series

$$F_\ell(\varepsilon) = \prod_{i=0}^{\ell-1} (1 - (i+1/2)\varepsilon)^{-1}, \quad G_r(x) = \prod_{i=0}^{r-1} (1 - ix). \quad (\text{II.35})$$

Each fixed coefficient is polynomial in the indices, by logarithmic expansion and Faulhaber sums. Their fixed-order remainders, when the indices are logarithmic in N , are a fixed polynomial in $1 + \ell$ times the requested power of N^{-1} . Define

$$\beta_{j,r} = s^{2j} \binom{j}{r} s^r (1-s)^{j-r}, \quad (\text{II.36})$$

$$Z_{j,r}(\varepsilon) = \sum_{a \geq 0} (\mu\varepsilon)^a C_a(j) (j-r)^a W_{j-a,r}(\mu\varepsilon). \quad (\text{II.37})$$

Only finitely many a contribute at a prescribed order. The identity

$$c_a(j) \binom{j-a}{r} = C_a(j) (j-r)^a \binom{j}{r}$$

justifies (II.37) when $j \geq a$ and $r \leq j-a$. For $j < a$ or $r > j-a$, its right side is zero at the integer indices used in the sum, so the boundary convention is exact.

The formal expansion of the opposite endpoint is

$$B(\varepsilon, s) = \frac{\varepsilon}{2} \sum_{j \geq 1} \sum_{r=0}^j \beta_{j,r} F_j(\varepsilon) G_r(\mu\varepsilon) V_{j,r}(\mu\varepsilon) Z_{j,r}(\varepsilon) = \sum_{q \geq 1} B_q(s) \varepsilon^q. \quad (\text{II.38})$$

Every B_q is rational in s . At fixed order each summand is a polynomial in j, r over $\mathbb{Q}(s)$, and its moments follow by Euler differentiation of

$$\sum_{j \geq 1, r} \beta_{j,r} x^j y^r = \frac{1}{1 - s^2 x(1 - s + sy)} - 1. \quad (\text{II.39})$$

All moments converge locally uniformly for $0 < s < 1$. In particular $B_1 = s^2/(2t)$.

We now justify the expansion with a genuine uniform remainder. For $j \leq A \log N$, $0 \leq d \leq j$, and sufficiently large N , the nonnegative monomial Newton coefficient is at most $\binom{d}{r} m^{d-r}$, since $0 \leq m - r + U \leq m$ in the integral formula. Lemma 15.1 and the factorial ratios then give

$$T_{j,d} := \frac{\mathcal{L}_m(k^d z_{N-j})}{Q_{N,m}} \leq C(2N)^{-j} m^d t^j (1 + h^{-1})^d = C s^{2j} M^{d-j}, \quad (\text{II.40})$$

$$M = m(1 + h^{-1}) \geq cN.$$

Coefficientwise positivity implies

$$0 \leq c_a(j) \leq 2j \gamma_j e_a(1, 3, \dots, 2j-1) \leq 2j \gamma_j \frac{j^{2a}}{a!}.$$

Hence losses $a \geq H$ contribute at most

$$\sum_{a=H}^j c_a(j) T_{j,j-a} \leq C s^{2j} j \frac{(C j^2/N)^H}{H!} \exp(C j^2/N) = O(s^{2j} (1+j)^{2H+1} N^{-H}). \quad (\text{II.41})$$

For the finitely many retained degree losses, apply Lemma 15.1, (II.34), and the finite-product expansions. Every error is bounded by the requisite power of N^{-1} times a fixed polynomial in j, r and the positive weight $\beta_{j,r}$. All polynomial moments of s^{2j} are uniformly summable on a compact s interval, so summation loses no logarithm. The extra factor $1/(2N)$ in (II.38) is retained. Lemma 16.1 disposes of $j > A \log N$. Thus for every fixed K ,

$$B_{N,m} = \sum_{q=1}^{K-1} B_q(s) N^{-q} + O_{J,K}(N^{-K}) \quad (\text{II.42})$$

on every compact interior ratio range.

17.4 The principal endpoint

Combining (II.28), (II.26), (II.35), and (II.15) gives, uniformly for $\ell \leq A \log N$,

$$a_{N,m,\ell,r} = a_s(\ell, r) \left(\sum_{q=0}^{K-1} A_q(\ell, r; s) N^{-q} + O(N^{-K} (1+\ell)^{M_K}) \right), \quad (\text{II.43})$$

$$a_s(\ell, r) = \gamma_\ell s^{2\ell} \binom{\ell}{r} s^r (1-s)^{\ell-r}, \quad (\text{II.44})$$

$$A_{\ell,r}(\varepsilon, s) = (1 - \ell\varepsilon) F_\ell(\varepsilon) G_r(\mu\varepsilon) U_{\ell,r}(\mu\varepsilon) V_{\ell,r}(\mu\varepsilon) = \sum_{q \geq 0} A_q(\ell, r; s) \varepsilon^q. \quad (\text{II.45})$$

All A_q are in $\mathbb{Q}(s)[\ell, r]$ and $A_0 = 1$. The normalization includes

$$A_{0,0}(\varepsilon, s) = 1, \quad A_q(0, 0; s) = 0 \quad (q \geq 1). \quad (\text{II.46})$$

The remainder in (II.43) is relative to the positive a_s . Its polynomially weighted sums are uniformly finite, since $\sum_r a_s(\ell, r) = \gamma_\ell s^{2\ell}$.

18 The exact coefficient algorithm and rationality

Let $H(\varepsilon, \alpha) = \sum_{q \geq 0} R_q(\alpha) \varepsilon^q$. Insert the endpoint series into the formal equation

$$H(\varepsilon, \alpha) = 1 - B(\varepsilon, \alpha) - \sum_{\ell \geq 1, r} a_s(\ell, r) A_{\ell, r}(\varepsilon, s) H\left(\frac{\varepsilon}{1 - \ell \varepsilon}, \frac{\alpha - r \varepsilon}{1 - \ell \varepsilon}\right). \quad (\text{II.47})$$

Every coefficient uses finitely many polynomial moments. The leading mass is

$$q_0(s) = \sum_{\ell \geq 1, r} a_s(\ell, r) = t^{-1/2} - 1. \quad (\text{II.48})$$

Thus $(1 + q_0)R_0 = 1$, giving $R_0 = \sqrt{t}$. At order $q \geq 1$, the only new unknown on the right is $-q_0 R_q$. If P_q^{known} denotes the known order- q principal sum after omitting that term, then

$$R_q = -\sqrt{t}(B_q + P_q^{\text{known}}). \quad (\text{II.49})$$

All other terms use only $R_0, \dots, R_{q-1}$ and finitely many α derivatives. The conversion is

$$\partial_\alpha = \frac{(1 + s)^2}{2s(s + 2)} \partial_s. \quad (\text{II.50})$$

This constructs unique analytic coefficient functions on the whole interior: all series moments converge locally uniformly and all necessary denominators $B, t, \alpha, 1 + q_0$ are positive there.

Proposition 18.1 (The zero index removes the radical obstruction). *Every R_q constructed by (II.49) belongs to $\sqrt{t}\mathbb{Q}(s)$.*

Proof. Suppose this holds for smaller indices. Every α derivative of $\sqrt{t}$ times a rational function has the same form. Divide the known principal residual summand by $\sqrt{t}$; it becomes a polynomial $P_q(\ell, r; s) \in \mathbb{Q}(s)[\ell, r]$. At $\ell = r = 0$ there is no kernel correction by (II.46), no argument shift, and no denominator shift. Because the unknown R_q contribution has been omitted, we have

$$P_q(0, 0; s) = 0. \quad (\text{II.51})$$

All moments including $\ell = 0$ are Euler derivatives of

$$\mathcal{G}(x, y) = (1 - s^2 x(1 - s + sy))^{-1/2}.$$

Evaluated at $x = y = 1$, each derivative is $t^{-1/2}$ times a rational function of s . Equation (II.51) means that removing $\ell = 0$ changes nothing. The principal residual, including its factor $\sqrt{t}$, is therefore rational in s . The opposite residual B_q is rational by (II.39). Equation (II.49) proves the result. Without (II.51), the removal of the zero index would introduce an extra radical term, so this cancellation is essential. $\square$

For clarity, the complete finite algorithm at a requested order is:

1. Form the necessary cumulants and Gaussian coefficients using (II.21), (II.24), and (II.25); then divide the two truncated series in (II.26).
2. Generate $U_{\ell,r}$ by Newton identities and uniform-sum moments, and generate F_ℓ, G_r by their finite-product logarithms.
3. Compute the finitely many C_a needed by the discrete antiderivative recurrence (II.33), and form (II.37).
4. Compute the required principal and opposite moments by Euler derivatives of $\mathcal{G}$ and (II.39).
5. Extract successive coefficients of (II.47), dividing only by $1 + q_0 = t^{-1/2}$ as in (II.49).

All these operations are exact rational-function algebra and finite series operations. The coefficients are regular in the full interior; any apparent rational pole there is removable by the analytic construction.

19 The uniform remainder on the full interior

The principal mass q_0 need not be less than one. We instead use an exact inverse for a frozen convolution, with an explicit correction for neighboring saddle changes.

19.1 The frozen inverse and its moments

For a fixed $s \in (0, 1)$ define operators on bounded functions on $\mathbb{Z}^2$:

$$(\mathcal{A}_s E)(N, m) = \sum_{\ell \geq 1} \sum_{r=0}^{\ell} a_s(\ell, r) E(N - \ell, m - r), \quad (\text{II.52})$$

$$(\mathcal{B}_s E)(N, m) = \sum_{\ell \geq 1} \sum_{r=0}^{\ell} b_s(\ell, r) E(N - \ell, m - r), \quad b_s(\ell, r) = \frac{a_s(\ell, r)}{2\ell - 1}. \quad (\text{II.53})$$

These b_s are inverse-kernel weights, distinct from the diagonal coefficients b_j . Their two-variable symbols satisfy

$$1 + A_s(u, v) = [1 - s^2 u(1 - s + sv)]^{-1/2}, \quad (\text{II.54})$$

$$B_s(u, v) = 1 - [1 - s^2 u(1 - s + sv)]^{1/2}, \quad (\text{II.55})$$

$$(I - \mathcal{B}_s)(I + \mathcal{A}_s) = I. \quad (\text{II.56})$$

The orientation and signs follow by multiplication of the displayed symbols; shifts compose as $T_\beta T_\eta = T_{\beta+\eta}$. Both kernels are nonnegative, and

$$\|\mathcal{A}_s\|_1 = t^{-1/2} - 1, \quad \|\mathcal{B}_s\|_1 = 1 - \sqrt{t} < 1. \quad (\text{II.57})$$

For bounded E , the double sum in the product is dominated by $\|E\|_\infty \|\mathcal{A}_s\|_1 \|\mathcal{B}_s\|_1$. Consequently (II.56) is an absolutely convergent identity on the entire lattice, not merely a formal one. Its inverse norm is at most $1 + \|\mathcal{B}_s\|_1 < 2$.

On any compact $S \subset (0, 1)$ every polynomial moment of both kernels is uniformly finite. Their first s derivatives also have exponentially decreasing, polynomially weighted tails. Indeed

$$\partial_s \log a_s(\ell, r) = \partial_s \log b_s(\ell, r) = \frac{2\ell + r}{s} - \frac{\ell - r}{1 - s}, \quad (\text{II.58})$$

so $|\partial_s a_s| \leq C_S \ell a_s$ and similarly for b_s . Summing in r leaves $\gamma_\ell s^{2\ell}$ or $\gamma_\ell s^{2\ell}/(2\ell - 1)$, both geometrically decreasing uniformly on S . In particular,

$$\|\mathcal{A}_s - \mathcal{A}_{s'}\|_1 \leq C_S |s - s'| \quad (\text{II.59})$$

when the intervening interval lies in S .

19.2 A bounded extension and its local residual

Fix K . Choose a compact interval J_* strictly inside $(0, 1)$, containing the desired ratio range and all the nested intervals to be used below, with additional margin. Put

$$H_K(N, m) = \sum_{j=0}^{K-1} R_j(m/N) N^{-j}.$$

Extend the finitely many coefficient functions smoothly and boundedly to $[0, 1]$, agreeing with their true values on a neighborhood of J_* . At valid states set $E_{N,m} = R_{N,m} - H_K(N, m)$; at every other lattice state, including $N \leq 0$, set $E = 0$. Equation (II.13) shows that this is a bounded function on $\mathbb{Z}^2$.

The local expansions, the coefficient cancellations in (II.47), and Lemma 16.1 prove the uniform residual

$$E(x) + (\mathcal{K}_x E)(x) = \rho(x), \quad |\rho(x)| \leq C N^{-K}, \quad x = (N, m), \quad m/N \in J_*. \quad (\text{II.60})$$

Here $\mathcal{K}_x$ is the finite exact kernel (II.18). To justify this subtraction independently of any remainder theorem, use Taylor expansion only for jumps at most $A \log N$, whose ratios remain within the extra compact margin. All coefficient-function derivatives are uniformly bounded there. Their remainders are a fixed polynomial in the jump times N^{-K} , summable against a_s . The long exact jumps act on bounded extensions and are $O(N^{-K})$ by (II.27); the formal tails have the same property by geometric decay. Equation (II.42) supplies the opposite residual. Inadmissible zero-kernel terms are omitted before evaluating H_K . No equation outside the local ratio domain is asserted.

Let $L_N = \lfloor A \log N \rfloor$, with A sufficiently large for all fixed-order tail bounds. From (II.43),

$$|a_{N,m,\ell,r} - a_{s(x)}(\ell, r)| \leq C N^{-1} (1 + \ell)^D a_{s(x)}(\ell, r) \quad (\ell \leq L_N). \quad (\text{II.61})$$

The exact and frozen tails acting on bounded E are $O(N^{-K})$. Writing $\mathcal{D}_x$ for $\mathcal{A}_{s(x)} - \mathcal{K}_x$ restricted to these short shifts, we obtain

$$((I + \mathcal{A}_{s(x)})E)(x) = \rho(x) + (\mathcal{D}_x E)(x) + O(N^{-K}). \quad (\text{II.62})$$

The sign of this perturbation is fixed by adding $\mathcal{A}_{s(x)} - \mathcal{K}_x$ to (II.60).

19.3 A one power improvement

Lemma 19.1. *Let $J \subset \text{int } J' \subset (0, 1)$ be compact intervals, with a further margin for the residual estimates. If, for an integer $p \geq 0$, the same error $E = R - H_K$ satisfies*

$$|E_{N,m}| \leq C_p N^{-p} \quad (m/N \in J'),$$

then it satisfies

$$|E_{N,m}| \leq C'_p N^{-\min(K,p+1)} \quad (m/N \in J). \quad (\text{II.63})$$

Proof. Fix $x = (N, m)$ in the smaller range and freeze $s_0 = s(x)$ for the entire following convolution. Define, at *every* lattice state y ,

$$F(y) = ((I + \mathcal{A}_{s_0})E)(y).$$

This function is globally bounded, uniformly for x in the fixed compact range. The exact identity (II.56) gives

$$E(x) = F(x) - \sum_{b \geq 1} \sum_{c=0}^b b_{s_0}(b, c) F(x - (b, c)). \quad (\text{II.64})$$

The inverse tail $b > L_N$ is $O(N^{-K})$ against this bounded function. For a retained shift $\beta = (b, c)$, including $(0, 0)$, set $y = x - \beta$ and $s_y = s(y)$. Its size is $N_y = N - O(\log N)$ and

$$|\alpha_y - \alpha_x| = \frac{|\alpha_x b - c|}{N - b} \leq C \frac{b}{N}, \quad |s_y - s_0| \leq C \frac{b}{N}. \quad (\text{II.65})$$

For sufficiently large N , y and every state reached by a second short shift remain in J' and have size comparable to N .

The natural neighboring equation has parameter s_y , not s_0 . Retain the difference explicitly:

$$\begin{aligned} F(y) &= ((I + \mathcal{A}_{s_y})E)(y) + ((\mathcal{A}_{s_0} - \mathcal{A}_{s_y})E)(y) \\ &= O(N^{-K}) + (\mathcal{D}_y E)(y) + ((\mathcal{A}_{s_0} - \mathcal{A}_{s_y})E)(y). \end{aligned} \quad (\text{II.66})$$

The first short perturbation is $O(N^{-p-1})$ by (II.61), the induction hypothesis on all short predecessors, and the finite weighted moments of a_{s_y} .

For the parameter-difference term, first discard shifts $\ell > L_N$ in both frozen kernels against globally bounded E , at cost $O(N^{-K})$. On the retained shifts the induction hypothesis applies. The mean value theorem, (II.58), and (II.65) give

$$|((\mathcal{A}_{s_0} - \mathcal{A}_{s_y})E)(y)| \leq C N^{-p} |s_0 - s_y| + O(N^{-K}) \leq C b N^{-p-1} + O(N^{-K}). \quad (\text{II.67})$$

The derivative norm is uniform on the full interval between the two saddles. There is no logarithmic loss in this estimate. Therefore each retained $F(y)$ is bounded by

$$C\{N^{-K} + (1 + b)N^{-p-1}\}.$$

Insert this in (II.64). The mass and first b moment of $\mathcal{B}_{s_0}$ are uniformly finite, while the discarded inverse tail is $O(N^{-K})$. This gives $|E(x)| \leq C(N^{-K} + N^{-p-1})$, proving the lemma. $\square$

The proof applies a translation-invariant inverse only after freezing the parameter at the base state. It neither substitutes the wrong saddle in a neighboring equation nor takes absolute values of an unproved signed cancellation.

19.4 Finite nested compact bootstrap

Choose $K + 1$ compact intervals

$$J_0 \subset \text{int } J_1 \subset \cdots \subset \text{int } J_K \subset (0, 1),$$

with the desired compact ratio set inside J_0 and the earlier residual margin outside J_K . Use the same H_K and the same bounded extension throughout. Initially $E = O(1)$ on J_K . Apply Lemma 19.1 to obtain $O(N^{-1})$ on J_{K-1} , then $O(N^{-2})$ on J_{K-2} , and so on, ending with $E = O(N^{-K})$ on J_0 . This proves Theorem 13.1. There are only K steps, independent of N ; no scaled error bound has been assumed in advance.

Finally, uniqueness as an asymptotic coefficient family follows by comparing two continuous families successively along integer pairs whose ratio tends to any prescribed interior value. At the first potentially different order, multiply by the corresponding power of N and pass to the limit on a compact neighborhood. The two coefficient values agree. This proves uniqueness on the whole interior, with no appeal to a contraction subinterval.

[write] Part I's question answered, 5 October 2026. This section answers Part I's question "Beyond the contraction interval" (Section 12): the limit and every fixed order hold on the whole interior, without $q_0(s) < 1$.

20 Diagonal normalization and the second correction

On the diagonal $N = 2n$, $m = n$, the saddle is $s_0 = (1 + \sqrt{17})/8$. From (II.15) and Theorem 13.1,

$$a_n = 4n \frac{(4n-1)!!}{n!} \frac{f(t)h^n t^{-2n}}{\sqrt{2\pi n B}} \sqrt{t} \left(\sum_{j \geq 0} D_j(0, 0; s_0) n^{-j} \right) \left(\sum_{j \geq 0} \frac{r_j(s_0)}{(2n)^j} \right),$$

where each series is truncated at the desired fixed order with its proved remainder. The factorial normalization is

$$\frac{(4n-1)!!}{(n!)^2} = \frac{16^n}{\pi \sqrt{2} n} \mathcal{S}(n^{-1}), \quad \mathcal{S}(x) = 1 - \frac{3}{16}x + \frac{9}{512}x^2 + \cdots. \quad (\text{II.68})$$

The logarithmic series of $\mathcal{S}$ has coefficient

$$\frac{B_{2v}}{2v(2v-1)} \left(4^{-(2v-1)} - 2^{-(2v-1)} - 2 \right) \quad (\text{II.69})$$

at x^{2v-1} and zero at positive even powers, where B_{2v} denotes a Bernoulli number. Consequently

$$\sum_{j \geq 0} b_j x^j = \mathcal{S}(x) \left(\sum_{j \geq 0} D_j(0, 0; s_0) x^j \right) \left(\sum_{j \geq 0} r_j(s_0) (x/2)^j \right). \quad (\text{II.70})$$

All its coefficients are in $\mathbb{Q}(s_0) = \mathbb{Q}(\sqrt{17})$. The leading exponential and amplitude simplify to

$$d = \frac{16h}{t^2}, \quad C = \frac{2\sqrt{t} f(t)}{\pi^{3/2} \sqrt{B}},$$

at $s = s_0$, which are precisely (II.9). Finite-order products of the established remainders prove Corollary 13.2.

For explicit second-order reproduction the algorithm yields

$$r_1(s) = \frac{s^4 + 6s^3 + 7s^2 + 3s + 4}{4(s-1)(s+1)(s+2)^2}, \quad (\text{II.71})$$

$$D_1(0, 0; s) = \frac{s^6 - 7s^5 - 5s^4 + 32s^3 + 32s^2 - 9s - 8}{12(s-1)(s+1)(s+2)^3}, \quad (\text{II.72})$$

$$B_2(s) = \frac{s^4 + 9s^3 + 13s^2 + 3s + 4}{4t^2(s+2)^2}. \quad (\text{II.73})$$

To keep the higher numerators legible, write

$$P_2(s) = s^{10} + 14s^9 + 22s^8 - 218s^7 - 1295s^6 - 3068s^5 - 3675s^4 - 2402s^3 - 1160s^2 - 500s - 4, \quad (\text{II.74})$$

$$T_2(s) = s^{12} - 14s^{11} + 759s^{10} + 1574s^9 - 1079s^8 - 4386s^7 - 1346s^6 + 4410s^5 + 4128s^4 - 368s^3 - 1871s^2 - 576s + 64. \quad (\text{II.75})$$

Then

$$r_2(s) = \frac{P_2(s)}{32s t^2(s+2)^5}, \quad D_2(0, 0; s) = \frac{T_2(s)}{288t^2(s+2)^6}. \quad (\text{II.76})$$

These formulas can be checked directly from the finite algorithm without inserting the proposed answers. One convenient Gaussian implementation uses $\exp(\sum_{j \geq 1} F_j z^j) = \sum_{n \geq 0} E_n z^n$ with

$$F_j = \frac{g_j - r\kappa_j + \ell \mathbf{1}_{j=1}}{j!} (iu)^j + \frac{\kappa_{j+2}}{(j+2)!} (iu)^{j+2}, \quad E_0 = 1, \quad E_n = \frac{1}{n} \sum_{j=1}^n j F_j E_{n-j}.$$

Apply (II.25) to E_2, E_4 and then divide by the unshifted series. This independently generates D_1, D_2 and the ratio coefficients needed in (II.45).

For a direct check of the residual bookkeeping, set $\delta = \alpha\ell - r$, $g = (\partial_\alpha \sqrt{t})/\sqrt{t}$, and

$$\mathcal{M}(P) = \sqrt{t} \sum_{\ell \geq 1, r} a_s(\ell, r) P(\ell, r).$$

The known shifted terms, divided by the current R_0 , are

$$H_1 = r_1 + g\delta, \\ H_2 = \ell r_1 + \delta(\partial_\alpha r_1 + g r_1) + \ell g\delta + \frac{\delta^2}{2}(\partial_\alpha g + g^2).$$

Thus the first two residual equations are

$$r_1 = -\mathcal{M}(A_1 + g\delta) - B_1, \quad r_2 = -\mathcal{M}(H_2 + A_1 H_1 + A_2) - B_2. \quad (\text{II.77})$$

This retains the derivatives of both R_0 and r_1 , the change in $1/(N - \ell)$, the cross term, and the opposite endpoint.

Finally, the second diagonal coefficient is

$$b_2 = \left[\frac{9}{512} + D_2 + \frac{r_2}{4} - \frac{3}{16} D_1 + \left(-\frac{3}{16} + D_1 \right) \frac{r_1}{2} \right]_{s=s_0}, \quad (\text{II.78})$$

which simplifies to (II.11). Its decimal value is $-0.583039199664459265278390709755\dots$. The saddle coefficients are in powers of m^{-1} , whereas r_j are in powers of N^{-1} ; the factors $1/2$ and $1/4$ on the diagonal are therefore necessary. The companion regenerates these identities exactly.

[write] Checks, 5 October 2026. b_1 is Part I's (84); r_1 is Part I's (58). At intake, exact values of a_n gave $n^3(a_n/(Cd^n n!/\sqrt{n}) - 1 - b_1/n - b_2/n^2) = -3.69, -2.96, -2.80, -2.73$ at $n = 25, 50, 75, 100$: bounded, as (II.8) with $K = 3$ requires. These numbers suggest, without certifying, a value of b_3 near -2.6 , which the article does not compute (Section 35, item 5).

21 All orders of the integer safe inverse

Set

$$c = \log(C\sqrt{2\pi}), \quad \Phi(x) = x(\log(dx) - 1) + c. \quad (\text{II.79})$$

Define λ_j by the formal identity

$$\sum_{j \geq 1} \lambda_j x^{-j} = \log \left(\sum_{j \geq 0} b_j x^{-j} \right) + \sum_{v \geq 1} \frac{B_{2v}}{2v(2v-1)} x^{-(2v-1)}. \quad (\text{II.80})$$

Each $\lambda_j \in \mathbb{Q}(\sqrt{17})$; specifically,

$$\lambda_1 = b_1 + \frac{1}{12} = -\frac{2423}{6528} - \frac{907\sqrt{17}}{36992}, \quad (\text{II.81})$$

$$\lambda_2 = b_2 - \frac{b_1^2}{2} = -\frac{4940681 + 601183\sqrt{17}}{10061824}. \quad (\text{II.82})$$

The forward theorem and Stirling's expansion imply, for every fixed K ,

$$\log a_n = \Phi(n) + \sum_{j=1}^{K-1} \lambda_j n^{-j} + O(n^{-K}). \quad (\text{II.83})$$

The $\frac{1}{2} \log n$ from $\log n!$ cancels the original $n^{-1/2}$ factor.

For large X put $L = \log X$, $T = L - c$, and

$$x_0 = \frac{T}{W(dT/e)}, \quad \Psi_K(x) = \Phi(x) + \sum_{j=1}^{K-1} \lambda_j x^{-j}, \quad (\text{II.84})$$

using the positive real Lambert W branch. Thus $\Phi(x_0) = L$ and $x_0 \sim \log X / \log \log X$.

Theorem 21.1 (All orders with the uncertainty inside the ceilings). *For every fixed $K \geq 1$, the equation $\Psi_K(x) = L$ has a unique root x_K in $[x_0/2, 2x_0]$ for sufficiently large X , and $x_K - x_0 = O((x_0 \log x_0)^{-1})$. There is a constant $M_K > 0$ such that*

$$\left[x_K - \frac{M_K}{x_0^K \log(dx_0)} \right] \leq N(X) := \min\{n : a_n \geq X\} \leq \left[x_K + \frac{M_K}{x_0^K \log(dx_0)} \right] \quad (\text{II.85})$$

for all sufficiently large X . Starting at $y_0 = x_0$, finitely many explicit Newton steps

$$y_{v+1} = y_v - \frac{\Psi_K(y_v) - L}{\Psi'_K(y_v)} \quad (\text{II.86})$$

may replace x_K in (II.85), after increasing M_K , whenever $2^{v+1} - 1 \geq K$.

Proof. On $[x_0/2, 2x_0]$, $\Psi'_K(x) \sim \log x_0$ uniformly, and its endpoints bracket L for large X . This proves existence and uniqueness there. The value $\Psi_K(x_0) - L$ is $O(x_0^{-1})$, giving the claimed root displacement by the mean value theorem.

No interpolation of the sequence is needed for integer rounding. For integers in this interval, (II.83) has absolute error at most $C_K x_0^{-K}$. Moving a real argument by $M_K x_0^{-K} / \log(dx_0)$ changes Ψ_K by more than this error when M_K is large enough. Every integer below the lower witness in (II.85) therefore has $a_n < X$, while the ceiling of the upper witness has $a_n \geq X$. The forward equivalent gives $a_{n+1}/a_n \sim dn$, so the sequence is eventually strictly increasing. All earlier integers, including the finite initial prefix, are below X when X is sufficiently large. These observations prove the bracket.

Throughout a neighborhood of x_0 , $|\Psi''_K/\Psi'_K| = O((x_0 \log x_0)^{-1})$. The initial error is $O((x_0 \log x_0)^{-1})$, and the Newton error recurrence therefore gives, for each fixed v ,

$$|y_v - x_K| = O\left((x_0 \log x_0)^{-(2^{v+1}-1)}\right). \quad (\text{II.87})$$

These iterates remain in the required neighborhood for sufficiently large X . If $2^{v+1} - 1 \geq K$, their error is $O(x_0^{-K} / \log x_0)$ and is absorbed by increasing the witness constant. For example, $v = \lceil \log_2(K+1) \rceil$ always suffices. $\square$

The constants and onset here are existence statements inherited from the asymptotic remainder, not effective numerical certificates. A real error tending to zero cannot be removed from the ceilings: arbitrarily close approach to an integer prevents an unrounded $o(1)$ formula for the integer threshold from following from these estimates.

[write] **Instances of the transseries apparatus; Part I superseded, 5 October 2026.** The inverse is the factorial core `p0:prop:factorial-core` of [15] (with $\kappa = 1$ and its d equal to $\log d - 1$), a form of the Lambert core `p0:thm:lambert-core`, followed by the separation pattern of `p0:thm:staircase`; see the front matter. λ_1 is Part I's β_1 , so $K = 1$ and $K = 2$ recover Part I's Theorem 10.1 and Corollary 10.3 (with half-width $O(x_0^{-1} / \log x_0)$ for $K = 1$, which is sharper than Part I's $o(1/\log \log X)$), and every larger K narrows the bracket further. λ_1 and λ_2 were checked to 20 digits at intake.

22 Reproducibility and scope

The accompanying archive includes this complete LaTeX source, its PDF, the unchanged Report 141 foundation, and an offline exact-arithmetic companion. The companion regenerates the second correction, polynomial boundary-zero identities, and frozen two-coordinate inverse checks from computational definitions. It distinguishes typed exact fixtures from public factual provenance. Its README specifies the runtime requirements, isolated entrypoints, closed-inventory verification, absent external output paths, semantic mutation tests, and deterministic PDF and ZIP workflows. Finite computations do not establish the asymptotic remainder; the proof is Sections 15–19.

The crucial analytic points are the shifted saddle remainder uniform for logarithmic shifts, summability of both endpoint expansions, the boundary zeros in Lemma 17.1, the zero-index cancellation (II.51), and the base-frozen identity (II.64) with the neighboring correction (II.67). Together these give every fixed order on every compact subset of the interior. They do not give convergence in the order, boundary asymptotics, growing-order uniformity, or a certified numerical threshold at a specified finite value of X .

[write] **The companion here, 5 October 2026.** Report 142’s companion is shipped as `142-interior-SOURCES.md`, `code/142-interior-*.py` and `data/142-interior-checks-fixtures.json`; its `foundation141/` copy is Part I. The report README gives the rerun route from the arrival commit.

22.1 Further questions and research

[write] **Added in the write, 5 October 2026.** Report 142 has no section of questions; its non-claims, stated after Corollary 13.2 and in Section 22, are recorded here as questions under the standing rule of 4 October 2026, and gathered with the other Parts’ in Section 35.

1. *Convergence and growing order.* The expansions (II.7) and (II.8) are fixed-order statements; their convergence or resummation, and uniformity for K growing with N , are not addressed (item 5).
2. *The two boundaries $\alpha = 0$ and $\alpha = 1$.* Not covered: the coefficient functions are regular on $(0, 1)$ but the remainder constants are uniform only on compact subsets. Part III treats $\alpha \rightarrow 1$ with $q = N - m = o(\sqrt{N})$ by another method; items 1–3.
3. *Effective constants and onsets,* and a certified threshold at a specified X (item 4).
4. *Higher coefficients in closed form.* The algorithm determines every r_j and b_j ; the companion computes them through $j = 2$ and makes no claim beyond (“no claim is made that this companion computes every r_j symbolically at arbitrary order”, delivered README).

Part III

Quantified upper edge transition of a weighted Dyck Newton triangle

Bundle Report 143, archive `Weighted_Dyck_Upper_Edge_Transition_All_Orders_and_Inverse_Source.zip`, delivered as `Report143.tex` (17 pages). Delivered title block “Quantified upper edge transition of a weighted Dyck Newton triangle / Report 143 / October 3, 2026”. Printed as delivered, with **[write]** notes. Delivered Section k is Section $k + 22$ here, its statement $k.j$ is $(k + 22).j$, and its equation (k) is (III. k). Section 35, which closes the report, is added in the write.

Reading conventions of Part III	Elsewhere in the report
$q = N - m$ the deficit; $G(x, k)$ the generating function	Part I: q_0 kernel mass, $F(x, k)$
L_a, V_a (A292186), K_q, C_q, F_q	Part II: $C_a(n)$ polynomials; Parts I, II: C amplitude, $C_{N,m}$
$h_n = n!L_n$; $h = 1/n$; $w_n, W(x)$; $r_n = w_n/w_{n-1}$	Parts I, II: $h(t) = f(t) - 1$; $w_\ell(s)$; W Lambert; r_j
λ_r Stirling coefficients; $\mu = 2q/3$; A, B binomial and Bernoulli variables	Part II: λ_j inverse coefficients; $\mu = 1/\alpha$; B the variance
$E_a(N)$; $D_\ell(h)$; $M_d(q)$; $\mathcal{G}(z)$	Part I: $E, D = t d/dt$; Part II: $\mathcal{G}(x, y)$
“Report 141”, “Report 142”, [13], [14]	Parts I and II

Abstract of this Part (as delivered). We study the upper edge $m = N - q$ of the falling-factorial triangle $P_{N,m} = [k^m]p_N(k)$ defined by the nonlinear recurrence $p_0 = 1$ and $p_N = (k + 2N - 2)p_{N-1} + \sum_{i=0}^{N-1} p_i p_{N-1-i}$. This is the weighted-Dyck triangle A258220, whose proportional diagonal $P_{2n,n}$ is A292692. A simultaneous comparison for every top-coefficient index a and size n gives exact positive finite-index sandwiches for $P_{N,N-q}$ and its hypergeometric auxiliary $Q_{N,N-q}$. Consequently the normalized connected ratio has a profile F_q uniformly for every growing deficit $q = o(\sqrt{N})$, with explicit exponential errors. The profile is a binomial smoothing of the classical four-regular-map sequence A292186. We prove $F_q = 2\sqrt{q/3}\{1 - 7/(16q) + O(q^{-2})\}$ with controlled convolution and binomial tails. The first correction is resolved by the joint bound when $q = o(N^{1/3})$. Finite rational recursions give every fixed order of the profile, with proved remainders, exact monotonicity, and a ceiling-safe threshold inverse. Classical map and generalized-Stirling asymptotics are credited separately from the nonlinear comparison. All boundary conventions and remainder ranges are explicit; finite computations are provided as reproducible checks rather than substitutes for proof.

23 The model and the contribution

A Dyck path of semilength N has up-steps $(1, 1)$ and down-steps $(1, -1)$, starts at $(0, 0)$, ends at $(2N, 0)$, and stays at nonnegative height. Give each peak at coordinates (a, b) the weight $k + a/b$ and multiply the peak weights of a path. Their sum is the polynomial $p_N(k)$ of OEIS A258219 [1]. Report 141 [13] proves the exact weighted-prefix identity that yields the recurrence

$$p_0(k) = 1, \quad p_N(k) = (k + 2N - 2)p_{N-1}(k) + \sum_{i=0}^{N-1} p_i(k)p_{N-1-i}(k) \quad (N \geq 1). \quad (\text{III.1})$$

The present article takes (III.1) as its starting definition, and proves its upper-edge results directly from it. An unchanged copy of Report 141 accompanies the source archive for the path interpretation. No analytic estimate from that report is needed in the edge proof.

[write] Reading Part III against Parts I and II, 5 October 2026. “Report 141” and [13] are Part I: the recurrence (III.1) is Part I’s (5), proved in Theorem 1.1. “Report 142” and [14] are Part II. The unchanged copies of both that accompanied the archive are byte-identical to the Parts printed here and are not shipped. Part III’s deficit q is not Part I’s kernel-mass q_0 ; see its reading conventions.

For a polynomial f , use the falling-factorial normalization

$$[k^m]f(k) = \frac{\Delta^m f(0)}{m!}, \quad k^m = k(k-1)\cdots(k-m+1), \quad k^0 = 1. \quad (\text{III.2})$$

Define

$$\begin{aligned} P_{N,m} &= [k^m]p_N(k), \\ \gamma_N &= 4^{-N} \binom{2N}{N}, \quad z_N(k) = \gamma_N \prod_{i=0}^{N-1} (k+2i+1), \\ Q_{N,m} &= [k^m]z_N(k), \quad R_{N,m} = \frac{P_{N,m}}{2NQ_{N,m}} \quad (N \geq 1). \end{aligned} \quad (\text{III.3})$$

Empty products are one. The triangle $P_{N,m}$ is A258220 [2]. Its Newton diagonal $P_{2n,n}$ is A292692 [3]; the ordinary parameter diagonal $p_n(n)$ is the different sequence A292693. Throughout this article the integer $q = N - m$ denotes the *deficit*. In particular the growing upper edge has $m/N \rightarrow 1$; it is not the fixed ratio $m/N = 1/2$ of A292692.

The principal result is an all- a, n nonlinear coefficient comparison, followed by exact P/Q sandwiches and a genuinely joint limiting theorem. It is stronger than proving a polynomial leading term separately for each fixed deficit and then taking a second limit of its coefficient: such iterated limits give no control over the error as q grows with N .

23.1 Prior work and claim boundary

The sequence that determines our edge constants is classical A292186, the rooted connected four-regular-map sequence [5, 9]. Its recurrence belongs to the Martin–Kearney self-convolutive family $S(4, -6, 1)$ with an index shift [6]; its Riccati equation, hypergeometric logarithmic derivative, and factorial leading asymptotic are prior art. We rederive the quantitative estimates needed below to make the proof self-contained and fix the normalization.

The auxiliary coefficients are classical generalized Stirling numbers. In the Hsu–Shiue convention,

$$\frac{Q_{N,m}}{\gamma_N} = S(N, m; -2, 1, 1). \quad (\text{III.4})$$

Indeed their connection identity between generalized factorials, with $z = k + 1$, becomes $\prod_{i=0}^{N-1} (k + 2i + 1) = \sum_m S(N, m; -2, 1, 1) k^m$. Hsu–Shiue’s Theorem 5 and fixed- r specialization [10, Section 5.2] cover a near-diagonal asymptotic regime with deficit $o(\sqrt{m})$. Thus the leading Q estimate by itself is not a new asymptotic phenomenon. Ordinary Stirling estimates in this range have a long history; effective error bounds and discussion of the Moser–Wyman expansions appear in Arratia–DeSalvo [11]. Neither a single factorial connection polynomial nor a fixed-parameter self-convolutive asymptotic supplies the nonlinear all- a, n comparison proved here.

A bounded primary-source check did not locate that comparison or an equivalent quantified growing-deficit theorem for (III.1). This article makes no exhaustive worldwide priority claim. Its substantive content is the explicit simultaneous comparison and its consequences; the classical sequence and the auxiliary asymptotic are identified as such.

[write] **Prior work, as checked, 5 October 2026.** OEIS A292186 names its terms “rooted unlabeled connected four-regular maps on a compact closed oriented surface” (of any genus), notes that it is Martin and Kearney’s $S(4, -6, 1)$ sequence with offset 1, and carries Peter Bala’s Riccati equation and S-fraction (August 22, 2023); see the front matter. The write read Martin–Kearney’s Section 2.4, Theorem 3, and Ciobanu–Kolpakov’s Theorem 4.1 and Examples 4.4 and 5.3 in their arXiv versions; the cited statements are where the manuscript places them (the write did not recompute their constants). It did not read Hsu and Shiue’s Theorem 5 and Section 5.2 or Arratia–DeSalvo; those attributions are the manuscript’s.

24 Exact bounds and the connected profile

Define positive rational constants L_a by

$$L_0 = 1, \quad L_a = \frac{L_{a-1}}{a} + \sum_{b+c=a-1} L_b L_c \frac{(2b)!(2c)!}{(2a)!} \quad (a \geq 1). \quad (\text{III.5})$$

Then $V_a = (2a)!L_a$ satisfies

$$V_0 = 1, \quad V_a = (4a - 2)V_{a-1} + \sum_{b+c=a-1} V_b V_c, \quad (\text{III.6})$$

so V_a begins 1, 3, 24, 297, 4896, 100278, $\dots$ and is A292186. Set

$$K_q = \sum_{a=0}^q \frac{L_a}{2^{q-a}(q-a)!}, \quad C_q = \frac{(3/2)^q}{q!}, \quad F_q = \frac{\sqrt{\pi}}{2} \frac{K_q}{C_q}. \quad (\text{III.7})$$

For example,

$$K_0 = 1, \quad K_1 = 2, \quad K_2 = \frac{15}{8}, \quad K_3 = \frac{269}{240}, \quad K_4 = \frac{6539}{13440}.$$

Theorem 24.1 (Simultaneous top-coefficient comparison). *Let $c_a(n) = [k^{n-a}]p_n(k)$ for $0 \leq a \leq n$, and set $c_a(n) = 0$ for $a > n$. For every pair of integers $a \geq 1$ and $n \geq 0$,*

$$L_a(n - 2a)_+^{2a} \leq c_a(n) \leq L_a(n + a)^{2a}, \quad x_+ = \max(x, 0). \quad (\text{III.8})$$

For $a = 0$, both bounding quantities are defined separately to be one, and $c_0(n) = 1$ exactly for every $n \geq 0$.

This is one assertion quantified over both indices. Its proof does not use a fixed- a asymptotic remainder with an unspecified dependence on a .

[write] **Relation to Part II, 5 October 2026.** Part II’s Lemma 17.1 proves that $c_a(n)$ is, for each fixed a , a polynomial in n of degree at most $2a$ divisible by n^a ; with the theorem above its leading coefficient is L_a . The two results are independent and complementary: polynomiality at fixed a (used by Part II’s opposite endpoint) and bounds uniform in a (used here).

Theorem 24.2 (Exact joint sandwiches). *For every integer $N \geq 1$ and $1 \leq q \leq \lfloor N/2 \rfloor$,*

$$K_q(N - 2q)^{2q} \leq P_{N,N-q} \leq K_q(N + q)^{2q}, \quad (\text{III.9})$$

$$\gamma_N C_q(N - 2q)^{2q} \leq Q_{N,N-q} \leq \gamma_N C_q N^{2q}. \quad (\text{III.10})$$

At $q = 0$ the separate exact statements are $P_{N,N} = 1$ and $Q_{N,N} = \gamma_N$. When $N = 2q \geq 2$, the lower bounds are zero with strictly positive exponent. No expression 0^0 is used.

For $q = 0$ define $A_{N,0} = B_{N,0} = 1$. For $q \geq 1$ put

$$A_{N,q} = \left(1 - \frac{2q}{N}\right)^{2q}, \quad B_{N,q} = \left(1 + \frac{q}{N}\right)^{2q}, \quad d_N = \frac{1}{\sqrt{\pi N} \gamma_N}. \quad (\text{III.11})$$

Corollary 24.3 (Explicit relative errors). *For $0 \leq q < N/2$,*

$$d_N A_{N,q} \leq \frac{\sqrt{N} R_{N,N-q}}{F_q} \leq d_N \frac{B_{N,q}}{A_{N,q}}. \quad (\text{III.12})$$

For $0 \leq q \leq N/4$ this implies

$$\exp\left(-\frac{8q^2}{N}\right) \leq \frac{\sqrt{N} R_{N,N-q}}{F_q} \leq \exp\left(\frac{10q^2 + 1/8}{N}\right). \quad (\text{III.13})$$

Consequently, simultaneously along every integer sequence $q = q(N) \geq 0$ with $q = o(\sqrt{N})$,

$$P_{N,N-q} = K_q N^{2q} \{1 + O(q^2/N)\}, \quad (\text{III.14})$$

$$Q_{N,N-q} = \gamma_N C_q N^{2q} \{1 + O(q^2/N)\}, \quad (\text{III.15})$$

$$\sqrt{N} R_{N,N-q} = F_q \{1 + O((q+1)^2/N)\}. \quad (\text{III.16})$$

The first two statements are exact at $q = 0$, interpreting their leading scales as one and γ_N . The implied constants can be chosen absolute on, for example, $(q+1)^2/N \leq 1$. More precisely the absolute relative error in (III.16) is at most $\exp((10q^2 + 1/8)/N) - 1$ whenever $q \leq N/4$.

Theorem 24.4 (First profile correction and overlap). *As $q \rightarrow \infty$,*

$$F_q = 2\sqrt{q/3} \left\{1 - \frac{7}{16q} + O(q^{-2})\right\}. \quad (\text{III.17})$$

Hence, jointly for $q \rightarrow \infty$ and $q = o(\sqrt{N})$,

$$R_{N,N-q} = 2\sqrt{\frac{q}{3N}} \left\{1 - \frac{7}{16q} + O\left(q^{-2} + \frac{(q+1)^2}{N}\right)\right\}. \quad (\text{III.18})$$

The displayed correction is resolved as a genuine first correction by this estimate in the narrower range $q = o(N^{1/3})$, where the remainder is $o(q^{-1})$.

24.1 Why this is a connected ratio

Write $G(x, k) = \sum_{N \geq 0} p_N(k) x^N$ and $Z(x, k) = \sum_{N \geq 0} z_N(k) x^N$, both as formal series. Recurrence (III.1) is equivalent to

$$G - 1 = kxG + xG^2 + 2x^2G_x. \quad (\text{III.19})$$

The product definition of z_N gives $2Nz_N = (2N - 1)(k + 2N - 1)z_{N-1}$, and therefore

$$2x^2Z_{xx} + ((k + 4)x - 1)Z_x + \frac{k + 1}{2}Z = 0.$$

Substitution of $1 + 2xZ_x/Z$ in (III.19), followed by the unique coefficientwise solution with constant term one, proves

$$G = 1 + 2x \frac{Z_x}{Z}, \quad 2Nz_N = \sum_{j=1}^N p_j z_{N-j}. \quad (\text{III.20})$$

Here $Z(0, k) = 1$ makes formal division legitimate. The generally factorially divergent series are never evaluated at nonzero x . All ordinary coefficients of p_N, z_N are nonnegative. Isolating the $j = N$ term yields $0 \preceq p_N \preceq 2Nz_N$ coefficientwise. Since $[k^m]k^j = \begin{Bmatrix} j \\ m \end{Bmatrix} \geq 0$, Newton extraction preserves this order. Thus $0 < R_{N,m} \leq 1$ for $0 \leq m \leq N, N \geq 1$. The logarithmic derivative explains the connected normalization used here.

[write] Restated from Part I, 5 October 2026. This subsection re-derives, to keep Report 143 self-contained, the equivalence of the recurrence with the Riccati equation (the “Equivalently” clause of Part I’s Theorem 1.1; here G is Part I’s F , and p_N is *defined* by (III.1), so the path identity and the continued fraction are not re-proved), Lemma 3.1, and the bound (20), here for $0 \leq m \leq N$ and with strict lower inequality. The strict inequality holds for $1 \leq m \leq N$ because p_N is monic of degree N with nonnegative coefficients, so $P_{N,m} \geq \begin{Bmatrix} N \\ m \end{Bmatrix} > 0$; at $m = 0$ the Stirling number vanishes, and instead $P_{N,0} = p_N(0) > 0$, since (III.1) gives $p_N(0) \geq p_0(0)p_{N-1}(0)$ (these are the terms 1, 4, 25, 208, ... of A005411). (Corrected after the independent check of 5 October 2026, recorded at the end of Section 35. The note first read: “This subsection re-derives, to keep Report 143 self-contained, Part I’s Theorem 1.1 (here G is Part I’s F) and Lemma 3.1, and the bound (20). The strict inequality $0 < R_{N,m}$ stated here holds because p_N is monic with nonnegative coefficients and the Stirling number $\begin{Bmatrix} N \\ m \end{Bmatrix}$ is positive for $0 \leq m \leq N$.” Both sentences were wrong: the subsection does not re-derive the path identity or the continued fraction, and $\begin{Bmatrix} N \\ 0 \end{Bmatrix} = 0$ for $N \geq 1$.)

25 A nonlinear bound uniform in both indices

Extracting the coefficient of k^{n-a} from (III.1) gives, for $a \geq 1$ and $n \geq 1$,

$$c_a(n) - c_a(n-1) = 2(n-1)c_{a-1}(n-1) + \sum_{b+c=a-1} \sum_{i+j=n-1} c_b(i)c_c(j).$$

The zero convention for indices above the degree makes this identity valid also when $a > n$. Telescoping from $c_a(0) = 0$ gives

$$c_a(n) = 2 \sum_{i=0}^{n-1} i c_{a-1}(i) + \sum_{b+c=a-1} \sum_{\substack{i,j \geq 0 \\ i+j \leq n-1}} c_b(i)c_c(j). \quad (\text{III.21})$$

Empty sums are zero. All coefficient indices on the right are below a . For $a > n$, a nonzero convolution term would require $a - 1 = b + c \leq i + j \leq n - 1$, which is impossible. The derivative terms vanish for the same degree reason. Thus the out-of-range convention is consistent with (III.21), rather than an extra assumption.

Proof of Theorem 24.1. Induct on a , proving the bounds for *every* $n \geq 0$ at each stage. The base $a = 0$ is the exact monic coefficient $c_0(n) = 1$.

For the upper bound, suppose $a \geq 1$ and $n \geq 1$. For the derivative term of (III.21), the induction hypothesis and $i \leq i + a - 1$ give

$$\begin{aligned} 2 \sum_{i=0}^{n-1} i c_{a-1}(i) &\leq 2L_{a-1} \sum_{i=0}^{n-1} (i + a - 1)^{2a-1} \\ &\leq 2L_{a-1} \int_0^n (x + a - 1)^{2a-1} dx \leq \frac{L_{a-1}}{a} (n + a)^{2a}. \end{aligned} \quad (\text{III.22})$$

At $a = 1$, the induction factor $c_0(i)$ is one, so the first step is simply $i \leq i$; the argument remains valid at $i = 0$ without any zero-th power. All integrands in this step have positive exponent. The integral bound follows from monotonicity on each unit interval.

For one convolution term, use $c_b(i) \leq L_b(i + b)^{2b}$ and $c_c(j) \leq L_c(j + c)^{2c}$, interpreting an index-zero factor as the constant one. On the unit square $[i, i + 1] \times [j, j + 1]$ the increasing integrand $(x + b)^{2b}(y + c)^{2c}$ dominates its value at (i, j) . The squares with $i + j \leq n - 1$ have disjoint interiors and lie in $x, y \geq 0, x + y \leq n + 1$. Shift $u = x + b, v = y + c$. As $b + c = a - 1$, the shifted union lies inside $u, v \geq 0, u + v \leq n + a$. Enlarging the domain, with nonnegative integrand, gives

$$\begin{aligned} \sum_{\substack{i, j \geq 0 \\ i + j \leq n - 1}} c_b(i) c_c(j) &\leq L_b L_c \int_{\substack{u, v \geq 0 \\ u + v \leq n + a}} u^{2b} v^{2c} du dv \\ &= L_b L_c \frac{(2b)!(2c)!}{(2a)!} (n + a)^{2a}. \end{aligned} \quad (\text{III.23})$$

The simplex integral is a beta integral; $2b + 2c + 2 = 2a$ is its total homogeneous degree including the two measures. Summing (III.22) and (III.23) gives the upper bound by (III.5). At $n = 0$, $c_a(0) = 0$ makes it immediate.

For the lower bound, positivity suffices if $n \leq 2a$. Assume $n > 2a$. The induction hypothesis gives

$$i c_{a-1}(i) \geq L_{a-1} (i - 2a + 2)_+^{2a-1}.$$

Where the positive part vanishes this follows from positivity; elsewhere use $i \geq i - 2a + 2$. At $a = 1$ it is exactly $i c_0(i) = i$. An increasing function sampled at the right endpoints dominates its integral over $[0, n - 1]$, so

$$2 \sum_{i=0}^{n-1} i c_{a-1}(i) \geq \frac{L_{a-1}}{a} (n - 2a + 1)^{2a} \geq \frac{L_{a-1}}{a} (n - 2a)^{2a}. \quad (\text{III.24})$$

For the convolution, integrate the lower-bound functions on the triangle $x, y \geq 0, x + y \leq n - 2$. Away from the coordinate integer lines, set $i = \lceil x \rceil, j = \lceil y \rceil$. Then $i + j < x + y + 2 \leq n$, whence the integer $i + j \leq n - 1$. Each assigned region lies in a distinct ceiling cell of area at most one. Monotonicity makes the lower-bound integrand there no greater than $c_b(i) c_c(j) / (L_b L_c)$. Integrating

over these cells therefore gives a lower bound for the discrete sum in (III.21). Restrict further to $x \geq 2b$, $y \geq 2c$ and set $u = x - 2b$, $v = y - 2c$. The resulting triangle is $u, v \geq 0$, $u + v \leq n - 2a$, since $b + c = a - 1$. Hence

$$\sum_{\substack{i, j \geq 0 \\ i+j \leq n-1}} c_b(i) c_c(j) \geq L_b L_c \frac{(2b)!(2c)!}{(2a)!} (n - 2a)^{2a}. \quad (\text{III.25})$$

For $b = 0$ or $c = 0$ its lower-bound function is identically one and its coordinate restriction is unchanged. These endpoint terms are therefore included, and measure-zero cell boundaries cause no problem. Adding (III.24) and (III.25) and using (III.5) proves the lower bound. The induction is complete. $\square$

For fixed a , division by n^{2a} shows $c_a(n)/n^{2a} \rightarrow L_a$. For orientation, $c_1(n) = n(3n - 1)/2$ and $c_2(n) = n^2(n - 1)^2$. Polynomiality for fixed a is consistent with the theorem but is not needed in its proof. In particular no growing index is substituted into an uncontrolled fixed-degree remainder.

26 Positive bounds for the Newton conversion

We record elementary estimates with constants convenient for common scales in the final sum. Their role is transparent and does not require a deep asymptotic theorem for Stirling numbers.

Lemma 26.1 (Near-diagonal Stirling bounds). *For integers $M \geq 2j$ and $j \geq 1$,*

$$\frac{M^{2j}}{2^j j!} \leq \left\{ \begin{matrix} M \\ M - j \end{matrix} \right\} \leq \frac{M^{2j}}{2^j j!}. \quad (\text{III.26})$$

In particular its lower bound is at least $(M - 2j)^{2j}/(2^j j!)$. For $j = 0$, both bounding quantities and $\left\{ \begin{matrix} M \\ M \end{matrix} \right\}$ are defined separately as one, including $M = 0$.

Proof. The lower bound counts partitions with exactly j doubleton blocks and all remaining blocks singletons; their number is $M!/((M - 2j)! 2^j j!)$. For the upper bound, associate to any set partition the graph joining the least element of each block to every other element of that block. This is a forest of exactly j edges. Its connected components recover the partition, so the map is injective into the j -subsets of the $\binom{M}{2}$ possible edges. Therefore

$$\left\{ \begin{matrix} M \\ M - j \end{matrix} \right\} \leq \binom{\binom{M}{2}}{j} \leq \frac{\binom{M}{2}^j}{j!} \leq \frac{M^{2j}}{2^j j!}.$$

Each of the $2j$ falling-factorial factors is at least $M - 2j$. At $M = 2j$ this last lower scale is zero with positive exponent. $\square$

Let $E_a(N) = e_a(1, 3, \dots, 2N - 1)$, so

$$\frac{z_N(k)}{\gamma_N} = \sum_{a=0}^N E_a(N) k^{N-a}. \quad (\text{III.27})$$

Lemma 26.2 (Odd-weight elementary symmetric bounds). *For $1 \leq a \leq N/2$,*

$$\frac{(N-2a)^{2a}}{a!} \leq E_a(N) \leq \frac{N^{2a}}{a!}. \quad (\text{III.28})$$

For $a = 0$, $E_0(N) = 1$ separately for all $N \geq 0$.

Proof. The sum of the odd weights is N^2 . Allowing repeated indices in the ordered weighted sample gives $a!E_a(N) \leq (N^2)^a$. After any r distinct choices, the remaining total weight is at least $N^2 - r(2N-1)$. Successive conditional summation of the positive weights therefore yields

$$a!E_a(N) \geq \prod_{r=0}^{a-1} \{N^2 - r(2N-1)\} \geq \{N(N-2a)\}^a \geq (N-2a)^{2a}. \quad (\text{III.29})$$

All factors are nonnegative in the stated range. The boundary $N = 2a$ is harmless, since $2a > 0$. $\square$

27 Assembly and explicit error bounds

Proof of Theorem 24.2. Since Newton extraction of a monomial is a Stirling number,

$$P_{N,N-q} = \sum_{a=0}^q c_a(N) \left\{ \begin{matrix} N-a \\ N-q \end{matrix} \right\}, \quad (\text{III.30})$$

$$\frac{Q_{N,N-q}}{\gamma_N} = \sum_{a=0}^q E_a(N) \left\{ \begin{matrix} N-a \\ N-q \end{matrix} \right\}. \quad (\text{III.31})$$

Put $j = q - a$ and $M = N - a$. If $q \leq N/2$, then $M \geq 2j$. Assume first $1 \leq q < N/2$, so $N - 2q > 0$. The lower bases in Theorem 24.1 and Lemma 26.1 are respectively $N - 2a$ and $M - 2j = N - 2q + a$, both at least $N - 2q$. Their exponents add to $2q$. Each summand of (III.30) is thus at least

$$\frac{L_a}{2^j j!} (N - 2q)^{2q}.$$

For the upper bound both bases $N + a$ and M are at most $N + q$, so the same coefficient multiplies the common upper scale $(N + q)^{2q}$. When $a = 0$ or $j = 0$ the associated factor has exact value one; this argument still applies. Summation gives (III.9) by (III.7).

Use Lemma 26.2 instead in (III.31). The lower scale is again $(N - 2q)^{2q}$ and the upper scale is N^{2q} . The coefficients sum to

$$\sum_{a=0}^q \frac{1}{a! 2^{q-a} (q-a)!} = \frac{(3/2)^q}{q!} = C_q,$$

proving (III.10). At $N = 2q \geq 2$ the lower claim follows directly from positivity: one need not multiply any zero-scale endpoint factors. The upper argument is unchanged. At $q = 0$, monicity gives the separate exact assertions. Thus no step requires a convention for 0^0 . $\square$

Proof of Corollary 24.3. The definitions give the exact normalization

$$\frac{\sqrt{N} R_{N,N-q}}{F_q} = d_N \frac{P_{N,N-q} / (K_q N^{2q})}{Q_{N,N-q} / (\gamma_N C_q N^{2q})},$$

where at $q = 0$ the powers mean the separate constant scales specified above. Theorem 24.2 immediately implies (III.12); $q < N/2$ makes its denominator positive.

Stirling's formula gives $d_N \rightarrow 1$. Directly from γ_N ,

$$\frac{d_{N+1}^2}{d_N^2} = 1 - \frac{1}{(2N+1)^2}.$$

Thus $d_N \geq 1$ and, using $-\log(1-x) \leq x/(1-x)$,

$$\begin{aligned} 0 \leq \log d_N &= -\frac{1}{2} \sum_{j=N}^{\infty} \log \left(1 - \frac{1}{(2j+1)^2} \right) \\ &\leq \frac{1}{2} \sum_{j=N}^{\infty} \frac{1}{(2j+1)^2 - 1} = \frac{1}{8N}. \end{aligned} \quad (\text{III.32})$$

The last sum telescopes because $(2j+1)^2 - 1 = 4j(j+1)$. For $q/N \leq 1/4$, the elementary logarithm inequalities imply

$$\begin{aligned} 2q \log(1 - 2q/N) &\geq -8q^2/N, \\ 2q \{\log(1 + q/N) - \log(1 - 2q/N)\} &\leq 10q^2/N. \end{aligned}$$

Together with (III.32), these prove (III.13). The larger upper exponential error dominates $1 - \exp(-8q^2/N)$, so it bounds the absolute relative error. The same logarithm bounds on (III.9) and (III.10) prove (III.14)–(III.16). If $(q+1)^2/N \leq 1$, then $N \geq 4q$ because $(q+1)^2 \geq 4q$; the constants converting small exponents to relative errors are consequently absolute. The statements along every $q = o(\sqrt{N})$ follow, including bounded and zero deficits. $\square$

28 The classical sequence with a controlled remainder

The leading factorial estimate below agrees with Martin–Kearney [6, Section 2.4, Theorem 3] and Ciobanu–Kolpakov [9, Theorem 4.1 and Examples 4.4, 5.3]. The proof here includes the remainder needed for binomial smoothing.

[write] **Prior art re-derived, 5 October 2026.** Equation (III.34) is Bala's Riccati equation for A292186 (August 22, 2023), $4x^2 A' = -1 + (1-2x)A - xA^2$, rearranged; (III.36) is the hypergeometric logarithmic derivative, the mechanism of Part I's Lemma 3.1; and the leading term of (III.42) is the classical factorial asymptotic. The manuscript presents the first correction $1 - 3/(16n)$ of Lemma 28.1 and the remainder control as its own; the write did not search the literature for them.

Define

$$w_n = \frac{4^n (1/4)_n (3/4)_n}{n!}, \quad W(x) = \sum_{n \geq 0} w_n x^n, \quad H(x) = \sum_{n \geq 0} V_n x^n, \quad (\text{III.33})$$

where $(u)_n$ is the rising factorial. These are formal series. From (III.6),

$$H - 1 = 2xH + xH^2 + 4x^2 H'. \quad (\text{III.34})$$

The coefficient ratio

$$r_n := \frac{w_n}{w_{n-1}} = 4n - 4 + \frac{3}{4n} \quad (\text{III.35})$$

gives $4x^2W'' + (8x - 1)W' + (3/4)W = 0$ coefficientwise. Substitution and uniqueness in (III.34) yield

$$H = 1 + 4x \frac{W'}{W}, \quad 4nw_n = \sum_{j=1}^n V_j w_{n-j}. \quad (\text{III.36})$$

All terms are positive; hence, for $n \geq 1$,

$$v_n := \frac{V_n}{4nw_n} \quad \text{satisfies} \quad 0 < v_n \leq 1. \quad (\text{III.37})$$

Lemma 28.1 (Convolution tail and first correction). *As $n \rightarrow \infty$,*

$$v_n = 1 - \frac{3}{16n} + O(n^{-2}). \quad (\text{III.38})$$

Proof. Put $U_{n,\ell} = w_\ell w_{n-\ell} / w_n$. Its adjacent ratio is

$$\frac{U_{n,\ell+1}}{U_{n,\ell}} = \frac{r_{\ell+1}}{r_{n-\ell}}.$$

The sequence r_j is strictly increasing, since $r_{j+1} - r_j = 4 - 3/(4j(j+1)) > 0$. Thus U decreases up to its midpoint. For $n \geq 8$ and $1 \leq \ell \leq \lfloor n/4 \rfloor$,

$$\frac{r_{\ell+1}}{r_{n-\ell}} \leq \frac{4\ell+1}{4(n-\ell-1)} \leq \frac{n+1}{3n-4} \leq \frac{2}{3}.$$

Moreover $w_1 = 3/4$, $w_2 = 105/32$, so

$$U_{n,1} = \frac{3}{16n} + O(n^{-2}), \quad U_{n,2} = O(n^{-2}).$$

The sum from $\ell = 2$ through the first quarter is bounded by a geometric series beginning at $O(n^{-2})$. The remaining part up to the midpoint has at most n terms, each bounded by an exponentially diminished quarter endpoint. By symmetry,

$$\sum_{\ell=2}^{n-2} U_{n,\ell} = O(n^{-2}). \quad (\text{III.39})$$

All sufficiently small exceptional n can be absorbed in a fixed constant.

Dividing (III.36) by $4nw_n$ and putting $\ell = n - j$ gives

$$v_n = 1 - \sum_{\ell=1}^{n-1} \left(1 - \frac{\ell}{n}\right) U_{n,\ell} v_{n-\ell}. \quad (\text{III.40})$$

First $0 < v_j \leq 1$, the two endpoint estimates, and (III.39) show $v_n = 1 + O(n^{-1})$. Retain the $\ell = 1$ term in a second use of the identity: its factor $(1 - 1/n)v_{n-1} = 1 + O(n^{-1})$. The opposite endpoint $\ell = n - 1$ has the extra factor $1/n$ and is $O(n^{-2})$; the middle is (III.39). This proves (III.38) with its stated sign and coefficient. $\square$

The gamma representation of w_n , fixed-parameter Stirling expansion, and $\Gamma(1/4)\Gamma(3/4) = \pi\sqrt{2}$ give

$$w_n = \frac{4^n n!}{\pi\sqrt{2}n} \left\{ 1 - \frac{3}{16n} + O(n^{-2}) \right\}. \quad (\text{III.41})$$

Combining (III.38) and (III.41), then dividing by $(2n)!$ and using the central binomial expansion, gives

$$V_n = \frac{2\sqrt{2}}{\pi} 4^n n! \left\{ 1 - \frac{3}{8n} + O(n^{-2}) \right\}, \quad (\text{III.42})$$

$$h_n := n! L_n = 2\sqrt{\frac{2}{\pi}} \sqrt{n} \left\{ 1 - \frac{1}{4n} + O(n^{-2}) \right\}. \quad (\text{III.43})$$

These are ordinary single-sequence asymptotics with absolute remainder constants. Absorbing finitely many small indices gives a global bound

$$0 < h_n \leq C\sqrt{n+1} \quad (n \geq 0), \quad h_0 = 1. \quad (\text{III.44})$$

29 Binomial smoothing and the profile correction

Let A have the binomial distribution with parameters q and $2/3$. An exact rewriting of (III.7) is

$$q! K_q = (3/2)^q \mathbb{E}[h_A], \quad F_q = \frac{\sqrt{\pi}}{2} \mathbb{E}[h_A]. \quad (\text{III.45})$$

This positive identity is the mechanism for passing from the classical sequence to the profile. Set $\mu = 2q/3$ and consider $\mathcal{C} = \{|A - \mu| \leq \mu/2\}$. The generating function $\mathbb{E}e^{tA} = (1/3 + (2/3)e^t)^q$ and Markov's inequality, with fixed positive and negative t , give

$$\mathbb{P}(\mathcal{C}^c) \leq 2e^{-cq} \quad (\text{III.46})$$

for an absolute $c > 0$. For example the usual fixed-fraction Chernoff bound permits a positive constant independent of q . By (III.44) and $A \leq q$, the contribution of this complement to $\mathbb{E}h_A$ is $O(\sqrt{q+1}e^{-cq})$. In particular $A = 0$ has probability 3^{-q} and value $h_0 = 1$; no negative power is evaluated there.

On $\mathcal{C}$, equation (III.43) holds with a uniform constant:

$$h_A = 2\sqrt{2/\pi} \left\{ \sqrt{A} - \frac{1}{4}A^{-1/2} + O(A^{-3/2}) \right\}. \quad (\text{III.47})$$

Taylor-expand $\sqrt{A}$ about μ through the cubic term. The fourth-derivative remainder on $\mathcal{C}$ has absolute value at most $C\mu^{-7/2}(A - \mu)^4$. The exact central moments are

$$\begin{aligned} \mathbb{E}(A - \mu) &= 0, & \mathbb{E}(A - \mu)^2 &= \mu/3, \\ \mathbb{E}(A - \mu)^3 &= -\mu/9, & \mathbb{E}(A - \mu)^4 &= \mu^2/3 - \mu/9. \end{aligned} \quad (\text{III.48})$$

Replacing central-event moments by full moments costs only a polynomial in q times e^{-cq} , since $|A - \mu| \leq q$. The cubic contribution and the fourth-derivative remainder both have relative order q^{-2} , whereas the quadratic contribution is $-1/(24\mu) = -1/(16q)$. Consequently

$$\mathbb{E}\sqrt{A} = \sqrt{\mu} \left\{ 1 - \frac{1}{16q} + O(q^{-2}) \right\}. \quad (\text{III.49})$$

The square root at zero is harmless in this equation. For the negative power, evaluated only at $A \geq 1$, the same central expansion gives

$$\mathbb{E}[A^{-1/2}; A \geq 1] = \mu^{-1/2} \{1 + O(q^{-1})\}. \quad (\text{III.50})$$

Outside $\mathcal{C}$ that integrand is at most one, so its tail is exponentially small. The remainder in (III.47) contributes $O(\mu^{-3/2})$ on $\mathcal{C}$; the complement for h_A is handled by the separate global bound, not by a large-index asymptotic there. Inserting these estimates into (III.45) gives

$$\begin{aligned}\mathbb{E}h_A &= 2\sqrt{2/\pi}\sqrt{\mu}\left\{1 - \frac{1}{16q} - \frac{1}{4\mu} + O(q^{-2})\right\} \\ &= 2\sqrt{2/\pi}\sqrt{2q/3}\left\{1 - \frac{7}{16q} + O(q^{-2})\right\}.\end{aligned}$$

This proves (III.17). Multiplying by the exact joint error of Corollary 24.3 proves (III.18). Finally $(q+1)^2/N = o(q^{-1})$ for growing q precisely under the sufficient scale $q^3/N \rightarrow 0$. This establishes the stated correction-resolving range and completes Theorem 24.4.

30 All fixed orders of the edge profile

The profile alone admits a complete fixed-order expansion. This section refines the one-variable analysis, while leaving the finite- N sandwich and its error unchanged.

Theorem 30.1 (Constructive fixed-order profile expansion). *For every fixed integer $K \geq 1$ there are explicitly constructible rational coefficients $f_0, \dots, f_{K-1}$, with $f_0 = 1$, such that*

$$F_q = 2\sqrt{q/3}\left\{\sum_{r=0}^{K-1} f_r q^{-r} + O_K(q^{-K})\right\} \quad (q \rightarrow \infty). \quad (\text{III.51})$$

The first coefficients are

$$\begin{aligned}f_0 &= 1, & f_1 &= -\frac{7}{16}, & f_2 &= -\frac{455}{512}, & f_3 &= -\frac{34717}{8192}, \\ f_4 &= -\frac{16688901}{524288}, & f_5 &= -\frac{2712316809}{8388608}.\end{aligned} \quad (\text{III.52})$$

Consequently, for every fixed K , jointly for $q \rightarrow \infty$ and $q = o(\sqrt{N})$,

$$R_{N,N-q} = 2\sqrt{\frac{q}{3N}}\left\{\sum_{r=0}^{K-1} f_r q^{-r} + O_K\left(q^{-K} + \frac{(q+1)^2}{N}\right)\right\}. \quad (\text{III.53})$$

For $K \geq 2$, the sufficient scale $q = o(N^{1/(K+1)})$ makes this remainder $o(q^{-(K-1)})$ and separates the last displayed correction. No convergence of the series or uniformity for growing K is asserted. Equation (III.53) is not an all-orders finite- N expansion.

30.1 Fixed-order control of the connected normalization

Write $\ell_0 = \lfloor n/2 \rfloor$. Splitting the exact positive identity (III.40) into disjoint halves gives

$$v_n + \sum_{\ell=1}^{\ell_0} \left(1 - \frac{\ell}{n}\right) U_{n,\ell} v_{n-\ell} = 1 - \sum_{j=1}^{n-\ell_0-1} \frac{V_j w_{n-j}}{4nw_n}. \quad (\text{III.54})$$

For every fixed $s \geq 1$ and $d \geq 0$, the ratio proof of Lemma 28.1 gives the stronger bound

$$\sum_{\ell=s}^{\lfloor n/2 \rfloor} (1 + \ell)^d U_{n,\ell} = O_{s,d}(n^{-s}). \quad (\text{III.55})$$

To see every part of the claim, $U_{n,s} = O_s(n^{-s})$ for fixed s . On the first quarter the adjacent ratio is at most $2/3$, so the polynomially weighted geometric sum starting at s has this order. On the remaining part up to the midpoint, monotonicity bounds each term by an exponentially small quarter-endpoint value. Any fixed polynomial in n is absorbed by that exponential. For fixed s this applies once n is sufficiently large; only the constant changes for finite exceptions. This proves (III.55). Furthermore

$$\frac{V_j w_{n-j}}{4n w_n} \leq \frac{j}{n} U_{n,j},$$

so the source tail starting at fixed s is $O_s(n^{-s-1})$. The entire source is $O(n^{-2})$, whereas the principal kernel mass in (III.54) is $O(n^{-1})$.

Put $h = 1/n$ and define rational products

$$D_\ell(h) = \prod_{i=0}^{\ell-1} \frac{1 - ih}{(1 - (i + 1/4)h)(1 - (i + 3/4)h)}. \quad (\text{III.56})$$

The exact ratio of hypergeometric coefficients is

$$\frac{w_{n-\ell}}{w_n} = 4^{-\ell} h^\ell D_\ell(h). \quad (\text{III.57})$$

A formal series $v(h) = \sum_{r \geq 0} \nu_r h^r$ is defined by

$$v(h) + \sum_{\ell \geq 1} w_\ell 4^{-\ell} h^\ell (1 - \ell h) D_\ell(h) v\left(\frac{h}{1 - \ell h}\right) + \frac{h}{4} \sum_{j \geq 1} V_j 4^{-j} h^j D_j(h) = 1. \quad (\text{III.58})$$

Every coefficient involves only finitely many terms. Set $\nu_0 = 1$. At order h^r , only $\ell \leq r$ and $j \leq r - 1$ contribute, and occurrences of ν_i in the first sum have $i \leq r - \ell < r$. Thus the finite recursion is

$$\begin{aligned} \nu_r = -[h^r] \left\{ \sum_{\ell=1}^r w_\ell 4^{-\ell} h^\ell (1 - \ell h) D_\ell(h) v_{<r} \left(\frac{h}{1 - \ell h} \right) \right. \\ \left. + \frac{h}{4} \sum_{j=1}^{r-1} V_j 4^{-j} h^j D_j(h) \right\}, \quad v_{<r}(h) = \sum_{i=0}^{r-1} \nu_i h^i. \end{aligned} \quad (\text{III.59})$$

It uses rational arithmetic and the finite exact recurrence for V_j . For example $\nu_1 = -3/16$, $\nu_2 = -183/512$, and $\nu_3 = -8313/8192$.

The formal calculation has a genuine remainder. Fix K and put $H_K(n) = \sum_{r=0}^{K-1} \nu_r n^{-r}$. Insert H_K in (III.54), retaining principal indices $\ell < K$ and source indices $j < K - 1$. These are finitely many fixed shifts; Taylor expansion of (III.57) and $H_K(n - \ell)$ has remainder $O_K(n^{-K})$, and (III.58) cancels all lower coefficients. The remaining principal tail is $O_K(n^{-K})$ by (III.55), since H_K is bounded on positive integers. For $K \geq 2$, the source tail begins at $K - 1$ and has order $O_K(n^{-K})$ from its extra $1/n$ factor. For $K = 1$, the whole source is $O(n^{-2})$, which is smaller still. Thus the residual of this approximation is $O_K(n^{-K})$. For $E_n = v_n - H_K(n)$, subtraction yields

$$|E_n| \leq C_K n^{-K} + \sum_{\ell=1}^{\ell_0} (1 - \ell/n) U_{n,\ell} |E_{n-\ell}|.$$

Each predecessor is at least $n/2$. The principal kernel mass is $O(1/n)$, so for sufficiently large n ,

$$\sum_{\ell=1}^{\ell_0} (1 - \ell/n) U_{n,\ell} \left(\frac{n}{n-\ell} \right)^K \leq \frac{1}{2}.$$

Strong induction, with M_K chosen to cover the finite initial indices and at least $2C_K$, proves $|E_n| \leq M_K n^{-K}$. We have proved

$$v_n = \sum_{r=0}^{K-1} \nu_r n^{-r} + O_K(n^{-K}). \quad (\text{III.60})$$

30.2 Gamma normalization and central-moment coefficients

Let $c = 2\sqrt{2/\pi}$. The exact gamma expression for $4nw_n n!/(2n)!$, with fixed-shift Stirling expansion [12, Eq. 5.11.8], gives

$$h_n = c\sqrt{n} \exp \left(\sum_{r \geq 1} \lambda_r n^{-r} \right) v_n \quad (\text{III.61})$$

as a fixed-order asymptotic identity, where

$$\lambda_r = \frac{(-1)^{r+1}}{r(r+1)} \{B_{r+1}(1/4) + B_{r+1}(3/4) - B_{r+1}(1/2) - B_{r+1}(1)\}. \quad (\text{III.62})$$

Here $B_j(x)$ is the Bernoulli polynomial, specified by $te^{xt}/(e^t - 1) = \sum_{j \geq 0} B_j(x)t^j/j!$. All even λ_r vanish; the first odd ones are $\lambda_1 = -1/16$, $\lambda_3 = 1/1536$, $\lambda_5 = -1/20480$. Define τ_r by the formal product

$$\sum_{r \geq 0} \tau_r h^r = \exp \left(\sum_{r \geq 1} \lambda_r h^r \right) \sum_{r \geq 0} \nu_r h^r. \quad (\text{III.63})$$

This is an explicit finite algorithm at any prescribed order: set $e_0 = 1$ and

$$ne_n = \sum_{r=1}^n r \lambda_r e_{n-r}, \quad \tau_n = \sum_{i=0}^n \nu_i e_{n-i}.$$

Combining (III.60) with the ordinary fixed-order Stirling remainder proves

$$h_n = c\sqrt{n} \left\{ \sum_{r=0}^{K-1} \tau_r n^{-r} + O_K(n^{-K}) \right\}. \quad (\text{III.64})$$

The first τ values are

$$1, \quad -\frac{1}{4}, \quad -\frac{11}{32}, \quad -\frac{127}{128}, \quad -\frac{9021}{2048}, \quad -\frac{215895}{8192}.$$

The global bound (III.44) remains available for all tail terms.

Put $p = 2/3$ and $M_d(q) = \mathbb{E}(A - pq)^d$ for $A \sim \text{Binomial}(q, p)$. These are exact rational polynomials in q , with $M_0 = 1$, $M_1 = 0$, and $\deg M_d \leq \lfloor d/2 \rfloor$ for $d \geq 2$. Indeed their exponential generating function is

$$\sum_{d \geq 0} M_d(q) \frac{z^d}{d!} = \exp\{q[\log(1 - p + pe^z) - pz]\}.$$

A finite recurrence is obtained from the Bernoulli cumulants $\kappa_1(p) = p$ and $\kappa_{j+1}(p) = p(1 - p) d\kappa_j(p)/dp$:

$$M_d(q) = \sum_{j=2}^d \binom{d-1}{j-1} q \kappa_j(2/3) M_{d-j}(q) \quad (d \geq 2). \quad (\text{III.65})$$

The absence of the first cumulant is responsible for the degree bound. The profile coefficients have the finite rational formula

$$f_r = [h^r] \sum_{j=0}^r \tau_j p^{-j} h^j \sum_{d=0}^{2(r-j)} \binom{1/2-j}{d} p^{-d} h^d M_d(1/h). \quad (\text{III.66})$$

The smallest possible degree of the j, d term is $j + \lceil d/2 \rceil$. Consequently no $d > 2(r - j)$ can contribute, and only $\tau_0, \dots, \tau_r$ and moments through $2r$ are needed. At $r = 1$, the two contributions are $\tau_1/p = -3/8$ and $-1/16$, recovering $f_1 = -7/16$. Applying the same exact recursion yields (III.52).

30.3 A smoothing remainder at every fixed order

Proof of Theorem 30.1. Fix K and use the same central event $\mathcal{C} = \{|A - \mu| \leq \mu/2\}$, $\mu = pq$, as in Section 29. The global bound for h_A and (III.46) make the noncentral contribution $O(\sqrt{q+1} e^{-cq})$; $A = 0$ is again handled by its exact value. On $\mathcal{C}$, the large- n expansion (III.64) is uniform:

$$h_A = c \sum_{j=0}^{K-1} \tau_j A^{1/2-j} + O_K(q^{1/2-K}).$$

For the j th retained term, Taylor-expand about μ through degree $2(K - j) - 1$. Its remainder is bounded by

$$C_{K,j} \mu^{1/2-j-2(K-j)} |A - \mu|^{2(K-j)}.$$

By (III.65), the even central moment of order $2(K - j)$ is $O_K(q^{K-j})$. After expectation and division by $\sqrt{\mu}$, this remainder is $O_K(q^{-K})$. Replacing central-event moments by full moments costs an exponentially small quantity times a fixed power of q , since $|A - \mu| \leq q$. The Taylor polynomials produce exactly (III.66), and any additional terms have powers at least q^{-K} . Inserting them in (III.45) proves (III.51) with a genuine remainder. The fixed- n expansion has only been used where A is a positive fixed fraction of q ; the global bound controls the entire small- A tail. Finally Corollary 24.3 gives (III.53). The condition $q^{K+1}/N \rightarrow 0$ separates the last correction as stated. $\square$

31 An increasing profile and a ceiling-safe inverse

Theorem 31.1 (Exact monotonicity and fixed-order inverse). *The sequence F_q is strictly increasing for all integers $q \geq 0$. For real Y , define*

$$\mathcal{Q}(Y) = \min\{q \geq 0 : F_q \geq Y\}.$$

It exists for every Y . For every fixed integer $M \geq 0$ there is a rational inverse polynomial

$$J_M(x) = x + \sum_{r=0}^M \xi_r x^{-r}, \quad x = 3Y^2/4, \quad (\text{III.67})$$

and constants $C_M, Y_M > 0$ such that, for $Y \geq Y_M$,

$$\left[J_M(x) - C_M x^{-M-1} \right] \leq \mathcal{Q}(Y) \leq \left[J_M(x) + C_M x^{-M-1} \right]. \quad (\text{III.68})$$

The first coefficients are

$$\xi_0 = \frac{7}{8}, \quad \xi_1 = \frac{203}{128}, \quad \xi_2 = \frac{3231}{512}. \quad (\text{III.69})$$

All higher coefficients are specified by the finite reversion below.

Proof. Multiply (III.6) by $n!/(2n)!$. Since $(4n-2)n!/(2n)! = (n-1)!/(2n-2)!$, one obtains the exact identity

$$h_n = h_{n-1} + \frac{n!}{(2n)!} \sum_{b+c=n-1} V_b V_c > h_{n-1} \quad (n \geq 1). \quad (\text{III.70})$$

Couple $A_q \sim \text{Binomial}(q, 2/3)$ to an independent Bernoulli(2/3) variable B by $A_{q+1} = A_q + B$. Strict increase of h_n gives $\mathbb{E}h_{A_{q+1}} > \mathbb{E}h_{A_q}$, and (III.45) proves strict increase of F_q . As $F_q \rightarrow \infty$, $\mathcal{Q}(Y)$ exists for every real Y ; for $Y \leq F_0 = \sqrt{\pi}/2$ it equals zero.

Square the formal profile expansion and write

$$\begin{aligned} \mathcal{G}(z) &= z \left(\sum_{r \geq 0} f_r z^{-r} \right)^2 = z + g_0 + g_1 z^{-1} + g_2 z^{-2} + \dots, \\ g_r &= \sum_{a+b=r+1} f_a f_b. \end{aligned} \quad (\text{III.71})$$

In particular $g_0 = -7/8$, $g_1 = -203/128$, and $g_2 = -7883/1024$. Define $\xi_0, \dots, \xi_M$ successively by

$$\mathcal{G} \left(x + \sum_{r=0}^M \xi_r x^{-r} \right) = x + O(x^{-M-1}) \quad (\text{III.72})$$

as a formal Laurent identity to the stated order. To compute ξ_r , substitute the preceding coefficients with ξ_r provisionally zero, and negate the coefficient of x^{-r} . The missing ξ_r enters that coefficient with multiplier one; every $g_j z^{-j}$ term with $j \geq 1$ changes only at higher order. This gives a finite and unique rational algorithm and yields (III.69).

Take $K = M + 2$ in (III.51). Squaring the estimate gives, for large integer q ,

$$\frac{3}{4} F_q^2 = \mathcal{G}_M(q) + O_M(q^{-M-1}), \quad \mathcal{G}_M(z) = z + g_0 + \sum_{r=1}^M g_r z^{-r}. \quad (\text{III.73})$$

Choose a constant that bounds this remainder. For all sufficiently large positive z , the two real envelopes $\mathcal{G}_M(z) \pm C z^{-M-1}$ have positive derivative tending to one. Their roots at level x are within $O_M(x^{-M-1})$ of $J_M(x)$, by (III.72) and the mean value theorem. The upper envelope has the smaller root y_- , and the lower envelope the larger root y_+ . At an integer $q < \lceil y_- \rceil$, one has $q < y_-$, so the upper envelope is below x and hence $F_q < Y$. At $q = \lceil y_+ \rceil$ the lower envelope is at least x , hence $F_q \geq Y$. These implications use $Y > 0$, valid in the stated large- Y range, so comparison of squares is equivalent to comparison with Y . The finitely many indices below the envelope domain also satisfy $F_q < Y$ once Y is large. Thus $\lceil y_- \rceil \leq \mathcal{Q}(Y) \leq \lceil y_+ \rceil$. Replacing the two roots by their bounds around $J_M(x)$ proves (III.68) after increasing C_M . $\square$

For $M = 0$, the continuous center is $3Y^2/4 + 7/8$ with $O(Y^{-2})$ uncertainty *inside* the two ceilings. Including one more term gives

$$\frac{3Y^2}{4} + \frac{7}{8} + \frac{203}{96Y^2}$$

with $O(Y^{-4})$ uncertainty inside the ceilings. There is no unrounded $O(Y^{-2})$ approximation to the integer-valued threshold in this statement: closeness to an integer can change a ceiling. The constants and sufficiently large onset in this inverse are existence statements, not certified numerical witnesses at a specified Y . This inverse concerns only the one-variable profile F_q ; it is not an inverse theorem for freely varying (N, q) .

[write] No Lambert function here, 5 October 2026. Unlike the inverses of Parts I and II, this one involves no Lambert W . After squaring, $\frac{3}{4}F_q^2 = q + g_0 + g_1q^{-1} + \dots$ has a linear core, the case $b = 0$ of `p0:thm:lambert-core` in [15], inverted by formal reversion; the ceiling bracket (III.68) has the separation form of `p0:thm:staircase`. The companion carries $\xi_0, \dots, \xi_7$ exactly (`data/143-edge-checks-fixtures.json`).

32 Comparison with the compact interior

Report 142 [14] gives a compact-interior expansion in the coordinates

$$\alpha = \frac{m}{N} = \frac{2s^2}{1+s}, \quad 0 < s < 1, \quad (\text{III.74})$$

with leading connected ratio $\sqrt{1-s^2}$ and first relative coefficient

$$r_1(s) = \frac{s^4 + 6s^3 + 7s^2 + 3s + 4}{4(s-1)(s+1)(s+2)^2}. \quad (\text{III.75})$$

Only these explicit functions are used in this section. An unchanged source/PDF pair of Report 142 is included for the provenance of this comparison; none of its companion code is needed here.

At $\alpha = 1 - q/N$, elementary expansion of (III.74) at $s = 1$ gives

$$1 - s = \frac{2q}{3N} + O((q/N)^2), \quad (\text{III.76})$$

$$\sqrt{1-s^2} = 2\sqrt{q/(3N)}\{1 + O(q/N)\}. \quad (\text{III.77})$$

The numerator of (III.75) equals 21 at $s = 1$, while its denominator is $72(s-1)$ to first order. Therefore

$$\frac{r_1(s)}{N} = -\frac{7}{16q} + O(N^{-1}). \quad (\text{III.78})$$

This matches the coefficient in (III.18). These are algebraic endpoint expansions of stated functions, not an application of the compact-interior theorem at a moving boundary. That theorem's remainder constants are only uniform on fixed compact subsets of $0 < \alpha < 1$ and cannot be carried to $s = 1$ by fiat. The actual error for the growing-deficit transition here comes from Theorems 24.1–24.2 and the smoothing proof.

[write] The gap between Parts II and III, 5 October 2026. The functions compared are Part II's (II.71) and the leading term of Theorem 13.1. As this section says, the matching is algebraic consistency only; the range $\sqrt{N} \lesssim q = o(N)$ is covered by neither theorem (Section 35, item 1).

33 Further questions and precise limitations

The estimates isolate several concrete questions without settling them.

1. **The scale** $q \asymp \sqrt{N}$. The exponential bounds remain informative there but their relative errors need not tend to zero. Is there a nontrivial two-parameter correction to $K_q N^{2q}$, or to the connected profile, on this scale? The present theorem neither identifies that limit nor asserts failure of a stronger approximation.
2. **Sharper nonlinear comparison.** The shifts $n - 2a$ and $n + a$ were selected for a uniform positive proof. A more precise expansion of the discrete convolution, with bounds uniform in a , could reveal a first q^2/N correction and enlarge the range in which profile corrections are separated. Fixed- a polynomial formulas alone would not justify such a conclusion. In particular, could first finite- N corrections in P and Q cancel in their connected ratio enough to improve the present $O(q^2/N)$ relative error? Such cancellation and any resulting larger range remain unproved here.
3. **A combinatorial meaning for K_q .** The exact binomial smoothing of A292186 gives a positive algebraic construction. An interpretation directly in terms of weighted paths or connected structures would clarify why the particular probability $2/3$ appears in the edge profile. No new counting interpretation is claimed here.
4. **A bridge across all moving ratios.** A uniform theorem linking this boundary range with the compact interior would require control of the remainder as the saddle approaches the boundary. Agreement of the displayed coefficients is evidence of compatibility, not a proof of that global statement.
5. **Effective onset constants.** The finite-index sandwich constants are explicit. Constants and onset indices in the asymptotic profile remainders are existence statements from the tail estimates. Computing usable, certified numerical onset bounds is a separate problem; finite tests do not provide them.

[write] **Status of these questions, 5 October 2026.** All five remain open after the merge. Item 1: Remark 35.1 records the order of magnitude $R_{N,N-q} \asymp \sqrt{(q+1)/N}$ on $q = O(\sqrt{N})$, which follows from Corollary 24.3; the limit question is unchanged. Items 1 and 4 are Section 35, items 1 and 3; item 2 is item 7; item 3 is item 6; item 5 is item 4.

34 Reproducibility and proof status

The archive contains this complete LaTeX source and PDF, the unchanged Report 141 foundation, the Report 142 comparison source/PDF pair, and an offline exact-arithmetic companion. The companion independently generates polynomials and Newton coefficients, tests the positive sandwiches and boundary cases, and regenerates exact profile and inverse coefficients. Its README specifies the checked ranges, fixture schemas, isolated entypoints, output containment, semantic mutation tests, and deterministic PDF and ZIP workflows. The arithmetic checks include $a = 0$, $a > n$, $n = 0$, $q = 0$, and $N = 2q$ as separate branches where necessary.

These computations verify finite identities and normalization. The simultaneous integral comparison, the positive Newton conversion, and the convolution and binomial tail proofs establish the

unbounded-index and limiting claims. No numerical fit, exchange of nonuniform iterated limits, or extension of a compact-interior remainder is used.

[write] **The companion here, 5 October 2026.** Report 143’s companion is shipped as `143-edge-SOU` `RCEs.md`, `code/143-edge-*.py` and `data/143-edge-checks-fixtures.json`; the copies of Reports 141 and 142 are Parts I and II. The report README gives the rerun route from the arrival commit.

35 Further questions and research (all three Parts)

[write] **Standing rule, 5 October 2026.** This section was added in the write. Under Vladimir’s standing rule of 4 October 2026 it collects every claim of the three manuscripts stated without proof and every question they leave open, with its source, the argument sketch it came with, and what is missing. Questions answered inside the merge are marked at their places (Part I’s Section 12) and listed at the end. No claim of the three manuscripts was found to be wrong.

1. **The intermediate upper range** $\sqrt{N} \lesssim q = o(N)$. (Part I, “Other moving regimes”; Part III, Section 33, items 1 and 4.) Part II covers m/N in compact subsets of $(0, 1)$; Part III covers $q = N - m = o(\sqrt{N})$. Between them the only estimates are $0 \leq R_{N,m} \leq 1$ and Part III’s exact bounds (III.12), valid for every $q < N/2$ but with relative error factors of size $\exp(\Theta(q^2/N))$; for $q \leq c\sqrt{N}$ these give the order of magnitude of Remark 35.1 below, and for $q \gg \sqrt{N}$ they do not even give the order of magnitude. *Sketch available:* at $\alpha = 1 - q/N$ the interior leading term and first correction agree with the edge profile to the displayed orders, (III.77) and (III.78). The natural conjecture is

$$R_{N,N-q} \sim 2\sqrt{q/(3N)} \quad (q \rightarrow \infty, q/N \rightarrow 0),$$

uniformly, with the corrections of both Parts. *Missing:* Part II’s remainders are uniform only on fixed compact subsets of $(0, 1)$ (its saddle variance $B = t(s+2)/(4s^4)$ tends to 0 and the kernel masses blow up as $s \rightarrow 1$), and Part III’s relative error $\exp(\Theta(q^2/N))$ is unbounded once $q \gg \sqrt{N}$. A two-parameter limit at $q \asymp \sqrt{N}$ (Part III’s item 1) is also open: Remark 35.1 gives only two-sided bounds there, not a limit, and asserts neither existence nor absence of a nontrivial correction to $K_q N^{2q}$.

2. **The lower edge** $m = o(N)$. (Part I, “Other moving regimes”.) Untouched by all three Parts; Part I notes that it approaches a different saddle limit. As $s \rightarrow 0$ the interior leading term $\sqrt{1-s^2}$ tends to 1. An uncertified observation of the write (exact rational arithmetic): at $m = 0$, $N(1 - R_{N,0}) = 0.3226, 0.2599, 0.2548, 0.2524, 0.2512$ for $N = 10, 50, 100, 200, 400$, suggesting $R_{N,0} = 1 - 1/(4N) + o(1/N)$. *Missing:* any estimate for growing $m = o(N)$, and a transition between fixed- k asymptotics (Martin–Kearney [6]) and the interior.
3. **A uniform bridge across all ratios.** (Part III, item 4.) A theorem uniform in $0 < m/N < 1$ up to both edges would need control of the interior remainders as the saddle approaches the boundary. The coefficient matching of Section 32 is consistency evidence, not a proof.
4. **Effective constants, onsets and certified thresholds.** (Part I, Section 12; Part II, Subsection 22.1; Part III, item 5.) Every asymptotic remainder constant and onset is an existence statement, so no threshold bracket is certified at a given X or Y . Part III’s finite-index sandwiches are explicit; its asymptotic remainders are not.

- 5. Convergence, resummation, growing order.** (Part II, Section 20 and Section 22; Part III, Theorem 30.1.) Whether the series in N^{-1} , n^{-1} or q^{-1} converge or are resummable, and whether the expansions hold uniformly for an order K growing with N or q , is not addressed. Uncertified indications: the dossier’s residuals at $n \leq 100$ (front matter) suggest b_3 near -2.6 ; the companion’s exact profile coefficients $f_6 = -1103681053651/268435456 \approx -4111.6$, f_7 , f_8 (in `data/143-edge-checks-fixtures.json`, computed by the algorithm of Section 30) grow roughly factorially (the ratios f_{r+1}/f_r for $r = 2, \dots, 7$ are about 4.8, 7.5, 10.2, 12.7, 15.2, 17.7), and the dossier’s floating-point q^6 residuals after f_5 , between -4455 and -4164 for $200 \leq q \leq 1200$, are consistent with that f_6 .
- 6. Combinatorial meanings.** (Part I, “Combinatorial meaning of the Newton triangle”; Part III, item 3.) A direct interpretation of $T(N, m)$, of the diagonal $T(2n, n)$ and its factor $\sqrt{t_0}$, and of K_q and the probability $2/3$ in the edge profile.
- 7. A sharper nonlinear comparison.** (Part III, item 2.) Bounds uniform in a that reveal a q^2/N correction, and possible cancellation of the first finite- N corrections of P and Q in their connected ratio, which would enlarge the range of Corollary 24.3.

Remark 35.1 ([**write**] The scale $q \asymp \sqrt{N}$: order of magnitude, 5 October 2026). There are absolute constants $0 < a_* < b_*$ such that for every $c > 0$, all $N \geq 1$ and all integers $0 \leq q \leq \min(c\sqrt{N}, N/4)$,

$$a_* e^{-8c^2} \sqrt{\frac{q+1}{N}} \leq R_{N, N-q} \leq b_* e^{10c^2+1/8} \sqrt{\frac{q+1}{N}}.$$

Proof. By (III.7) every F_q is positive, since every L_a in (III.5) is, and $F_0 = \sqrt{\pi}/2$; by Theorem 24.4 $F_q/\sqrt{q+1} \rightarrow 2/\sqrt{3}$. (Trimmed after the independent check of 5 October 2026, recorded at the end of this section; the first wording read “By Theorem 31.1 the profile F_q is strictly increasing, with $F_0 = \sqrt{\pi}/2 > 0$ ”, but monotonicity is not needed, only positivity.) A sequence of positive numbers with a positive limit is bounded above and below by positive constants, so $a_* \sqrt{q+1} \leq F_q \leq b_* \sqrt{q+1}$ for all $q \geq 0$ with absolute $0 < a_* < b_*$. For $0 \leq q \leq N/4$, (III.13) gives $e^{-8q^2/N} F_q \leq \sqrt{N} R_{N, N-q} \leq e^{(10q^2+1/8)/N} F_q$, and $q^2/N \leq c^2$, $1/(8N) \leq 1/8$. Dividing by $\sqrt{N}$ proves the claim. $\square$

The remark settles the order of magnitude of $R_{N, N-q}$ on the whole scale $q = O(\sqrt{N})$, which Part III’s item 1 calls informative but does not state; it does not identify a limit there, and it says nothing for $q \gg \sqrt{N}$, where the exponential factors are unbounded.

Answered inside the merge (dated notes at the questions): Part I’s “Higher relative orders” and “Beyond the contraction interval” (by Part II, Theorem 13.1); Part I’s “Other moving regimes” for the upper edge with $q = o(\sqrt{N})$ (by Part III, Corollary 24.3); Part I’s statement that no coefficient beyond b_1 is asserted (by Part II, Corollary 13.2). Still open: items 1–7 above.

Independent check of the write (dated note, 5 October 2026). An independent adversarial check of the write, made by the intake after the write (42721c521), reread the front matter against the Parts, the merge (the restatements, the “answered” notes, the transseries citations), the write’s deductions in the dated notes and Remark 35.1, and recomputed the numbers. It confirmed the live OEIS entries (A258219 revision #30, A292692 #12, A292186 #57, with the histories: Bala’s edits of July 28, 2022, Kotěšovec’s of December 1, 2017 and March 17, 2024) and every quotation from them; that Bala’s equation is (4) rearranged, and that Theorem 1.1 and Proposition 3.2 prove the conjecture for indeterminate k (Bala’s form and the S-fraction agree with the recurrence through x^{24} at $k = 0, \dots, 24$); that Kotěšovec’s form and rate are proved in the same normalization;

C and d to 45 digits, also from their saddle definitions, and, independently of the derivation, C to 27 digits, b_1 to 17 and b_2 to 15 by Richardson extrapolation of the OEIS b-file ($n \leq 290$); the first 91 terms of A258219 and the terms of A292692 for $n \leq 39$ from the path definition by a dynamic program over Dyck prefixes, and longer ranges from the recurrence; Remark 35.1 (with F_q increasing and $F_q/\sqrt{q+1} \in [0.8355, 1.1520]$ for $q \leq 400$, and no violation of (III.13) for $16 \leq N \leq 220$, $q \leq \min(2\sqrt{N}, N/4)$); the leading coefficient L_a of $c_a(n)$; the factorial-core identity and the half-width of $K = 1$; the correction $1 - 3/(8n)$ of V_n ; and the delivered suites on fresh extractions. No mathematical claim of the three manuscripts was found wrong. One statement of the write was false: the note in Subsection 24.1 justified $0 < R_{N,m}$ by $\left\{ \begin{smallmatrix} N \\ m \end{smallmatrix} \right\} > 0$ for $0 \leq m \leq N$, but $\left\{ \begin{smallmatrix} N \\ 0 \end{smallmatrix} \right\} = 0$ for $N \geq 1$; the claim holds at $m = 0$ because $P_{N,0} = p_N(0) > 0$ (A005411), and the same note said that the subsection re-derives Part I's Theorem 1.1, whereas Part III defines p_N by the recurrence and re-derives only its equivalence with the Riccati equation. That note is corrected. Also corrected, each with a dated note keeping the first wording: the Guide's "Each Part ends with open questions" (Part III ends with its reproducibility section); the lists of the Restatements item of the merge decisions; the transseries citations of the front matter and of Part I's note in Section 10 (the factorial core is the Lambert core only after taking logarithms, with $a = b = 1$ in the variable $\log x + d'$, and the ceiling brackets are proved directly, not as instances of the staircase theorem, which presupposes an interpolation); "bounded" for the interior residuals at $\alpha = 0.9$, which are $-5565, -7052, -5496$ at $N = 100, 200, 300$ ("of stable size"); and the proof of Remark 35.1, which cited the monotonicity of Theorem 31.1 without needing it. No other statement or number changed. This was a careful reading with numerical tests, not a formal verification.

References

- [1] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A258219. Weighted Dyck peak array; entry by Alois P. Heinz, May 23, 2015, and Riccati/continued-fraction conjecture by Peter Bala, July 28, 2022. <https://oeis.org/A258219>. Inspected October 2, 2026. [Part III: A258219, weighted Dyck-path polynomials; accessed October 2026.]
- [2] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A258220. Newton-transform triangle; entry by Alois P. Heinz, May 23, 2015. <https://oeis.org/A258220>. Inspected October 2, 2026. [Part III: Newton triangle of the weighted Dyck-path polynomials; accessed October 2026.]
- [3] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A292692. Diagonal $T(2n, n)$ and asymptotic conjecture. <https://oeis.org/A292692>. Inspected October 2, 2026. [Part III: the Newton diagonal $P_{2n,n}$; accessed October 2026.]
- [4] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A005411. <https://oeis.org/A005411>. Inspected October 2, 2026.
- [5] OEIS Foundation, A292186, rooted connected four-regular maps. <https://oeis.org/A292186>. Accessed October 2026. [Cited by Part III.]
- [6] R. J. Martin and M. J. Kearney, An exactly solvable self-convolutive recurrence, *Aequationes Mathematicae* **80** (2010), 291–318. <https://arxiv.org/abs/1103.4936>. [Cited by Parts I and III.]

- [7] M. Borinsky, Renormalized asymptotic enumeration of Feynman diagrams, *Annals of Physics* **385** (2017), 95–135. <https://arxiv.org/abs/1703.00840>.
- [8] A. N. Stokes, Continued fraction solutions of the Riccati equation, *Bulletin of the Australian Mathematical Society* **25** (1982), 207–214. <https://doi.org/10.1017/S0004972700005219>.
- [9] L. Ciobanu and A. Kolpakov, Free subgroups of free products and combinatorial hypermaps, *Discrete Mathematics* **342** (2019), 1415–1433. <https://doi.org/10.1016/j.disc.2019.01.014>; <https://arxiv.org/abs/1708.03842>.
- [10] L. C. Hsu and P. J.-S. Shiue, A unified approach to generalized Stirling numbers, *Advances in Applied Mathematics* **20** (1998), 366–384. <https://doi.org/10.1006/aama.1998.0586>.
- [11] R. Arratia and S. DeSalvo, Completely effective error bounds for Stirling numbers of the first and second kinds via Poisson approximation, 2014. <https://arxiv.org/abs/1404.3007>.
- [12] NIST Digital Library of Mathematical Functions, *Gamma function asymptotic expansions*, Section 5.11, especially Eq. 5.11.8 and Section 5.11(ii). <https://dlmf.nist.gov/5.11>.
- [13] *A weighted Dyck path Newton diagonal*, Report 141, October 2, 2026. [Part II:] Complete unchanged source and PDF included as the foundation of this continuation. Contains the exact weighted-path bridge, the formal convolution, source normalization for OEIS A258219, A258220, and A292692, and the leading and first-corrected diagonal estimates. [Part III:] Unchanged source and PDF included in the archive. Used here for the exact path interpretation of (III.1); the upper-edge proof starts from the recurrence. **[[write]** Printed in this report as Part I; the included copies are byte-identical to it and are not shipped.]
- [14] *All orders for the weighted Dyck Newton diagonal*, Report 142, October 2, 2026. Unchanged source and PDF included in the archive. Used only for the explicit compact-interior coefficient in Section 32; its boundary remainder is not invoked. **[[write]** Printed in this report as Part II; the included copy is byte-identical to it and is not shipped.]
- [15] **[[write]]** *Transseries and inversion*, ProveIt volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: Theorem `p0:thm:lambert-core` (exact solution of the dominant block), Proposition `p0:prop:factorial-core` (exact inversion of a log-linear phase), Theorem `p0:thm:staircase` (staircase recovery and the separation condition).